

Project Data Page:





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CCW Expansion Vessel (V8801)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : D:\QUEP\CCW Expansion Vessel (V8801)-7%.pvdb

Date of Analysis : Sep 16, 2026 3:00pm

PV Elite 28, January 2026

**Vessel Design Summary:****ASME Code, Section VIII Division 1, 2025**

Diameter Spec : 4500.000 mm ID	
Vessel Design Length, Tangent to Tangent	11300.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	82 °C
Internal Design Pressure	0.350 MPa
External Design Temperature	82 °C
External Design Pressure	0.052 MPa
Maximum Allowable Working Pressure	0 MPa
External Max. Allowable Working Pressure	0.1 MPa
Shop Test Pressure	0.455 MPa
Field Test Pressure	0.455 MPa
Required Minimum Design Metal Temperature	-1.0 °C
Warmest Computed Minimum Design Metal Temperature	-22.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Rings	SA-516 70	...	...	K02700	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
left Dish	0.389	0.05	0.000	1.6000	No	No
Shell-1	0.389	0.05	1.000	1.6000	N/A	No
Shell2	0.389	0.05	1.000	1.6000	N/A	No
Shell3	0.389	0.05	1.000	1.6000	N/A	No
shell4	0.389	0.05	1.000	1.6000	N/A	No
Right dish	0.389	0.05	0.000	1.6000	No	No

Liquid Level: 4000.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.989**Stiffener Ring Specifications:**

Elevation mm	Selected Type	Ring Description
1400.00	Bar 150.0 x 12.0	STF Ring-1
4200.00	Bar 150.0 x 12.0	STF Ring-2
7000.00	Bar 150.0 x 12.0	STF Ring-3



9850.00

Bar 150.0 x 12.0

STF Ring-4

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	12.0	10.0	9.1	...	0.85	0.85
Cylinder	3050.0	3000.0	12.0	12.0	9.1	...	0.85	0.85
Cylinder	6050.0	3000.0	12.0	12.0	9.1	...	0.85	0.85
Cylinder	8650.0	2600.0	12.0	12.0	9.1	...	0.85	0.85
Cylinder	11250.0	2600.0	14.0	14.0	9.1	...	0.85	0.85
Ellipse	11300.0	50.0	12.0	10.0	9.1	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	10	9.87542	0.051714	0.053285
20	Ring	12	9.09731	0.051714	0.11938
Ring	30	12	10.6688	0.051714	0.073861
30	Ring	12	10.6688	0.051714	0.073861
Ring	40	12	10.6688	0.051714	0.073861
40	Ring	12	10.6688	0.051714	0.073861
Ring	50	12	10.7289	0.051714	0.072509
50	Ring	14	10.7339	0.051714	0.11367
Ring	70	14	9.19022	0.051714	0.18192
70	80	10	9.87542	0.051714	0.053285

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	1775	3947.05	No Calc	No Calc
Ring	30	2800	3947.29	7543757	10107544
30	Ring	2800	3947.29	No Calc	No Calc
Ring	40	2800	3947.29	9231119	10107544
40	Ring	2800	3947.29	No Calc	No Calc
Ring	50	2850	3947.4	9308742	10107544
50	Ring	2850	6094.5	No Calc	No Calc
Ring	70	1825	6092.63	7701523	11039288
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Trunnion	WRC-107/537	0.922	n/a	Passed
Trunnion	WRC-107/537	0.751	n/a	Passed
N3-4"	WRC-297	0.935	Shell	Passed
N4-4"	WRC-297	0.935	Shell	Passed
N1-24	WRC-297	0.950	Nozzle	Passed
N2-8"	WRC-297	0.482	Nozzle	Passed
N5-4"	WRC-297	0.935	Shell	Passed
N6-4"	WRC-297	0.935	Shell	Passed

Foundation/Support Summary**Saddle Parameters:**

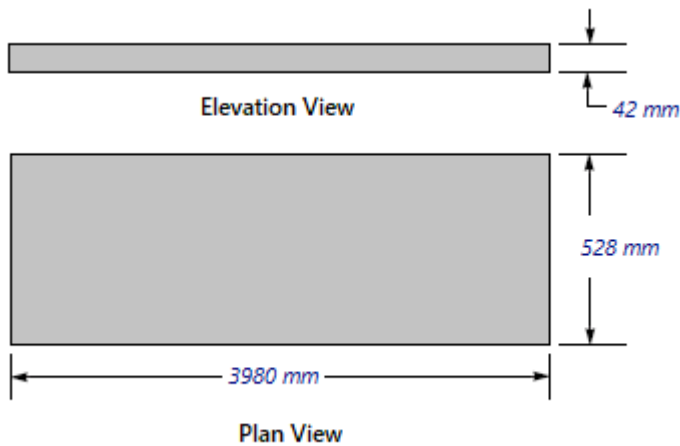
Saddle Width

350.000 mm



Saddle Bearing Angle	150.000	deg.
Centerline Dimension	2550.000	mm
Wear Pad Width	700.000	mm
Wear Pad Thickness	20.000	mm
Wear Pad Bearing Angle	164.000	deg.
Distance from Saddle to Tangent	2240.000	mm
Baseplate Length	3980.000	mm
Baseplate Thickness	42.000	mm
Baseplate Width	528.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	25.000	mm
Web Thickness	25.000	mm
Height of Center Web	264.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	1303352.62	N
Maximum Transverse Saddle Shear Load	72838.75	N
Maximum Longitudinal Saddle Shear Load	145677.48	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	1214921.88	N
Maximum Transverse Saddle Shear Load	9534.02	N
Maximum Longitudinal Saddle Shear Load	11675.81	N

Weights:

Fabricated - Bare w/o removable internals	30029.3	kgm
Shop Test - Fabricated + water (full)	233516.7	kgm
Shipping - Fab. + rem. intls.+ shipping app.	30029.3	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	39594.3	kgm
Empty - Fab. + intls. + details + wghts.	39594.3	kgm
Operating - Empty + operating liquid (no ca)	229282.0	kgm
Field Test - Empty weight + test medium (full)	243081.7	kgm

**Nozzle Calculation Summary (MPa & mm):**

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1-24"	...	OK	...	...		OK	Passed
K2B-2"	1	OK	...	OK	6.40	OK	Passed
K2A-2"	1	OK	...	OK	6.40	OK	Passed
N3-4"	1	OK	...	OK	6.86	OK	Passed
N4-4"	1	OK	...	OK	6.86	OK	Passed
K3-2"	1	OK	...	OK	6.40	OK	Passed
K4-2"	1	OK	...	OK	6.40	OK	Passed
N1-24	1	OK	...	OK	8.76	OK	Passed
N2-8"	5	OK	...	OK	4.10	OK	Passed
N5-4"	1	OK	...	OK	6.86	OK	Passed
N6-4"	1	OK	...	OK	6.86	OK	Passed
K1A-2"	1	OK	...	OK	6.40	OK	Passed
K1B-2"	1	OK	...	OK	6.40	OK	Passed
M2-24"	...	OK	...	...		OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 0 Nozzle : M2-24"
 Minimum MAWP Shells/Flanges : 0 Element : Right dish
 Minimum MAPnc Shells/Flanges : 1 Element : Right dish

Computed Vessel M.A.W.P. : 0 MPa

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	M1-24"	0.000	0.000	1113.600
20	K2B-2"	650.000	180.000	108.000
20	K2A-2"	550.000	0.000	108.000
20	N3-4"	1050.000	0.000	175.081
20	N4-4"	2550.000	0.000	175.081
30	K3-2"	3450.000	0.000	108.000
30	K4-2"	4050.000	0.000	108.000
40	N1-24	7550.000	180.000	1000.000
40	N2-8"	0.000	180.000	380.631
40	N5-4"	7250.000	0.000	175.081
40	N6-4"	7650.000	0.000	175.081
50	K1A-2"	10149.999	0.000	109.200
50	K1B-2"	10149.999	180.000	109.200
70	M2-24"	0.000	0.000	1113.600

The nozzle spacing is computed by the following:

$$= \sqrt{(l_l^2 + l_c^2)} \text{ where}$$

l_l - Arc length along the inside vessel surface in the long. direction.

l_c - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
K2B-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	218.77	300
K2A-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	490.04	300
K3-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	490.04	300
K4-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	490.04	300
K1A-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	492.04	300
K1B-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	220.77	300
N3-4"	4.000 in	XXS	WNF	4.500	17.120	...	...	426.87	300
N4-4"	4.000 in	XXS	WNF	4.500	17.120	...	...	426.87	300
N5-4"	4.000 in	XXS	WNF	4.500	17.120	...	...	426.87	300
N6-4"	4.000 in	XXS	WNF	4.500	17.120	...	...	426.87	300
N2-8"	8.000 in	120	WNF	8.625	18.263	...	...	241.91	150
M1-24"	24.000 in	Actual	WNF	24.000	28.000	909.60	12.00	271.46	150
N1-24	24.000 in	Actual	WNF	24.000	20.000	...	...	794.74	150
M2-24"	24.000 in	Actual	WNF	24.000	28.000	909.60	12.00	271.46	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K2B-2"	SA-105	12.000	10.000	...	...	...
K2A-2"	SA-105	12.000	10.000	...	...	...
K3-2"	SA-105	12.000	10.000	...	...	...
K4-2"	SA-105	12.000	10.000	...	...	...
K1A-2"	SA-105	12.000	10.000	...	...	...
K1B-2"	SA-105	12.000	10.000	...	...	...
N3-4"	SA-106 B	12.000	10.000	...	...	...
N4-4"	SA-106 B	12.000	10.000	...	...	...
N5-4"	SA-106 B	12.000	10.000	...	...	...
N6-4"	SA-106 B	12.000	10.000	...	...	...
N2-8"	SA-106 B	12.000	10.000	...	...	...
M1-24"	SA-516 70	10.000	10.000	12.000	12.000	...
N1-24	SA-516 70	12.000	10.000	...	...	...
M2-24"	SA-516 70	10.000	10.000	12.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K2B-2"	650.000	180.0	206.38	0.00	Shell-1
K2A-2"	550.000	0.0	477.65	0.00	Shell-1
K3-2"	3450.000	0.0	477.65	0.00	Shell12
K4-2"	4050.000	0.0	477.65	0.00	Shell12
K1A-2"	10149.999	0.0	477.65	0.00	shell14
K1B-2"	10149.999	180.0	206.38	0.00	shell14
N3-4"	1050.000	0.0	414.15	0.00	Shell-1
N4-4"	2550.000	0.0	414.15	0.00	Shell-1

CCW Expansion Vessel (V8801)

PV Elite 28

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FileName : CCW Expansion Vessel (V8801)-7%

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Nozzle Schedule:

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HEXAGON

N5-4"	7250.000	0.0	414.15	0.00	Shell3
N6-4"	7650.000	0.0	414.15	0.00	Shell3
N2-8"	6550.000	180.0	200.00	0.00	N1-24
M1-24"	...	0.0	250.00	0.00	left Dish
N1-24	7550.000	180.0	750.00	0.00	Shell3
M2-24"	...	0.0	250.00	0.00	Right dish

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**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm THK X 4500.0mm ID X 50.0mm	SA-516 70
2	SADDLE: 350.0mm X 150 DEG	SA-516 70
1	PLATFORM: 1000mm X 4524mm WIDE	
1	LADDER: 6600.0mm	...
3	BAR RING STIFFENER: 4524.0mm ID X 4824.0mm OD X 12.0mm	SA-516 70
4	TRUNNIONS:	SA-516 70
4	TRUNNION PADS: 750.0mm OD X 355.6mm ID X 12.0mm THK	
1	BAR RING STIFFENER: 4528.0mm ID X 4828.0mm OD X 12.0mm	SA-516 70
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
9	CYLINDRICAL SEGMENT 12.0mm THK X 2491.1mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 1681.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 11645.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 7473.2mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 12.0mm THK X 2491.1mm X 2600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 11645.9mm X 2600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 7473.2mm X 2600.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 14.0mm THK X 2491.1mm X 2600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 11645.9mm X 2600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 7473.2mm X 2600.0mm	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
left Dish	[10]	B	-29	-37	-29	0.850	10.000	0.85	No
left Dish	[7]	B	-23	-40	-29	0.688	12.000	0.85	No
Shell-1	[8]	B	-23	-40	-29	0.688	12.000	0.85	No
Shell2	[8]	B	-23	-40	-29	0.688	12.000	0.85	No
Shell3	[8]	B	-23	-40	-29	0.688	12.000	0.85	No
shell4	[8]	B	-18	-43	-29	0.577	14.000	0.85	No
Right dish	[10]	B	-29	-37	-29	0.850	10.000	0.85	No
Right dish	[7]	B	-23	-40	-29	0.688	12.000	0.85	No
STF Ring-1	[23]	B	-23	-40	-29	0.688	12.000	0.85	No
STF Ring-2	[23]	B	-23	-40	-29	0.688	12.000	0.85	No
STF Ring-3	[23]	B	-23	-40	-29	0.688	12.000	0.85	No
STF Ring-4	[23]	B	-23	-47	-29	0.577	12.000	0.85	No
Left Saddle	[24]	B	-23	-40	-29	0.688	12.000	0.85	No
Right Saddle	[24]	B	-18	-43	-29	0.577	14.000	0.85	No
M1-24"	[1]	B	-29	-44	-29	0.727	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2B-2"	[1]	A	-3	-29	-29	0.688	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2A-2"	[1]	A	-3	-29	-29	0.627	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-4"	[1]	B	-23	-44	-29	0.627	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4-4"	[1]	B	-23	-44	-29	0.627	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3-2"	[1]	A	-3	-29	-29	0.627	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4-2"	[1]	A	-3	-29	-29	0.627	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-24	[1]	B	12	-104		0.052	20.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N2-8"	[1]	B	6	-104		0.028	15.980	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N5-4"	[1]	B	-23	-44	-29	0.627	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6-4"	[1]	B	-23	-44	-29	0.627	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A-2"	[1]	A	2	-27		0.526	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1B-2"	[1]	A	2	-22		0.577	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M2-24"	[1]	B	-29	-44	-29	0.727	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 12 -22

Required Minimum Design Metal Temperature -1.0 °C

Warmest Computed Minimum Design Metal Temperature -22.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable



- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

CCW Expansion Vessel (V8801)

Design Internal Pressure (for Hydrotest)	0.35	MPa
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	500	mm
Projection of Nozzle from Vessel Bottom	500	mm
Minimum Design Metal Temperature	-1.0	°C
Type of Construction	Welded	
Special Service	Air/Water/Steam	
Degree of Radiography	RT-3	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	21500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.770
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes



Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	left Dish
Distance "FROM" to "TO"	50 mm
Inside Diameter	4500 mm
Element Thickness	10 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.05171 MPa
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	4000 mm
Liquid Density	0.0009874 kg/cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	M1-24"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	10
Detail Type	Weight
Detail ID	Davit-1
Dist. from "FROM" Node / Offset dist	-900 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	300 mm

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	3000 mm



Inside Diameter	4500	mm
Element Thickness	12	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.35	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.05171	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	2190	mm
Width of Saddle	350	mm
Height of Saddle at Bottom	2550	mm
Saddle Contact Angle	150.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	700	mm
Thickness of Wear Plate	20	mm
Contact Angle, Wear Plate (degrees)	164.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	13982201	mm ²
Platform Weight	87658	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	4524	mm
Platform Height	1000	mm
Platform Clearance or End Offset	150	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	-2000	mm
Ladder End Elevation	4600	mm
Unit Weight of Ladder	0.3628	N/mm
Platform Length (top head platform)	11300	mm

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	4000	mm
Liquid Density	0.0009874	kg/cm ³

Element From Node	20	
Detail Type	Ring	
Detail ID	STF Ring-1	
Dist. from "FROM" Node / Offset dist	1350	mm
Inside Diameter of Ring	4524	mm
Thickness of Ring	12	mm
Outside Diameter of Ring	4824	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm



Using Custom Stiffener Section	No
Element From Node	20
Detail Type	Nozzle
Detail ID	K2B-2"
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K2A-2"
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	N3-4"
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	4 in.
Nozzle Schedule	XXS
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	321.56 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	20
Detail Type	Nozzle
Detail ID	N4-4"
Dist. from "FROM" Node / Offset dist	2500 mm
Nozzle Diameter	4 in.
Nozzle Schedule	XXS
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell12
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4500 mm
Element Thickness	12 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	82 °C



Design External Pressure	0.05171	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	4000	mm
Liquid Density	0.0009874	kg/cm³

Element From Node	30	
Detail Type	Ring	
Detail ID	STF Ring-2	
Dist. from "FROM" Node / Offset dist	1150	mm
Inside Diameter of Ring	4524	mm
Thickness of Ring	12	mm
Outside Diameter of Ring	4824	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K3-2"	
Dist. from "FROM" Node / Offset dist	400	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K4-2"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	30	
Detail Type	Weight	
Detail ID	Clip	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	1078.7	N
Offset from Element Centerline	0	mm

Element#	4/6	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell3	
Distance "FROM" to "TO"	2600	mm
Inside Diameter	4500	mm
Element Thickness	12	mm



Internal Corrosion Allowance	1.6	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.35	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.05171	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	4000	mm
Liquid Density	0.0009874	kg/cm³
Element From Node	40	
Detail Type	Ring	
Detail ID	STF Ring-3	
Dist. from "FROM" Node / Offset dist	950	mm
Inside Diameter of Ring	4524	mm
Thickness of Ring	12	mm
Outside Diameter of Ring	4824	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N1-24	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	3409.1	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2-8"	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	8	in.
Nozzle Schedule	120	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	446.36	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N5-4"	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	4	in.
Nozzle Schedule	XXS	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	321.56	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



Element From Node	40
Detail Type	Nozzle
Detail ID	N6-4"
Dist. from "FROM" Node / Offset dist	1600 mm
Nozzle Diameter	4 in.
Nozzle Schedule	XXS
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	321.56 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	40
Detail Type	Weight
Detail ID	N1-I.P. wt
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	196.12 N
Offset from Element Centerline	0 mm

Element#	5/6
Element From Node	50
Element To Node	70
Element Type	Cylinder
Description	shell4
Distance "FROM" to "TO"	2600 mm
Inside Diameter	4500 mm
Element Thickness	14 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.05171 MPa
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Right Saddle
Dist. from "FROM" Node / Offset dist	410 mm
Width of Saddle	350 mm
Height of Saddle at Bottom	2550 mm
Saddle Contact Angle	150.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	700 mm
Thickness of Wear Plate	20 mm
Contact Angle, Wear Plate (degrees)	164.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	4000 mm
Liquid Density	0.0009874 kg/cm³



Element From Node 50
 Detail Type Ring
 Detail ID STF Ring-4
 Dist. from "FROM" Node / Offset dist 1200 mm
 Inside Diameter of Ring 4528 mm
 Thickness of Ring 12 mm
 Outside Diameter of Ring 4828 mm
 Material Name SA-516 70
 Height of Section Ring 0 mm
 Using Custom Stiffener Section No

Element From Node 50
 Detail Type Nozzle
 Detail ID K1A-2"
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element From Node 50
 Detail Type Nozzle
 Detail ID K1B-2"
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element# 6/6
 Element From Node 70
 Element To Node 80
 Element Type Elliptical
 Description Right dish
 Distance "FROM" to "TO" 50 mm
 Inside Diameter 4500 mm
 Element Thickness 10 mm
 Internal Corrosion Allowance 1.6 mm
 Nominal Thickness 12 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.35 MPa
 Design Temperature Internal Pressure 82 °C
 Design External Pressure 0.05171 MPa
 Design Temperature External Pressure 82 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 0.85
 Efficiency, Circumferential Seam 0.85
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 70
 Detail Type Liquid
 Detail ID Liquid 70
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 4000 mm
 Liquid Density 0.0009874 kg/cm³

Element From Node 70



Detail Type	Nozzle
Detail ID	M2-24"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	70
Detail Type	Weight
Detail ID	Davit-2
Dist. from "FROM" Node / Offset dist	900 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	300 mm

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
left Dish		50	...	50	...
Shell-1		3050	...	3000	...
Shell2		6050	...	3000	...
Shell3		8650	...	2600	...
shell4		11250	...	2600	...
Right dish		11300	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
left Dish		0.3888	12	1.6	4500	117.3	1
Shell-1		0.3888	12	1.6	4500	117.3	2
Shell2		0.3888	12	1.6	4500	117.3	3
Shell3		0.3888	12	1.6	4500	117.3	4
shell4		0.3888	14	1.6	4500	117.3	5
Right dish		0.3888	12	1.6	4500	117.3	6

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
left Dish		0.35002	0.39906	0.52109	10	9.05892	1
Shell-1		0.35002	0.5015	0.62359	12	9.07839	2
Shell2		0.35002	0.5015	0.62359	12	9.07839	3
Shell3		0.35002	0.5015	0.62359	12	9.07839	4
shell4		0.35002	0.60506	0.72714	14	9.07839	5
Right dish		0.35002	0.39906	0.52109	10	9.05892	6

Minimum 0 1

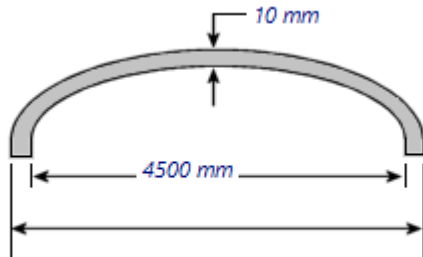
MAWP: 0 MPa, limited by: Right dish.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: left Dish

Elliptical Head Design Information:					
Design Pressure	P	0.389 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4500.000 mm	Actual Stress		122.5 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		22896100.0 mm²	SG of Contents		0.989
Element Volume		12725.7 ltr	Weight of Contents		119108.6 N
Empty Weight		22868.3 N	Operating Weight		141976.9 N
			Comp. All. Ope. Stress		55.7 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 56.250 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.125 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.389 \cdot 4503.2002 \cdot 0.999) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 0.389)$$

$$= 7.4589 + 1.6000 = 9.0589 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 8.4) / (0.999 \cdot 4503.2002 + 0.2 \cdot 8.4)$$

$$= 0 - 0 = 0 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.0 \cdot 4500.0 + 0.2 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.389 \cdot (0.999 \cdot 4503.2002 + 0.2 \cdot 8.4)) / (2 \cdot 0.85 \cdot 8.4)$$

$$= 122.542 \text{ MPa}$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (0.389 \cdot 2251.6001) / (138.0 \cdot 0.85 - 0.6 \cdot 0.389) + 1.6$$

$$= 9.078 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.4) / (2251.6001 + 0.6 \cdot 10.4)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (4503.2 / (2 \cdot 1126.6))^2) / 6$$

$$= 0.999054$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.167 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10.0$, $t_r = 8.4$, $c = 1.6$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 8.424$, $c = 1.6$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -40 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.438 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-37 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 56.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.125 mm

EXTERNAL PRESSURE RESULTS for: left Dish

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
8.400	4520.00	538.10	0.0002581	25.81

$$MAEP = B / (K_0 \cdot D_o / t) = 25.8054 / (0.9 \cdot 538.0953) = 0.0533 \text{ MPa}$$



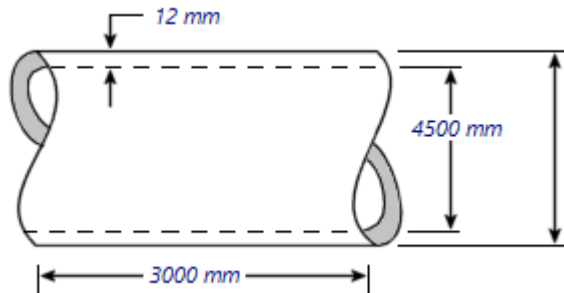
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
8.275	4520.00	546.20	0.0002543	25.42

$$MAEP = B / (K0 * Do / t) = 25.4227 / (0.9 * 546.1957) = 0.0517 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.389 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4500.000 mm	Actual Stress		99.3 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		42637700.0 mm²	SG of Contents		0.989
Element Volume		47721.5 ltr	Weight of Contents		434426.7 N
Empty Weight		38783.3 N	Operating Weight		473210.0 N
			Comp. All. Ope. Stress		69.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.389 \cdot 2251.6001) / (138.0 \cdot 0.85 \cdot 0.6 \cdot 0.389) \\
 &= 7.4784 + 1.6000 = 9.0784 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 10.4) / (2251.6001 + 0.6 \cdot 10.4) \\
 &= 1 - 0 = 1 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.0) / (2250.0 + 0.6 \cdot 12.0) \\
 &= 1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.389 \cdot (2251.6001 + 0.6 \cdot 10.4)) / (0.85 \cdot 10.4) \\
 &= 99.308 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.266 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 12.0$, $t_r = 8.424$, $c = 1.6$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.438 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.688	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-40 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 20 to STF Ring-1 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	1775.00	435.00	0.3924	0.0003896	38.95

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.9491)/(3*435.0) = 0.1194 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.497	4524.00	1775.00	603.42	0.3924	0.0002341	23.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.4051)/(3*603.4167) = 0.0517 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	3947.05	435.00	0.8725	0.0001689	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8823)/(3*435.0) = 0.0517 \text{ MPa}$$

Cylindrical Shell From STF Ring-1 to 30 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	2800.00	435.00	0.6189	0.0002410	24.10

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.097)/(3*435.0) = 0.0739 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.069	4524.00	2800.00	498.85	0.6189	0.0001935	19.35



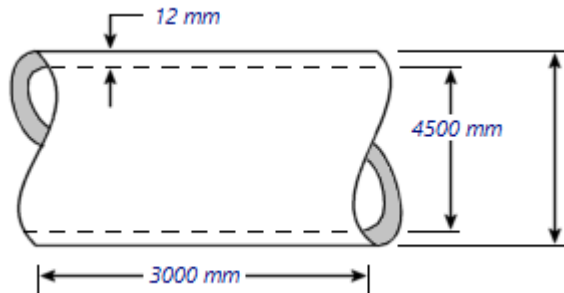
$$MAEP = (4*B)/(3*(Do/t)) = (4*19.3487)/(3*498.8507) = 0.0517 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	3947.29	435.00	0.8725	0.0001689	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8812)/(3*435.0) = 0.0517 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell2

Cylindrical Shell Design Information:					
Design Pressure	P	0.389 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4500.000 mm	Actual Stress		99.3 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		42637700.0 mm²	SG of Contents		0.989
Element Volume		47721.5 ltr	Weight of Contents		434426.7 N
Empty Weight		38783.3 N	Operating Weight		473210.0 N
			Comp. All. Ope. Stress		69.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.389 \cdot 2251.6001) / (138.0 \cdot 0.85 - 0.6 \cdot 0.389) \\
 &= 7.4784 + 1.6000 = 9.0784 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 10.4) / (2251.6001 + 0.6 \cdot 10.4) \\
 &= 1 - 0 = 1 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.0) / (2250.0 + 0.6 \cdot 12.0) \\
 &= 1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.389 \cdot (2251.6001 + 0.6 \cdot 10.4)) / (0.85 \cdot 10.4) \\
 &= 99.308 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.266 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 12.0$, $t_r = 8.424$, $c = 1.6$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.438 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.688	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-40 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Shell2

Cylindrical Shell From 30 to STF Ring-2 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	2800.00	435.00	0.6189	0.0002410	24.10

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.097)/(3*435.0) = 0.0739 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.069	4524.00	2800.00	498.85	0.6189	0.0001935	19.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.3487)/(3*498.8507) = 0.0517 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	3947.29	435.00	0.8725	0.0001689	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8812)/(3*435.0) = 0.0517 \text{ MPa}$$

Cylindrical Shell From STF Ring-2 to 40 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	2800.00	435.00	0.6189	0.0002410	24.10

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.097)/(3*435.0) = 0.0739 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.069	4524.00	2800.00	498.85	0.6189	0.0001935	19.35



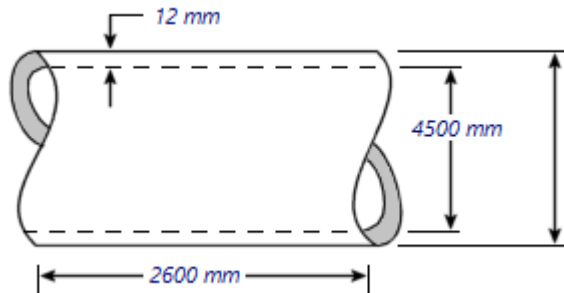
$$MAEP = (4*B)/(3*(Do/t)) = (4*19.3487)/(3*498.8507) = 0.0517 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	3947.29	435.00	0.8725	0.0001689	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8812)/(3*435.0) = 0.0517 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell3

Cylindrical Shell Design Information:					
Design Pressure	P	0.389 MPa	Int. Design Temperature	82 °C	
Material		SA-516 70	Ext. Design Temperature	82 °C	
Length		2600.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.6000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4500.000 mm	Actual Stress	99.3 MPa	
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.	12.000 mm	
Surface Area		36952672.0 mm²	SG of Contents	0.989	
Element Volume		41358.6 ltr	Weight of Contents	376503.2 N	
Empty Weight		33612.2 N	Operating Weight	410115.4 N	
			Comp. All. Ope. Stress	69.0 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.389 \cdot 2251.6001) / (138.0 \cdot 0.85 - 0.6 \cdot 0.389) \\
 &= 7.4784 + 1.6000 = 9.0784 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 10.4) / (2251.6001 + 0.6 \cdot 10.4) \\
 &= 1 - 0 = 1 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.0) / (2250.0 + 0.6 \cdot 12.0) \\
 &= 1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.389 \cdot (2251.6001 + 0.6 \cdot 10.4)) / (0.85 \cdot 10.4) \\
 &= 99.308 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.266 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 12.0$, $t_r = 8.424$, $c = 1.6$ mm, $E^* = 0.85$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.438 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.688	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-40 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Shell3

Cylindrical Shell From 40 to STF Ring-3 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	2800.00	435.00	0.6189	0.0002410	24.10

$$MAEP = (4*B) / (3*(Do/t)) = (4*24.097) / (3*435.0) = 0.0739 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.069	4524.00	2800.00	498.85	0.6189	0.0001935	19.35

$$MAEP = (4*B) / (3*(Do/t)) = (4*19.3487) / (3*498.8507) = 0.0517 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	3947.29	435.00	0.8725	0.0001689	16.88

$$MAEP = (4*B) / (3*(Do/t)) = (4*16.8812) / (3*435.0) = 0.0517 \text{ MPa}$$

Cylindrical Shell From STF Ring-3 to 50 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	2850.00	435.00	0.6300	0.0002366	23.66

$$MAEP = (4*B) / (3*(Do/t)) = (4*23.6562) / (3*435.0) = 0.0725 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.129	4524.00	2850.00	495.57	0.6300	0.0001923	19.22



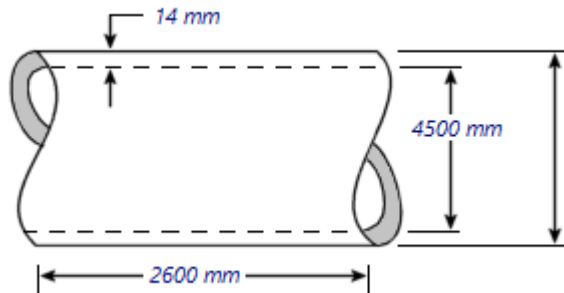
$$MAEP = (4*B)/(3*(Do/t)) = (4*19.2214)/(3*495.5679) = 0.0517 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4524.00	3947.40	435.00	0.8725	0.0001688	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8807)/(3*435.0) = 0.0517 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: shell4

Cylindrical Shell Design Information:					
Design Pressure	P	0.389 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		2600.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4500.000 mm	Actual Stress		83.3 MPa
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		36985340.0 mm²	SG of Contents		0.989
Element Volume		41358.6 ltr	Weight of Contents		376503.2 N
Empty Weight		39231.4 N	Operating Weight		415734.5 N
			Comp. All. Ope. Stress		82.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.389 \cdot 2251.6001) / (138.0 - 0.6 \cdot 0.389) \\
 &= 7.4784 + 1.6000 = 9.0784 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.4) / (2251.6001 + 0.6 \cdot 12.4) \\
 &= 1 - 0 = 1 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 14.0) / (2250.0 + 0.6 \cdot 14.0) \\
 &= 1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.389 \cdot (2251.6001 + 0.6 \cdot 12.4)) / (0.85 \cdot 12.4) \\
 &= 83.335 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.310 %

Minimum Design Metal Temperature Results:

Govern. thk, tg = 14.0, tr = 8.424, c = 1.6 mm, E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.577, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.438 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.577	Temperature Difference	24 °C
UCS-66(c) red.	No	Computed Min. Temp.	-43 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: shell4

Cylindrical Shell From 50 to STF Ring-4 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.400	4528.00	2850.00	365.16	0.6294	0.0003114	31.13

$$MAEP = (4*B)/(3*(Do/t)) = (4*31.1321)/(3*365.1613) = 0.1137 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.134	4528.00	2850.00	495.73	0.6294	0.0001923	19.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.2277)/(3*495.7336) = 0.0517 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.400	4528.00	6094.50	365.16	1.3460	0.0001417	14.17

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.1707)/(3*365.1613) = 0.0517 \text{ MPa}$$

Cylindrical Shell From STF Ring-4 to 70 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.400	4528.00	1825.00	365.16	0.4030	0.0004983	49.82

$$MAEP = (4*B)/(3*(Do/t)) = (4*49.8228)/(3*365.1613) = 0.1819 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.590	4528.00	1825.00	596.56	0.4030	0.0002314	23.14



$$MAEP = (4*B)/(3*(Do/t)) = (4*23.139)/(3*596.5571) = 0.0517 \text{ MPa}$$

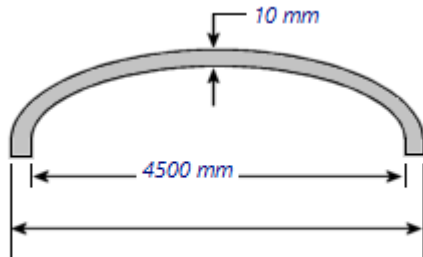
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.400	4528.00	6092.63	365.16	1.3455	0.0001418	14.18

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.1752)/(3*365.1613) = 0.0518 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Right dish

Elliptical Head Design Information:					
Design Pressure	P	0.389 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4500.000 mm	Actual Stress		122.5 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		22896100.0 mm²	SG of Contents		0.989
Element Volume		12725.7 ltr	Weight of Contents		119108.6 N
Empty Weight		22868.3 N	Operating Weight		141976.9 N
			Comp. All. Ope. Stress		82.1 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 56.250 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.125 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.389 \cdot 4503.2002 \cdot 0.999) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 0.389)$$

$$= 7.4589 + 1.6000 = 9.0589 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 8.4) / (0.999 \cdot 4503.2002 + 0.2 \cdot 8.4)$$

$$= 0 - 0 = 0 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.0 \cdot 4500.0 + 0.2 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.389 \cdot (0.999 \cdot 4503.2002 + 0.2 \cdot 8.4)) / (2 \cdot 0.85 \cdot 8.4)$$

$$= 122.542 \text{ MPa}$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (0.389 \cdot 2251.6001) / (138.0 \cdot 0.85 - 0.6 \cdot 0.389) + 1.6$$

$$= 9.078 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.4) / (2251.6001 + 0.6 \cdot 10.4)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (4503.2 / (2 \cdot 1126.6))^2) / 6$$

$$= 0.999054$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.167 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10.0$, $t_r = 8.4$, $c = 1.6$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 8.424$, $c = 1.6$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -40 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.438 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-37 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 56.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.125 mm

EXTERNAL PRESSURE RESULTS for: Right dish

Elliptical Head From 70 to 80 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
8.400	4520.00	538.10	0.0002581	25.81

$$MAEP = B / (K_0 \cdot D_o / t) = 25.8054 / (0.9 \cdot 538.0953) = 0.0533 \text{ MPa}$$



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
8.275	4520.00	546.20	0.0002543	25.42

$$MAEP = B / (K0 * Do / t) = 25.4227 / (0.9 * 546.1957) = 0.0517 \text{ MPa}$$

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.519 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.623 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.439 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.501 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.501 MPa
App 27-4	= M.A.W.P.	0.399 MPa

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 0.35 • 1.0

= 0.455 MPa

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
left Dish	133.5	234.0	0.570	0.50
Shell-1	111.5	234.0	0.477	0.50
Shell12	111.5	234.0	0.477	0.50
Shell13	111.5	234.0	0.477	0.50
shell14	95.7	234.0	0.409	0.50
Right dish	133.5	234.0	0.570	0.50

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K2B-2"	Nozzle	138.00	138.00	1.000
K2A-2"	Nozzle	138.00	138.00	1.000
N3-4"	Nozzle	118.00	118.00	1.000
N4-4"	Nozzle	118.00	118.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
N1-24	Nozzle	138.00	138.00	1.000
N2-8"	Nozzle	118.00	118.00	1.000
N5-4"	Nozzle	118.00	118.00	1.000
N6-4"	Nozzle	118.00	118.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000
K1B-2"	Nozzle	138.00	138.00	1.000
M2-24"	Nozzle	138.00	138.00	1.000
M2-24"	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
STF Ring-1	138.00	138.00	1.000
STF Ring-2	138.00	138.00	1.000
STF Ring-3	138.00	138.00	1.000
STF Ring-4	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):



Description	Ambient	Operating	Ratio
left Dish	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell2	138.00	138.00	1.000
Shell3	138.00	138.00	1.000
shell4	138.00	138.00	1.000
Right dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
M1-24"	5.29	179.40	0.029
K2B-2"	1.06	179.40	0.006
K2A-2"	1.06	179.40	0.006
N3-4"	1.72	216.00	0.008
N4-4"	1.72	153.40	0.011
K3-2"	1.06	179.40	0.006
K4-2"	1.06	179.40	0.006
N1-24	7.48	179.40	0.042
N2-8"	3.25	153.40	0.021
N5-4"	1.72	153.40	0.011
N6-4"	1.72	153.40	0.011
K1A-2"	1.06	179.40	0.006
K1B-2"	1.06	179.40	0.006
M2-24"	5.29	179.40	0.029

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.519 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.623 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.439 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.501 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.501 MPa
App 27-4	= M.A.W.P.	0.399 MPa

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 0.35 • 1.0

= 0.455 MPa

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
left Dish	158.9	234.0	0.679	0.50
Shell-1	128.7	234.0	0.550	0.50
Shell12	128.7	234.0	0.550	0.50
Shell13	128.7	234.0	0.550	0.50
shell14	108.0	234.0	0.462	0.50
Right dish	158.9	234.0	0.679	0.50

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K2B-2"	Nozzle	138.00	138.00	1.000
K2A-2"	Nozzle	138.00	138.00	1.000
N3-4"	Nozzle	118.00	118.00	1.000
N4-4"	Nozzle	118.00	118.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
N1-24	Nozzle	138.00	138.00	1.000
N2-8"	Nozzle	118.00	118.00	1.000
N5-4"	Nozzle	118.00	118.00	1.000
N6-4"	Nozzle	118.00	118.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000
K1B-2"	Nozzle	138.00	138.00	1.000
M2-24"	Nozzle	138.00	138.00	1.000
M2-24"	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
STF Ring-1	138.00	138.00	1.000
STF Ring-2	138.00	138.00	1.000
STF Ring-3	138.00	138.00	1.000
STF Ring-4	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):



Description	Ambient	Operating	Ratio
left Dish	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell2	138.00	138.00	1.000
Shell3	138.00	138.00	1.000
shell4	138.00	138.00	1.000
Right dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
M1-24"	5.29	179.40	0.029
K2B-2"	1.06	179.40	0.006
K2A-2"	1.06	179.40	0.006
N3-4"	1.72	216.00	0.008
N4-4"	1.72	153.40	0.011
K3-2"	1.06	179.40	0.006
K4-2"	1.06	179.40	0.006
N1-24	7.48	179.40	0.042
N2-8"	3.25	153.40	0.021
N5-4"	1.72	153.40	0.011
N6-4"	1.72	153.40	0.011
K1A-2"	1.06	179.40	0.006
K1B-2"	1.06	179.40	0.006
M2-24"	5.29	179.40	0.029

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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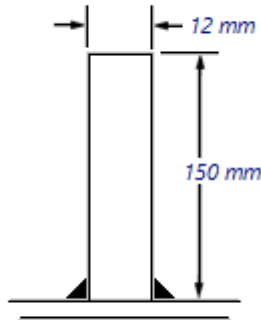
MDMT Calculations for: STF Ring-1, Shell to Ring Junction, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 8.424$, $c = 1.6$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: STF Ring-1, SA-516 70 , Bar Ring: 150.0 x 12.0 mm

Material treatment options:



Effective Length of Shell: 239 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	2481.441	5.2000	12903.496
Ring :	1799.999	85.4000	153719.906
Total:	4281.441		166623.406

Centroid of Ring plus Shell: 39 mm

	Inertia	Distance	A*Dist ²
Shell:	22366.039	33.7176	2821092.500
Ring :	3374993.750	-46.4824	3889099.000
Total:	3397359.750		6710191.500

Available Moment of Inertia, Ring + Shell: 0.101E+08 mm⁴

Required Stress in Ring plus Shell Breq 15.68 MPa
 Required Strain in Ring plus Shell Areq 0.0001570

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4524.0^2 \cdot 2287.4998(10.4 + 1800.0/2287.4998)0.000157) / 10.9$$

$$= 7543758 \text{ mm}^4$$

MDMT Calculations for: STF Ring-2, Shell to Ring Junction, Curve: B

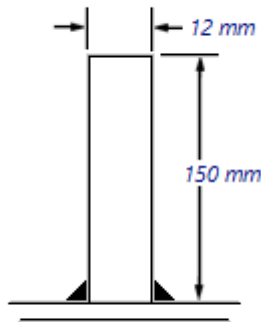
Govrn. thk, $t_g = 12.0$, $t_r = 8.424$, $c = 1.6$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: STF Ring-2, SA-516 70 , Bar Ring: 150.0 x 12.0 mm

Material treatment options:



Effective Length of Shell: 239 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	2481.441	5.2000	12903.496
Ring :	1799.999	85.4000	153719.906
Total:	4281.441		166623.406

Centroid of Ring plus Shell: 39 mm

	Inertia	Distance	A*Dist ²
Shell:	22366.039	33.7176	2821092.500
Ring :	3374993.750	-46.4824	3889099.000
Total:	3397359.750		6710191.500

Available Moment of Inertia, Ring + Shell: 0.101E+08 mm⁴

Required Stress in Ring plus Shell Breq 15.89 MPa
 Required Strain in Ring plus Shell Areq 0.0001590

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4524.0^2 \cdot 2800.0(10.4 + 1800.0/2800.0)0.000159) / 10.9$$

$$= 9231119 \text{ mm}^4$$

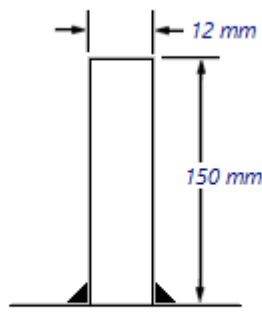
MDMT Calculations for: STF Ring-3, Shell to Ring Junction, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 8.424$, $c = 1.6$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: STF Ring-3, SA-516 70 , Bar Ring: 150.0 x 12.0 mm

Material treatment options:



Effective Length of Shell: 239 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	2481.441	5.2000	12903.496
Ring :	1799.999	85.4000	153719.906
Total:	4281.441		166623.406

Centroid of Ring plus Shell: 39 mm

	Inertia	Distance	A*Dist ²
Shell:	22366.039	33.7176	2821092.500
Ring :	3374993.750	-46.4824	3889099.000
Total:	3397359.750		6710191.500

Available Moment of Inertia, Ring + Shell: 0.101E+08 mm⁴

Required Stress in Ring plus Shell Breq 15.90 MPa
Required Strain in Ring plus Shell Areq 0.0001590

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4524.0^2 \cdot 2825.0(10.4 + 1800.0/2825.0)0.000159) / 10.9$$

$$= 9308742 \text{ mm}^4$$

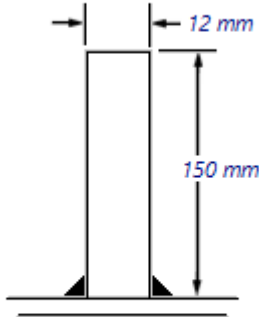
MDMT Calculations for: STF Ring-4, Shell to Ring Junction, Curve: B

Govern. thk, tg = 12.0, c = 1.6 mm, E* = 0.85
Thickness Ratio = tr * E*/(tg - c) = 0.577, Temp. Reduction = 24 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: STF Ring-4, SA-516 70 , Bar Ring: 150.0 x 12.0 mm

Material treatment options:



Effective Length of Shell: 261 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	3232.053	6.2000	20038.730
Ring :	1800.002	87.4001	157320.281
Total:	5032.055		177359.016

Centroid of Ring plus Shell: 35 mm

	Inertia	Distance	A*Dist ²
Shell:	41413.340	29.0458	2726757.250
Ring :	3375006.500	-52.1542	4896118.500
Total:	3416419.750		7622876.000

Available Moment of Inertia, Ring + Shell: 0.110E+08 mm⁴

Required Stress in Ring plus Shell Breq 13.33 MPa
Required Strain in Ring plus Shell Areq 0.0001330

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4528.0^2 \cdot 2337.5(12.4 + 1800.0017/2337.5)0.000133) / 10.9$$

$$= 7701523 \text{ mm}^4$$

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	4520	8.4	0.00025811	25.8054
20	Ring	1775	4524	10.4	0.00038958	38.9491
Ring	30	2800	4524	10.4	0.00024102	24.097
30	Ring	2800	4524	10.4	0.00024102	24.097
Ring	40	2800	4524	10.4	0.00024102	24.097
40	Ring	2800	4524	10.4	0.00024102	24.097
Ring	50	2850	4524	10.4	0.00023661	23.6562
50	Ring	2850	4528	12.4	0.00031139	31.1321
Ring	70	1825	4528	12.4	0.00049834	49.8228
70	80	No Calc	4520	8.4	0.00025811	25.8054

External Pressure Calculations:

From	To	External Actual T.	External Required T.	External Design Pressure	External M.A.W.P.
------	----	--------------------	----------------------	--------------------------	-------------------



		mm	mm	MPa	MPa
10	20	10	9.87542	0.051714	0.053285
20	Ring	12	9.09731	0.051714	0.11938
Ring	30	12	10.6688	0.051714	0.073861
30	Ring	12	10.6688	0.051714	0.073861
Ring	40	12	10.6688	0.051714	0.073861
40	Ring	12	10.6688	0.051714	0.073861
Ring	50	12	10.7289	0.051714	0.072509
50	Ring	14	10.7339	0.051714	0.11367
Ring	70	14	9.19022	0.051714	0.18192
70	80	10	9.87542	0.051714	0.053285

Minimum

0.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	1775	3947.05	No Calc	No Calc
Ring	30	2800	3947.29	7543757	10107544
30	Ring	2800	3947.29	No Calc	No Calc
Ring	40	2800	3947.29	9231119	10107544
40	Ring	2800	3947.29	No Calc	No Calc
Ring	50	2850	3947.4	9308742	10107544
50	Ring	2850	6094.5	No Calc	No Calc
Ring	70	1825	6092.63	7701523	11039288
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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**Element and Detail Weights:**

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	2332.07	12725.7	2021.13	12752.3	163.245
20	30	3955.06	47721.5	3428.93	47789.4	276.854
30	40	3955.06	47721.5	3428.93	47789.4	276.854
40	50	3427.72	41358.6	2971.73	41417.5	239.941
50	70	4000.75	41358.6	3544.76	41417.5	280.053
70	80	2332.07	12725.7	2021.13	12752.3	163.245
Total		20002	203611.70	17416	203918.38	1400

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	12146.5	-375	250	...	Liquid 10
10	Nozl	486.018	-1125	...	...	M1-24"
10	Wght	50	-1125	...	300	Davit-1
20	Sadl	2737.99	2190	2384	...	Left Saddle
20	Plat	9565.02	5650	2338.2	...	PLAT:[1 OF 1]
20	Liqd	44302.1	1500	250	...	Liquid 20
20	Ring	219.181	1350	...	...	STF Ring-1
20	Nozl	11.0391	600	-2275.4	...	K2B-2"
20	Nozl	19.0652	500	2275.4	...	K2A-2"
20	Nozl	35.0876	1000	2307.15	...	N3-4"
20	Nozl	30.3234	2500	2307.15	...	N4-4"
30	Liqd	44302.1	1500	250	...	Liquid 30
30	Ring	219.181	1150	...	...	STF Ring-2
30	Nozl	19.0652	400	2275.4	...	K3-2"
30	Nozl	19.0652	1000	2275.4	...	K4-2"
30	Wght	110	...	...	...	Clip
40	Liqd	38395.2	1300	250	...	Liquid 40
40	Ring	219.181	950	...	...	STF Ring-3
40	Nozl	371.986	1500	-2554.8	...	N1-24
40	Nozl	48.7051	500	-2359.54	...	N2-8"
40	Nozl	35.0876	1200	2307.15	...	N5-4"
40	Nozl	35.0876	1600	2307.15	...	N6-4"
40	Wght	20	...	...	...	N1-I.P. wt
50	Sadl	2737.71	410	2383	...	Right Saddle
50	Liqd	38395.2	1300	250	...	Liquid 50
50	Ring	219.368	1200	...	...	STF Ring-4
50	Nozl	19.1243	1500	2275.4	...	K1A-2"
50	Nozl	11.0983	1500	-2275.4	...	K1B-2"
70	Liqd	12146.5	425	250	...	Liquid 70
70	Nozl	486.018	1175	...	...	M2-24"
70	Wght	50	1175	...	300	Davit-2
20	Trun	208.469	950	...	...	Trunnions
50	Trun	208.469	1550	...	...	Trunnions

Total Weight of Each Detail Type:

Saddles	5475.7
Platforms	9565.0
Liquid	189687.7
Stiffeners	876.9
Nozzles	1626.8
Weights	230.0
Trunnions	416.9

Sum of the Detail Weights 207879.0 kgm

**Weight Summation Results: (kgm)**

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	21402.9	21402.9	21402.9	21402.9	21402.9	21402.9
Saddles	5475.7	5475.7	5475.7	5475.7	5475.7	5475.7
Stif. Rings	876.9	876.9	876.9	876.9	876.9	876.9
Nozzles	1626.8	1626.8	1626.8	1626.8	1626.8	1626.8
Wld Weights	230.0	230.0	230.0	230.0	230.0	230.0
Trunnions	416.9	416.9	416.9	416.9	416.9	416.9
Platforms	...	...	...	9565.0	9565.0	9565.0
Ope. Liquid	...	...	...	...	...	189687.7
Test Liquid	...	203487.4	...	...	...	...
Totals	30029.3	233516.7	30029.3	39594.3	39594.3	229282.0

Field Installation Options:

Platforms installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	30029.3 kgm
Shop Test Wt.	- Fabricated weight + water (full)	233516.7 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	30029.3 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	39594.3 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	39594.3 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	229282.0 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	226514.8 kgm
Field Test Wt.	- Empty weight + test medium (full)	243081.7 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	22896100	22657562
20	30	42637700	42411500
30	40	42637700	42411500
40	50	36952672	36756636
50	70	36985340	36756636
70	80	22896100	22657562

Total	205005616.0	203651392.0 mm ²
Total	205.0	203.7 m ²

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**Nozzle Flange MAWP Results: (MPa & °C)**

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
M1-24"	1.82	1.96	82	150	GR 1.1	...	...	...	...
K2B-2"	4.78	5.10	82	300	GR 1.1	...	...	...	...
K2A-2"	4.78	5.10	82	300	GR 1.1	...	...	...	...
N3-4"	4.78	5.10	82	300	GR 1.1	...	...	...	...
N4-4"	4.78	5.10	82	300	GR 1.1	...	...	...	...
K3-2"	4.78	5.10	82	300	GR 1.1	...	...	...	...
K4-2"	4.78	5.10	82	300	GR 1.1	...	...	...	...
N1-24	1.82	1.96	82	150	GR 1.1	1.411	1.821	1.821	1.320
N2-8"	1.82	1.96	82	150	GR 1.1	2.936	1.070	1.821	-0.205
N5-4"	4.78	5.10	82	300	GR 1.1	...	...	...	...
N6-4"	4.78	5.10	82	300	GR 1.1	...	...	...	...
K1A-2"	4.78	5.10	82	300	GR 1.1	...	...	...	...
K1B-2"	4.78	5.10	82	300	GR 1.1	...	...	...	...
M2-24"	1.82	1.96	82	150	GR 1.1	...	...	...	...
Min Rating	1.070	1.960, [for Core Elements]					...	...	-0.205

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1-24	657.225	1.20	71279976	45000	0.399	1.821	Pass
N2-8"	248.539	1.20	7919597	15000	0.399	1.070	Pass

Evaluating Flange on Nozzle [N1-24]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(71279976) + 4 \cdot \text{abs}(45000)657.225 \leq \pi \cdot 657.225^3 [(1.821 - 0.399) + 1.200 \cdot 1.821]$$

$$= 0.12588E+10 \leq 0.32172E+10 \quad \text{Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-24 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.821, 1.200 \cdot 1.821 + 1.821 - 1258827904.0 / (\pi \cdot 657.225^3))$$

$$= \min(1.821, 2.594)$$

$$= 1.821 \text{ MPa}$$

Evaluating Flange on Nozzle [N2-8"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(7919597) + 4 \cdot \text{abs}(15000)248.539 \leq \pi \cdot 248.539^3 [(1.821 - 0.399) + 1.200 \cdot 1.821]$$

$$= 0.14163E+09 \leq 0.17399E+09 \quad \text{Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-8" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.821, 1.200 \cdot 1.821 + 1.821 - 141631856.0 / (\pi \cdot 248.539^3))$$

$$= \min(1.821, 1.070)$$

$$= 1.070 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	97.2	km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	21500	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
= $\min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
= $\min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.192 \cdot 0.913) / (1 + 1.7 \cdot 3.4 \cdot 0.192)))$
= $\min(0.85, 0.883)$
= 0.850

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.530
Structure Height to Diameter ratio	2.803

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm

Effective Height $[z]$:
= Centroid Height + Vessel Base Elevation
= $2550.0 + 21500.0 = 24050.0$ mm
= 78.904 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
Because z (78.904 ft.) > 15 ft.
= $2.01(z / Zg)^{2/\alpha}$
= $2.01(78.904/900.0)^{2/9.5}$
= 1.204

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
= 1, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.204 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf [0.766] kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.53 \cdot 46.29$$

$$= 333.7 \text{ lbs. [1484.3] N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
left Dish	24050.0	0.000	0.000	0.000	1.204	1.000	0.766
Shell-1	24050.0	0.000	0.000	0.000	1.204	1.000	0.766
Shell2	24050.0	0.000	0.000	0.000	1.204	1.000	0.766
Shell3	24050.0	0.000	0.000	0.000	1.204	1.000	0.766
shell4	24050.0	0.000	0.000	0.000	1.204	1.000	0.766
Right dish	24050.0	0.000	0.000	0.000	1.204	1.000	0.766

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT:[1 OF 1]	13982201.00	24050.00	0.77	10711.01	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Davit-1	0.00	24050.00	0.77	0.00
Clip	0.00	24050.00	0.77	0.00
N1-I.P. wt	0.00	24050.00	0.77	0.00
Davit-2	0.00	24050.00	0.77	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	24050	5424	4300450	0.76606	1632.68
20	30	24050	5428.8	16286402	0.76606	17965.3
30	40	24050	5428.8	16286402	0.76606	6183.18
40	50	24050	5428.8	14114880	0.76606	5358.76
50	70	24050	5433.6	14127360	0.76606	5363.5
70	80	24050	5424	4300450	0.76606	1632.68

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.770
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(15.7480.75)$
= 0.158 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/33.0)
= 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.158$, 0.03) per 12.8.2
= 0.030

Loads on Building/Structure Supported Equipment:

Seismic Force per equation 13.3-1, [Fp]:
= $0.4 \cdot A_p \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / I_e)$
= $0.4 \cdot 1.0 \cdot 0.157 \cdot 2194644(1 + 2 \cdot 0.77) / (3.0/1.25)$
= 145677.500 N

Maximum value of Fp per equation 13.3-2:
= $1.6 \cdot Sds \cdot I \cdot W$
= $1.6 \cdot 0.157 \cdot 1.25 \cdot 2194644 = 688240$ N

Minimum value of Fp per equation 13.3-3:
= $0.3 \cdot 0.16 \cdot 1.25 \cdot 2194644$
= 129045.047 N

Total Base Shear $V = F_p$, [V]:
= 145677.500 N

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 2194644$
= 68824.0 N

**Earthquake Load Calculation:**

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N
10	20	2250	274330	18209.7
20	30	2250	274330	18209.7
20	30	2250	274330	18209.7
30	40	2250	274330	18209.7
40	50	2250	274330	18209.7
50	0	2250	274330	18209.7
50	70	2250	274330	18209.7
70	80	2250	274330	18209.7

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5649.826 mm
Center of Gravity of the Platforms	5700.000 mm
Center of Gravity of the Liquid	5650.000 mm
Center of Gravity of the Stiffening Rings	5613.407 mm
Center of Gravity of the Nozzles	5977.424 mm
Center of Gravity of the Added Weights (Operating)	4441.304 mm
Center of Gravity of the Added Weights (Empty)	4441.304 mm
Center of Gravity of the Trunnions	5600.000 mm
Center of Gravity of Bare Shell New and Cold	5773.185 mm
Center of Gravity of Bare Shell Corroded	5791.473 mm
Vessel CG in the Operating Condition	5664.637 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	5744.483 mm
Vessel CG in the Test Condition	5662.150 mm

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	11300.00 mm
Total vessel weight (No Liquid) Twt	388261.44 N
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	582392.12 N
Distance to the Lug on the Left End	1000.00 mm
Distance to the Lug on the Right End	10200.00 mm
Design Reaction force at the Left End Trunnion	282049.78 N
Design Reaction force at the Right End Trunnion	300342.38 N
CG distance during field lifting operation	5744.48 mm
CG distance from the left end lug/trunnion	4744.48 mm
CG distance from the nearer lifting lug/trunnion	4455.52 mm
Compressive Allowable Increase Factor	1.00

Critical Values: (MPa, mm)

	Max Stress	Elevation	Allowables
Bending	0.95	6050.00	66.30 (UG-23)
Shear	0.97	650.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment N-mm	Bending Stress MPa	Shear Force N	Shear Stress MPa
0.00	0.0	0.0	126185.0	0.9
50.00	38041052.0	0.2	135891.6	0.8
3050.00	88084568.0	0.5	28592.1	0.2
6050.00	181178512.0	0.9	44206.9	0.3
8650.00	72505680.0	0.3	117005.9	0.6
11250.00	34156588.0	0.2	9706.5	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0143 at elevation 6050.0 mm - Must be <=1



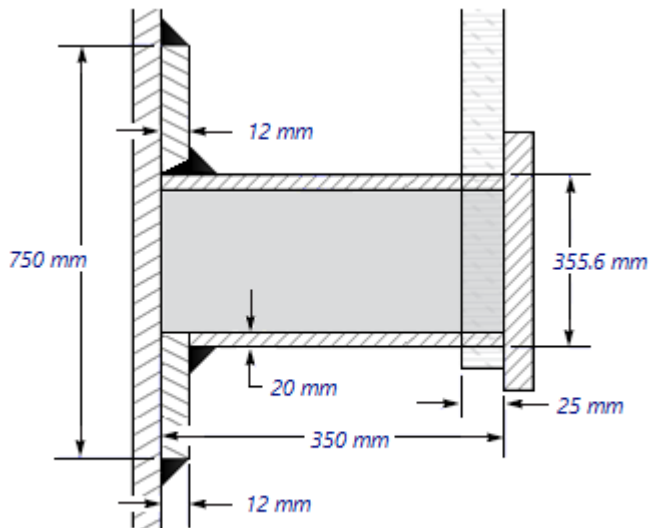
Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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**Trunnion Analysis: Left End of Vessel****Input Data:**

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 MPa
Trunnion Outside Diameter	Tod 355.600 mm
Trunnion Thickness	Tthk 20.0000 mm
Trunnion Length	Tl 350.0000 mm
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 750.000 mm
Pad Thickness	Thkp 12.0000 mm
Pad Edge Fillet Weld Thk	Tfp 12.0000 mm
Bail/Sling Width	Tt 25.0000 mm
Sling Angle from Horizontal	90.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.50
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 94016.6 • 1.5
 = 141024.9 N

Trunnion Section Modulus [Z]:

$$= \pi (Tod^4 - Tid^4) / (32 \cdot Tod)$$

$$= \pi (355.6^4 - 315.6^4) / (32 \cdot 355.6)$$

 = 1675574.1 mm³

Trunnion Bending Moment [Tmom]:

$$= (Tl - 0.5 \cdot Tt) / 1.000 \cdot Tlf$$

$$= (350.0 - 0.5 \cdot 25.0) / 1.000 \cdot 141024.89$$

 = 47615196.0 N-mm

Trunnion Calculated Bending Stress [Sbc]:

$$= Tmom \cdot 1.000 / Z$$

$$= 47615196 \cdot 1.000 / 1675574$$

 = 28.41 MPa [Passed] Sbc < min(0.66 • SY • Occfac, 0.75 • Sy) (172.92)

Trunnion Cross Section Area [Txsa]:

$$= \pi / 4 (Tod^2 - Tid^2)$$

$$= \pi / 4 (355.6^2 - 315.6^2)$$



$$= 21086.4 \text{ mm}^2$$

Trunnion Calculated Shear Stress [Ssc]:

$$= Tlf/Txsa$$

$$= 141024.89/21086.37$$

$$= 6.69 \text{ MPa [Passed] } Ssc < 0.4 \cdot Sy \cdot Occfac (104.80)$$

Trunnion Unity Check [Uchk]:

$$= Sbc/Sba + Ssc/Ssa$$

$$= 28.41/172.92 + 6.69/104.8$$

$$= 0.228 \text{ [Passed]}$$

Weld Section Modulus at Edge of Trunnion [Sw]:

$$= \pi (T_{od}/2)^2$$

$$= \pi (355.6/2)^2$$

$$= 99314.6 \text{ mm}^2$$

Shear Stress in Weld due to Bending Moment [fs]:

$$= T_{mom}/Sw$$

$$= 47615196/99314.6$$

$$= 479.244 \text{ N/mm}$$

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:

$$= \max(10 \text{ mm}, fs/(0.4 \cdot Sy))$$

$$= \max(10 \text{ mm}, 479.244/(0.4 \cdot 262.0))$$

$$= 10.000 \text{ mm}$$

Trunnion Bearing Contact Area [Tbca]:

$$= T_{od} \cdot T_t$$

$$= 355.6 \cdot 25.0$$

$$= 8890.0 \text{ mm}^2$$

Trunnion Bearing Stress [Sbec]:

$$= Tlf/Tbca$$

$$= 141024.89/8890.0$$

$$= 15.86 \text{ MPa [Passed] } Sbec < \min(0.75 \cdot SY \cdot Occfac, 0.9 \cdot Sy) (196.50)$$

Summary of Results:

Trunnion Unity Check Value: 0.2281 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (MPa)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	28.41	172.92	Pass
Shear Stress	6.69	104.80	Pass
Bearing Stress	15.86	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm
Vessel Diameter	Dv	4500.000 mm
Vessel Thickness	Tv	12.000 mm

Design Temperature	T1	37.8	C
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	355.600	mm
Nozzle Thickness	Tn	20.000	mm
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Thickness of Reinforcing Pad	Tpad	12.000	mm
Diameter of Reinforcing Pad	Dpad	774.000	mm
Design Internal Pressure	Dp	0.000	MPa
Include Pressure Thrust		No	

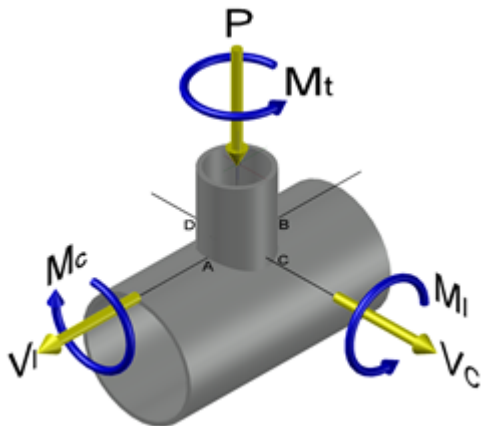
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	N
Longitudinal Shear (SUS)	VL	0.0	N
Circumferential Shear (SUS)	Vc	141024.9	N
Circumferential Moment (SUS)	Mc	47615196.0	N-mm
Longitudinal Moment (SUS)	ML	0.0	N-mm
Torsional Moment (SUS)	Mt	0.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 355.6 + 2 \cdot 1.65 \cdot \sqrt{2256.0 (12.0 - 0)} \\
 &= 898.568 \text{ mm}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	0.0	N
Circumferential Shear	VC	141024.9	N
Longitudinal Shear	VL	0.0	N
Circumferential Moment	MC	47615196.0	N-mm
Longitudinal Moment	ML	0.0	N-mm



Torsional Moment MT 0.0 N-mm

Dimensionless Parameters used : Gamma = 94.25

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.069	4C	15.842	(A,B)
N(PHI) / (P/Rm)	0.069	3C	14.521	(C,D)
M(PHI) / (P)	0.069	2C1	0.084	(A,B)
M(PHI) / (P)	0.069	1C	0.120	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.069	3A	2.841	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.069	1A	0.093	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.069	3B	9.602	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.069	1B	0.045	(A,B,C,D)
N(x) / (P/Rm)	0.069	3C	14.521	(A,B)
N(x) / (P/Rm)	0.069	4C	15.842	(C,D)
M(x) / (P)	0.069	1C1	0.125	(A,B)
M(x) / (P)	0.069	2C	0.084	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.069	4A	4.110	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.069	2A	0.053	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.069	4B	2.812	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.069	2B	0.070	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-16.0	-16.0	16.0	16.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-295.9	295.9	295.9	-295.9
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-312.0	279.9	312.0	-279.9
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-23.2	-23.2	23.2	23.2
Long. Bend. MC		0.0	0.0	0.0	0.0	-169.5	169.5	169.5	-169.5
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-192.6	146.3	192.6	-146.3
Shear VC		10.5	10.5	-10.5	-10.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		10.5	10.5	-10.5	-10.5	0.0	0.0	0.0	0.0
Str. Int.		21.0	21.0	21.0	21.0	312.0	279.9	312.0	279.9

Dimensionless Parameters used : Gamma = 188.00

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.150	4C	20.648	(A,B)
N(PHI) / (P/Rm)	0.150	3C	9.108	(C,D)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-16.0	-16.0	16.0	16.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-295.9	295.9	295.9	-295.9
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0



Long. P1 (SUS)	0.0	0.0	0.0	0.0	-23.2	-23.2	23.2	23.2
Long. Q (SUS)	0.0	0.0	0.0	0.0	-169.5	169.5	169.5	-169.5
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	10.5	10.5	-10.5	-10.5	0.0	0.0	0.0	0.0
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)	21.0	21.0	21.0	21.0	23.2	23.2	23.2	23.2
Pm+P1+Q (Total)	21.0	21.0	21.0	21.0	312.0	279.9	312.0	279.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+P1 (SUS)	23.17	207.00	Passed
Pm+P1+Q (TOTAL)	311.95	413.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (MPa)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. P1 (SUS)		0.0	0.0	0.0	0.0	-33.2	-33.2	33.2	33.2
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-348.6	348.6	348.6	-348.6
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. P1 (SUS)		0.0	0.0	0.0	0.0	-81.5	-81.5	81.5	81.5
Long. Q (SUS)		0.0	0.0	0.0	0.0	-142.4	142.4	142.4	-142.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)		9.7	9.7	-9.7	-9.7	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)		19.3	19.3	19.3	19.3	81.5	81.5	81.5	81.5
Pm+P1+Q (Total)		19.3	19.3	19.3	19.3	381.8	315.3	381.8	315.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+P1 (SUS)	81.54	207.00	Passed
Pm+P1+Q (TOTAL)	381.80	413.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

CCW Expansion Vessel (V8801)

PV Elite 28 Licensee: SL User

FileName : CCW Expansion Vessel (V8801)-7%

Trunnion Calcs: Left End



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Step: 18 3:00pm Sep 16, 2026



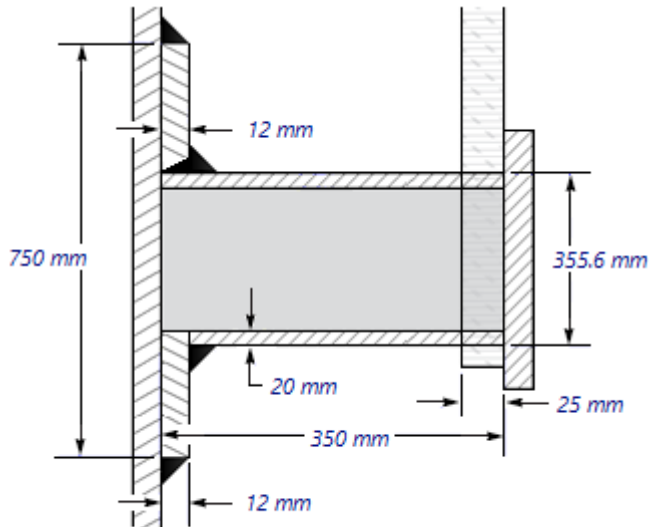
HEXAGON

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Trunnion Analysis: Right End of Vessel

Input Data:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 MPa
Trunnion Outside Diameter	Tod 355.600 mm
Trunnion Thickness	Tthk 20.0000 mm
Trunnion Length	Tl 350.0000 mm
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 750.000 mm
Pad Thickness	Thkp 12.0000 mm
Pad Edge Fillet Weld Thk	Tfp 12.0000 mm
Bail/Sling Width	Tt 25.0000 mm
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.50
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Calculation Results:

Sling Angle from Horizontal: 0.0 deg

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 100114.1 • 1.5
 = 150171.2 N

Trunnion Section Modulus [Z]:
 = $\pi (Tod^4 - Tid^4) / (32 \cdot Tod)$
 = $\pi (355.6^4 - 315.6^4) / (32 \cdot 355.6)$
 = 1675574.1 mm³

Trunnion Bending Moment [Tmom]:
 = $(Tl - 0.5 \cdot Tt) / 1.000 \cdot Tlf$
 = $(350.0 - 0.5 \cdot 25.0) / 1.000 \cdot 150171.19$
 = 50703324.0 N-mm

Trunnion Calculated Bending Stress [Sbc]:
 = $Tmom \cdot 1.000 / Z$
 = $50703324 \cdot 1.000 / 1675574$
 = 30.25 MPa [Passed] $Sbc < \min(0.66 \cdot SY \cdot Occfac, 0.75 \cdot Sy)$ (172.92)

Trunnion Cross Section Area [Txa]:
 = $\pi / 4 (Tod^2 - Tid^2)$



$$= \pi/4 (355.6^2 - 315.6^2)$$

$$= 21086.4 \text{ mm}^2$$

Trunnion Calculated Shear Stress [Ssc]:

$$= Tlf/Txsa$$

$$= 150171.19/21086.37$$

$$= 7.12 \text{ MPa [Passed] } Ssc < 0.4 \cdot Sy \cdot Occfac (104.80)$$

Trunnion Unity Check [Uchk]:

$$= Sbc/Sba + Ssc/Ssa$$

$$= 30.25/172.92 + 7.12/104.8$$

$$= 0.243 \text{ [Passed]}$$

Weld Section Modulus at Edge of Trunnion [Sw]:

$$= \pi (T_{od}/2)^2$$

$$= \pi (355.6/2)^2$$

$$= 99314.6 \text{ mm}^2$$

Shear Stress in Weld due to Bending Moment [fs]:

$$= T_{mom}/Sw$$

$$= 50703324/99314.6$$

$$= 510.326 \text{ N/mm}$$

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:

$$= \max(10 \text{ mm}, fs/(0.4 \cdot Sy))$$

$$= \max(10 \text{ mm}, 510.326/(0.4 \cdot 262.0))$$

$$= 10.000 \text{ mm}$$

Trunnion Bearing Contact Area [Tbca]:

$$= T_{od} \cdot T_t$$

$$= 355.6 \cdot 25.0$$

$$= 8890.0 \text{ mm}^2$$

Trunnion Bearing Stress [Sbec]:

$$= Tlf/Tbca$$

$$= 150171.19/8890.0$$

$$= 16.89 \text{ MPa [Passed] } Sbec < \min(0.75 \cdot SY \cdot Occfac, 0.9 \cdot Sy) (196.50)$$

Summary of Results:

Trunnion Unity Check Value: 0.2429 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (MPa)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	30.25	172.92	Pass
Shear Stress	7.12	104.80	Pass
Bearing Stress	16.89	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).**Input Echo, WRC107/537 Item 1, Description: Trunnion**

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm
Vessel Diameter	Dv	4500.000 mm
Vessel Thickness	Tv	14.000 mm

Design Temperature	T1	37.8	C
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	355.600	mm
Nozzle Thickness	Tn	20.000	mm
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Thickness of Reinforcing Pad	Tpad	12.000	mm
Diameter of Reinforcing Pad	Dpad	774.000	mm
Design Internal Pressure	Dp	0.000	MPa
Include Pressure Thrust	No		

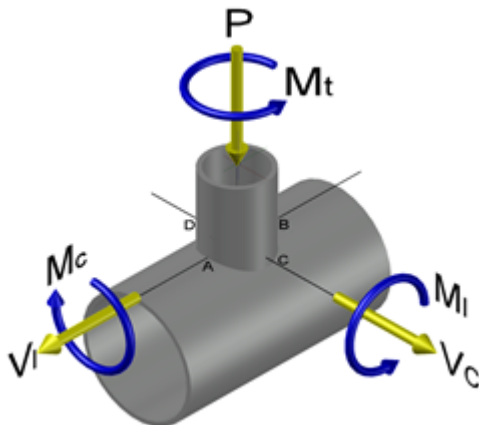
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	N
Longitudinal Shear (SUS)	VL	0.0	N
Circumferential Shear (SUS)	Vc	150171.2	N
Circumferential Moment (SUS)	Mc	50703324.0	N-mm
Longitudinal Moment (SUS)	ML	0.0	N-mm
Torsional Moment (SUS)	Mt	0.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(2257.0 (14.0 - 0))}$$

$$= 942.202 \text{ mm}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	0.0	N
Circumferential Shear	VC	150171.2	N
Longitudinal Shear	VL	0.0	N
Circumferential Moment	MC	50703324.0	N-mm



Longitudinal Moment ML 0.0 N-mm
Torsional Moment MT 0.0 N-mm

Dimensionless Parameters used : Gamma = 87.04

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.069	4C	14.787	(A,B)
N(PHI) / (P/Rm)	0.069	3C	13.548	(C,D)
M(PHI) / (P)	0.069	2C1	0.088	(A,B)
M(PHI) / (P)	0.069	1C	0.124	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.069	3A	2.530	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.069	1A	0.094	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.069	3B	8.741	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.069	1B	0.046	(A,B,C,D)
N(x) / (P/Rm)	0.069	3C	13.548	(A,B)
N(x) / (P/Rm)	0.069	4C	14.787	(C,D)
M(x) / (P)	0.069	1C1	0.129	(A,B)
M(x) / (P)	0.069	2C	0.088	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.069	4A	3.611	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.069	2A	0.054	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.069	4B	2.517	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.069	2B	0.072	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-14.0	-14.0	14.0	14.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-272.4	272.4	272.4	-272.4
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-286.4	258.4	286.4	-258.4
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-20.0	-20.0	20.0	20.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-155.8	155.8	155.8	-155.8
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-175.8	135.8	175.8	-135.8
Shear VC		10.3	10.3	-10.3	-10.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		10.3	10.3	-10.3	-10.3	0.0	0.0	0.0	0.0
Str. Int.		20.7	20.7	20.7	20.7	286.4	258.4	286.4	258.4

Dimensionless Parameters used : Gamma = 161.21

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.150	4C	18.388	(A,B)



N(PHI) / (P/Rm)	0.150	3C	8.683	(C,D)
M(PHI) / (P)	0.150	2C1	0.021	(A,B)
M(PHI) / (P)	0.150	1C !	0.070	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.150	3A	5.648	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.150	1A	0.062	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.150	3B	12.545	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.150	1B	0.014	(A,B,C,D)
N(x) / (P/Rm)	0.150	3C	8.683	(A,B)
N(x) / (P/Rm)	0.150	4C	18.388	(C,D)
M(x) / (P)	0.150	1C1	0.046	(A,B)
M(x) / (P)	0.150	2C !	0.039	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.150	4A	13.468	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.150	2A	0.026	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.150	4B	5.731	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.150	2B	0.017	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-26.8	-26.8	26.8	26.8
Circ. Bend. MC		0.0	0.0	0.0	0.0	-284.1	284.1	284.1	-284.1
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-310.9	257.4	310.9	-257.4
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-63.8	-63.8	63.8	63.8
Long. Bend. MC		0.0	0.0	0.0	0.0	-118.2	118.2	118.2	-118.2
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-182.0	54.4	182.0	-54.4
Shear VC		8.8	8.8	-8.8	-8.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		8.8	8.8	-8.8	-8.8	0.0	0.0	0.0	0.0
Str. Int.		17.6	17.6	17.6	17.6	310.9	257.4	310.9	257.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-14.0	-14.0	14.0	14.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-272.4	272.4	272.4	-272.4



Long. Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. P1 (SUS)	0.0	0.0	0.0	0.0	-20.0	-20.0	20.0	20.0
Long. Q (SUS)	0.0	0.0	0.0	0.0	-155.8	155.8	155.8	-155.8
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	10.3	10.3	-10.3	-10.3	0.0	0.0	0.0	0.0
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)	20.7	20.7	20.7	20.7	20.0	20.0	20.0	20.0
Pm+P1+Q (Total)	20.7	20.7	20.7	20.7	286.4	258.4	286.4	258.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+P1 (SUS)	20.68	207.00	Passed
Pm+P1+Q (TOTAL)	286.41	413.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. P1 (SUS)		0.0	0.0	0.0	0.0	-26.8	-26.8	26.8	26.8
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-284.1	284.1	284.1	-284.1
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. P1 (SUS)		0.0	0.0	0.0	0.0	-63.8	-63.8	63.8	63.8
Long. Q (SUS)		0.0	0.0	0.0	0.0	-118.2	118.2	118.2	-118.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)		8.8	8.8	-8.8	-8.8	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)		17.6	17.6	17.6	17.6	63.8	63.8	63.8	63.8
Pm+P1+Q (Total)		17.6	17.6	17.6	17.6	310.9	257.4	310.9	257.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+P1 (SUS)	63.80	207.00	Passed
Pm+P1+Q (TOTAL)	310.89	413.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

CCW Expansion Vessel (V8801)

PV Elite 28 Licensee: SL User

FileName : CCW Expansion Vessel (V8801)-7%

Trunnion Calcs: Right End



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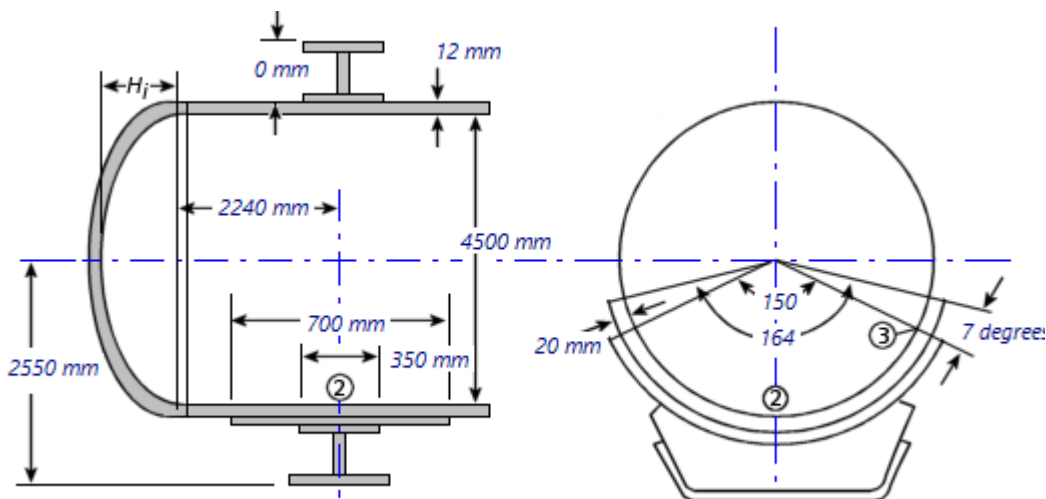
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	2256.80	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	11300.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2240.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	164.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	1130.80	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2550.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	1266969.38	N
Pressure used in Analysis	P	0.367 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	36.76	138.00
Long. Stress at Bottom of Midspan	42.87	138.00
Long. Stress at Top of Saddles	63.82	138.00
Long. Stress at Bottom of Saddles	26.01	138.00
Tangential Shear in Shell	22.96	110.40
Circ. Stress at Horn of Saddle	120.61	172.50
Circ. Compressive Stress in Shell	4.76	138.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (38136.1/2 + 0) \cdot 2550.0/3980.0$
 $= 36650.9 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(23317.11, 0) \cdot 2550.0/6820.0$
 $= 8718.3 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(145677.48, 0) \cdot 2550.0/6820.0$
 $= 54468.9 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (145677/2 + 0) \cdot 2550.0/3980.0$
 $= 140004.1 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 1126965 + \max(8718, 36651, 54469, 140004)$
 $= 1266969.4 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08 (4.524 + \pi/4 (4.524 \cdot 1.2)^2)$
 $= 23318.318 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1293818.12	N
Transverse Shear Load	72838.75	N
Longitudinal Shear Load	145677.48	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 3980.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0313	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0243
K7p = 0.0240			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -1266969 \cdot 2240.0 [1 - (1 - 2240.0/11300.0 + (2256.8^2 - 1130.8^2) / (2 \cdot 2240.0 \cdot 11300.0)) / (1 + (4 \cdot 1130.8)/(3 \cdot 11300.0))]$
 $= -642038208.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/(3L)) - 4a/L)$
 $= 1266969 \cdot 11300/4 ((1 + 2(2257^2 - 1131^2)/(11300^2)) / (1 + (4 \cdot 1131)/(3 \cdot 11300)) - 4 \cdot 2240/11300)$
 $= 508705760.0 \text{ N-mm}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 0.367 \cdot 2256.8/(2 \cdot 10.4) - 0.50871 \text{E}+09/(\pi \cdot 2256.8^2 \cdot 10.4)$
 $= 36.76 \text{ MPa}$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.367 \cdot 2256.8 / (2 \cdot 10.4) + 0.50871E+09 / (\pi \cdot 2256.8^2 \cdot 10.4)$$

$$= 42.87 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.367 \cdot 2256.8 / (2 \cdot 10.4) - 0.6E+09 / (0.1607 \cdot \pi \cdot 2256.8^2 \cdot 10.4)$$

$$= 63.82 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.367 \cdot 2256.8 / (2 \cdot 10.4) + 0.6E+09 / (0.2792 \cdot \pi \cdot 2256.8^2 \cdot 10.4)$$

$$= 26.01 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 1266969(11300.0 - 2 \cdot 2240.0) / (11300.0 + (4 \cdot 1130.8 / 3))$$

$$= 674649.6 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 674650 / (2256.8003 \cdot 10.4)$$

$$= 22.96 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2256.8 \cdot 10.4)}$$

$$= 119.497 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{(2256.8 \cdot 10.4)}, 2 \cdot 2240.0)$$

$$= 588.99 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.6733 \cdot 1266969 \cdot 0.1 / (588.994(10.4 + 1.0 \cdot 20.0))$$

$$= -4.76 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2)$$

$$= -1266969 / (4(10.4 + 1.0 \cdot 20.0) 588.994) -$$

$$12 \cdot 0.031 \cdot 1266969 \cdot 2256.8 / (11300.0(10.4 + 1.0 \cdot 20.0)^2)$$

$$= -120.61 \text{ MPa}$$

Distance between Saddle Supports [Ls]:

$$= 6820.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 6820.0(82.0 - 21.1)$$

$$= 4.977 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3980.0000	mm
Baseplate Thickness	Bpthk	42.0000	mm
Baseplate Width	Bpwid	528.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	25.0000	mm
Web Thickness	Webtk	25.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C



Saddle Yield Stress Sy 242.6 MPa
Height of Web at Center Hw,c 264.0 mm
Friction Coefficient mu 0.000

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(2251.6 \cdot 10.4)} + 700.0)^3) / 12$
 $= 87994.180 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 87994.18 + 9762.677(141.469 - 5.2)^2$
 $= 181373232.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	938.7	10.4	5.2	9762.7	50765.9	0.181E+09
Wearplate	700.0	20.0	20.4	14000.0	285600.0	0.206E+09
Web	25.0	226.0	143.4	5650.0	810209.4	0.241E+08
BasePlate	528.0	42.0	277.4	22176.0	6151620.0	0.413E+09
Totals	...	...	...	51588.7	7298195.5	0.824E+09

Distance to Centroid [C1]:
 $= AY / A$
 $= 7298196 / 51588.668$
 $= 141.469 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 150.0 / 2$
 $= 105.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$
 $= 0.2594$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.259 \cdot 1266969$
 $= 328616.8438 \text{ N}$

Tension Stress, St = (Fh/As) = 7.8574 MPa
Allowed Stress, Sa = 0.6 * Yield Str = 145.5600 MPa

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 2550.0 - 2251.6 \cdot \sin(150.0/2) / 1.309$
 $= 888.515 \text{ mm}$

Bending Moment, M = Fh * d = 292099360.0000 N-mm

Bending Stress, Sb = (M * C1 / I) = 50.1254 MPa
Allowed Stress, Sa = 2/3 * Yield Str = 161.7334 MPa

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt[3]{(3(1266969 + 26849)528.0 / (2 \cdot 3980.0 \cdot 161.733))}$
 $= 39.900 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 150.0/2) (2250.0 + 12.0 + 20.0)$
 $= 4408.485 \text{ mm}$



Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 4408.4854/(5 - 1)
 = 1102.121 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 1102.1213 • 528.0/2
 = 290960.031 mm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 1293818/(3980.0 • 528.0)
 = 0.616 N/mm²

Axial Load [P]:
 = Ap • Bp
 = 290960.0 • 0.62
 = 179138.766 N

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 8125.0 + 13776.518
 = 21901.518 mm²

Compressive Stress [Sc]:
 = P/Ar
 = 179138.8/21901.5176
 = 8.180 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.170E+09
Web	551.1	25.0	12.5	13776.5	172206.5	0.588E+08
Totals	...	...	...	21901.5	1695643.9	0.229E+09

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 350.0 - 25.0
 = 325.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 1695644/21901.518
 = 77.421 mm

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(77.421, 350.0 - 77.421)
 = 272.579 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{0.22875\text{E}+09/21901.518}$
 = 102.199 mm

Length of Outer Rib, step 1, [Lv]:
 = Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 2550.0 - cos(150.0/2)(2250.0 + 12.0 + 20.0) - 42.0
 = 1917.375 mm

Length of Outer Rib, step 2, [Lw]:
 = 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(150.0/2)(2250.0 + 12.0 + 20.0)



$$= 4408.485 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3980.0 - 50.8$$

$$= 3929.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (4408.485 - 3929.2) / 2$$

$$= 239.643 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1917.375^2 + 239.643^2}$$

$$= 1932.293 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 242.6)}$$

$$= 127.552$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1932.293 / 102.199$$

$$= 18.907$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 145677.5 / (2 \cdot 3980.0) \cdot 1102.121 \cdot 1932.29 / 2$$

$$= 19495206.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($18.9072 < 127.5515$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.91)^2 / (2 \cdot 127.55^2)) 243 /$$

$$(5/3 + 3 \cdot (18.91) / (8 \cdot 127.55) - (18.91^3) / (8 \cdot 127.55^3))$$

$$= 139.3 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.18 / 139.35 + (19495206 \cdot 272.579 / .22875E+09) / 161.73$$

$$= 0.202$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.220E+09
Web	1102.1	25.0	12.5	27553.0	344412.9	0.452E+08
Totals	...	...	...	35678.0	1867850.4	0.265E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1867850 / 35678.035$$

$$= 52.353 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(52.353, \text{abs}(350.0 - 52.353))$$

$$= 297.647 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2550.0 - \cos(2282.0/2) (42.0 + 0.0 + 0.0) - 0.0$$

$$= 226.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$



$$= \sqrt{0.26511E+09/35678.035}$$

$$= 86.202 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 226.0/86.202$$

$$= 2.622$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 145677.484 / (2 \cdot 3980.0)$$

$$= 18.301 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 18.301 \cdot 1102.121 \cdot 226.0$$

$$= 4560297.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4560297/890700$$

$$= 5.118 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (2.6218 < 127.5515) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (2.62)^2 / (2 \cdot 127.55^2)) 243 / (5/3 + 3 \cdot (2.62) / (8 \cdot 127.55) - (2.62^3) / (8 \cdot 127.55^3))$$

$$= 144.9 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 9.07/144.86 + (4560297 \cdot 297.647/0.26511E+09)/161.73$$

$$= 0.094$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	138.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	40.0 mm
Nominal Bolt Diameter	Bnd	48.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	1155.3340 mm ²
Saddle Load Q0 (Weight)	Q0	1153814.0 N
Saddle Load QL (Wind/Seismic contribution)	QL	54468.9 N
Maximum Transverse Force	Ft	72838.8 N
Maximum Longitudinal Force	Fl	145677.5 N
Saddle Bolted to Steel Foundation		Yes

Friction force due to thermal growth, [Ff]:

$$= 349004.6 \text{ N}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 349004.6 / (1155.334 \cdot 4)$$

$$= 75.5 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 72838.8 / (1155.334 \cdot 4)$$

$$= 15.8 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarear]:



= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
= $145677.48 / (82.8 \cdot 4.0)$
= 439.8849 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $2550.0 \cdot 72838.75 + 0.0$
= 185814096.00 N-mm

Eccentricity (e):
= R_{mom} / QO
= $0.18581E+09 / 1153814$
= 160.98 mm < $B_{\text{plen}}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 439.8849, 0.0]$
= 439.8849 mm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 8.424$, $c = 1.6$ mm, $E^* = 0.85$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	2257.80	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	11300.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2240.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	164.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	1130.80	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2550.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		1276506.62	N
Pressure used in Analysis	P	0.367	MPa



Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	36.76	138.00
Long. Stress at Bottom of Midspan	42.91	138.00
Long. Stress at Top of Saddles	53.67	138.00
Long. Stress at Bottom of Saddles	21.75	138.00
Tangential Shear in Shell	19.40	110.40
Circ. Stress at Horn of Saddle	107.39	172.50
Circ. Compressive Stress in Shell	4.34	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (38136.1/2 + 0) \cdot 2550.0/3980.0$
 $= 36650.9 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(23351.62, 0) \cdot 2550.0/6820.0$
 $= 8731.2 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(145677.48, 0) \cdot 2550.0/6820.0$
 $= 54468.9 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (145677/2 + 0) \cdot 2550.0/3980.0$
 $= 140004.1 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 1136503 + \max(8731, 36651, 54469, 140004)$
 $= 1276506.6 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08 (4.524 + \pi/4 (4.528 \cdot 1.2)^2)$
 $= 23352.828 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1303352.62	N
Transverse Shear Load	72838.75	N
Longitudinal Shear Load	145677.48	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 3980.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0313	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0243
K7p = 0.0240			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -1276507 \cdot 2240.0 [1 - (1 - 2240.0/11300.0 + (2257.8^2 - 1130.8^2) /$



$$(2 \cdot 2240.0 \cdot 11300.0) / (1 + (4 \cdot 1130.8) / (3 \cdot 11300.0)) \\ = -646646144.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ = 1276507 \cdot 11300 / 4 \cdot ((1 + 2(2258^2 - 1131^2) / (11300^2)) / (1 + (4 \cdot 1131) / (3 \cdot 11300)) - 4 \cdot 2240 / 11300) \\ = 512760256.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\ = 0.367 \cdot 2257.8 / (2 \cdot 10.4) - 0.51276E+09 / (\pi \cdot 2257.8^2 \cdot 10.4) \\ = 36.76 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = 0.367 \cdot 2257.8 / (2 \cdot 10.4) + 0.51276E+09 / (\pi \cdot 2257.8^2 \cdot 10.4) \\ = 42.91 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\ = 0.367 \cdot 2257.8 / (2 \cdot 12.4) - 0.6E+09 / (0.1607 \cdot \pi \cdot 2257.8^2 \cdot 12.4) \\ = 53.67 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = 0.367 \cdot 2257.8 / (2 \cdot 12.4) + 0.6E+09 / (0.2792 \cdot \pi \cdot 2257.8^2 \cdot 12.4) \\ = 21.75 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 1276507(11300.0 - 2 \cdot 2240.0) / (11300.0 + (4 \cdot 1130.8 / 3)) \\ = 679728.1 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 0.7988 \cdot 679728 / (2257.8 \cdot 12.4) \\ = 19.40 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(2257.8 \cdot 12.4)} \\ = 130.511 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(350.0 + 1.56 \cdot \sqrt{(2257.8 \cdot 12.4)}, 2 \cdot 2240.0) \\ = 611.02 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1) \\ = \min(137.997/137.997, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ = -0.6733 \cdot 1276507 \cdot 0.1 / (611.023(12.4 + 1.0 \cdot 20.0)) \\ = -4.34 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\ = -1276507 / (4(12.4 + 1.0 \cdot 20.0)611.023) - 12 \cdot 0.031 \cdot 1276507 \cdot 2257.8 / (11300.0(12.4 + 1.0 \cdot 20.0)^2) \\ = -107.39 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3980.0000	mm
Baseplate Thickness	Bpthk	42.0000	mm



Baseplate Width	Bpwid	528.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	25.0000	mm
Web Thickness	Webtk	25.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.6	MPa
Height of Web at Center	Hw,c	264.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(2251.6 \cdot 12.4)} + 700.0)12.4^3)/12$$

$$= 152635.297 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 152635.297 + 11912.234(136.859 - 6.2)^2$$

$$= 203515232.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	960.7	12.4	6.2	11912.2	73855.9	0.204E+09
Wearplate	700.0	20.0	22.4	14000.0	313600.0	0.184E+09
Web	25.0	224.0	144.4	5600.0	808638.9	0.237E+08
BasePlate	528.0	42.0	277.4	22176.0	6151618.5	0.441E+09
Totals	...	...	...	53688.2	7347713.0	0.852E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7347713/53688.227$$

$$= 136.859 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 150.0/2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0)\cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 1276507$$

$$= 331090.5312 \text{ N}$$

Tension Stress, St = (Fh/As) = 7.9261 MPa
 Allowed Stress, Sa = 0.6 * Yield Str = 145.5600 MPa

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 2550.0 - 2251.6 \cdot \sin(150.0/2) / 1.309$$

$$= 888.515 \text{ mm}$$

Bending Moment, M = Fh * d = 294298144.0000 N-mm

Bending Stress, Sb = (M * C1 / I) = 47.2364 MPa
 Allowed Stress, Sa = 2/3 * Yield Str = 161.7334 MPa

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt})\text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt[3]{(3(1276507 + 26846)528.0) / (2 \cdot 3980.0 \cdot 161.733)}$$

$$= 40.047 \text{ mm}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (2250.0 + 14.0 + 20.0)$$

$$= 4412.349 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 4412.3486 / (5 - 1)$$

$$= 1103.087 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1103.0872 \cdot 528.0/2$$

$$= 291215.000 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1303353 / (3980.0 \cdot 528.0)$$

$$= 0.620 \text{ N/mm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 291215.0 \cdot 0.62$$

$$= 180617.031 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8125.0 + 13788.59$$

$$= 21913.590 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/\text{Ar}$$

$$= 180617.0/21913.5898$$

$$= 8.243 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.170E+09
Web	551.5	25.0	12.5	13788.6	172357.4	0.588E+08
Totals	...	...	...	21913.6	1695794.8	0.229E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350.0 - 25.0$$

$$= 325.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 1695795/21913.59$$

$$= 77.386 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(77.386, 350.0 - 77.386)$$

$$= 272.614 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(0.22880\text{E}+09/21913.59)}$$

$$= 102.182 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$



$$= 2550.0 - \cos(150.0/2) (2250.0 + 14.0 + 20.0) - 42.0$$

$$= 1916.857 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(150.0/2) (2250.0 + 14.0 + 20.0)$$

$$= 4412.349 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3980.0 - 50.8$$

$$= 3929.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (4412.349 - 3929.2) / 2$$

$$= 241.574 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1916.857^2 + 241.574^2}$$

$$= 1932.020 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 242.6)}$$

$$= 127.552$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1932.02 / 102.182$$

$$= 18.908$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 145677.5 / (2 \cdot 3980.0) \cdot 1103.087 \cdot 1932.02 / 2$$

$$= 19509530.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($18.9076 < 127.5515$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (18.91)^2 / (2 \cdot 127.55^2)) 243 /$$

$$(5/3 + 3 \cdot (18.91) / (8 \cdot 127.55) - (18.91^3) / (8 \cdot 127.55^3))$$

$$= 139.3 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.24 / 139.35 + (19509530 \cdot 272.614 / .22880E+09) / 161.73$$

$$= 0.203$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.220E+09
Web	1103.1	25.0	12.5	27577.2	344714.8	0.452E+08
Totals	...	...	...	35702.2	1868152.1	0.265E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1868152 / 35702.18$$

$$= 52.326 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(52.326, \text{abs}(350.0 - 52.326))$$

$$= 297.674 \text{ mm}$$

Length of Inner Rib [L]:



$$\begin{aligned}
 &= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk} \\
 &= 2550.0 - \cos(2284.0/2)(42.0 + 0.0 + 0.0) - 0.0 \\
 &= 224.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{\text{Total Inertia} / \text{Total Area}} \\
 &= \sqrt{0.26515\text{E}+09/35702.18} \\
 &= 86.179 \text{ mm}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 224.0/86.179 \\
 &= 2.599
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Baseplate Length}) \\
 &= 145677.484 / (2 \cdot 3980.0) \\
 &= 18.301 \text{ N/mm}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 18.301 \cdot 1103.087 \cdot 224.0 \\
 &= 4523903.500 \text{ N-mm}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 4523904/890752 \\
 &= 5.077 \text{ MPa}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ (2.5992 < 127.5515) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (2.6)^2 / (2 \cdot 127.55^2)) 243 / \\
 &\quad (5/3 + 3 \cdot (2.6) / (8 \cdot 127.55) - (2.6^3) / (8 \cdot 127.55^3)) \\
 &= 144.9 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba} \\
 &= 9.14/144.87 + (4523904 \cdot 297.674/0.26515\text{E}+09)/161.73 \\
 &= 0.094
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	138.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	40.0 mm
Nominal Bolt Diameter	Bnd	48.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	1155.3340 mm ²
Saddle Load Q0 (Weight)	Q0	1163348.5 N
Saddle Load QL (Wind/Seismic contribution)	QL	54468.9 N
Maximum Transverse Force	Ft	72838.8 N
Maximum Longitudinal Force	Fl	145677.5 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$\begin{aligned}
 &= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 145677.5 / (1155.334 \cdot 4) \\
 &= 31.5 \text{ MPa. Must be less than } 82.8 \text{ MPa.}
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$\begin{aligned}
 &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 72838.8 / (1155.334 \cdot 4) \\
 &= 15.8 \text{ MPa. Must be less than } 82.8 \text{ MPa.}
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 145677.48/(82.8 • 4.0)
= 439.8849 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 2550.0 • 72838.75 + 0.0
= 185814096.00 N-mm

Eccentricity (e):
= Rmom / QO
= 0.18581E+09/1163349
= 159.66 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 439.8849, 0.0]
= 439.8849 mm²

MDMT Calculations for: Right Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 14.0, tr = 8.424, c = 1.6 mm, E* = 0.85
Thickness Ratio = tr • E*/(tg - c) = 0.577, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

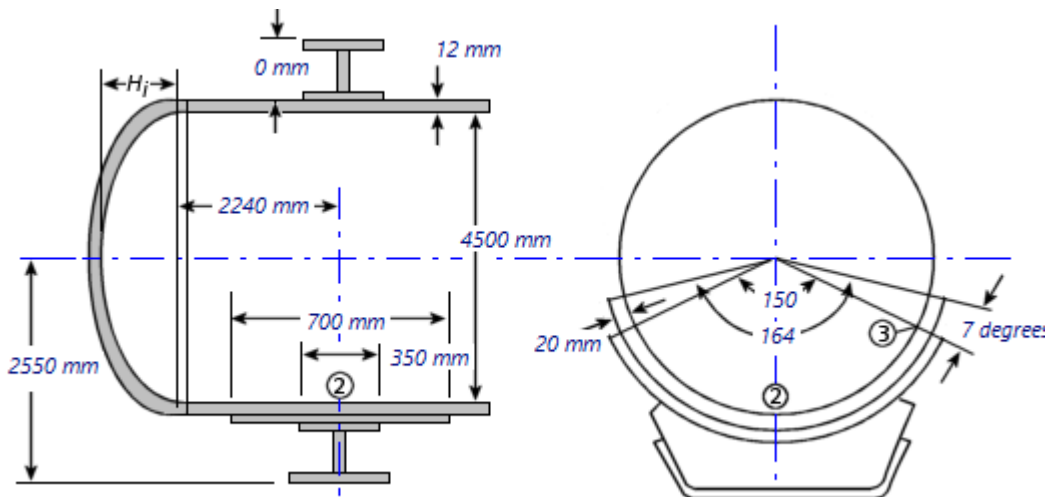
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	2256.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	11300.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2240.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	164.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	1130.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2550.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	1178538.75	N
Pressure used in Analysis	P	0.477 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	42.38	234.00
Long. Stress at Bottom of Midspan	47.32	234.00
Long. Stress at Top of Saddles	64.21	234.00
Long. Stress at Bottom of Saddles	33.71	234.00
Tangential Shear in Shell	18.52	138.58
Circ. Stress at Horn of Saddle	101.59	350.99
Circ. Compressive Stress in Shell	4.09	234.00

**Intermediate Results: Saddle Reaction Q due to Wind or Seismic:**

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (19068.0/2 + 0) \cdot 2550.0/3980.0$$

$$= 18325.4 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(11658.56, 0) \cdot 2550.0/6820.0$$

$$= 4359.1 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 1160213 + \max(4359, 18325, 0, 0)$$

$$= 1178538.8 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (4.524 + \pi/4 (4.524 \cdot 1.2)^2)$$

$$= 23318.318 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1205387.38	N
Transverse Shear Load	9534.02	N
Longitudinal Shear Load	11658.56	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3980.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0313	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0243
K7p = 0.0240			

*The suffix 'p' denotes the values for a wear plate if it exists.***Note: Dimension a is greater than or equal to Rm/2.**

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -1178539 \cdot 2240.0 [1 - (1 - 2240.0/11300.0 + (2256.0^2 - 1130.0^2) / (2 \cdot 2240.0 \cdot 11300.0)) / (1 + (4 \cdot 1130.0)/(3 \cdot 11300.0))]$$

$$= -597138624.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 1178539 \cdot 11300 / 4 ((1 + 2(2256^2 - 1130^2)/(11300^2)) / (1 + (4 \cdot 1130)/(3 \cdot 11300)) - 4 \cdot 2240/11300)$$

$$= 473376096.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 0.477 \cdot 2256.0 / (2 \cdot 12.0) - 0.47338E+09 / (\pi \cdot 2256.0^2 \cdot 12.0)$$

$$= 42.38 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.477 \cdot 2256.0 / (2 \cdot 12.0) + 0.47338E+09 / (\pi \cdot 2256.0^2 \cdot 12.0)$$

$$= 47.32 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.477 \cdot 2256.0 / (2 \cdot 12.0) - 0.6E+09 / (0.1607 \cdot \pi \cdot 2256.0^2 \cdot 12.0)$$

$$= 64.21 \text{ MPa}$$



Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.477 \cdot 2256.0 / (2 \cdot 12.0) + 0.6E+09 / (0.2792 \cdot \pi \cdot 2256.0^2 \cdot 12.0)$$

$$= 33.71 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 1178539 (11300.0 - 2 \cdot 2240.0) / (11300.0 + (4 \cdot 1130.0 / 3))$$

$$= 627613.4 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 627613 / (2256.0 \cdot 12.0)$$

$$= 18.52 \text{ MPa}$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2256.0 \cdot 12.0)}$$

$$= 128.338 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{(2256.0 \cdot 12.0)}, 2 \cdot 2240.0)$$

$$= 606.68 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma6, r$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.6733 \cdot 1178539 \cdot 0.1 / (606.676(12.0 + 1.0 \cdot 20.0))$$

$$= -4.09 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma7, r^*$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -1178539 / (4(12.0 + 1.0 \cdot 20.0)606.676) -$$

$$12 \cdot 0.031 \cdot 1178539 \cdot 2256.0 / (11300.0(12.0 + 1.0 \cdot 20.0)^2)$$

$$= -101.59 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3980.0000	mm
Baseplate Thickness	Bpthk	42.0000	mm
Baseplate Width	Bpwid	528.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	25.0000	mm
Web Thickness	Webtk	25.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	264.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(2250.0 \cdot 12.0)} + 700.0)12.0^3) / 12$$

$$= 137711.984 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 137711.984 + 11476.01(138.517 - 6.0)^2$$

$$= 201663456.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	956.3	12.0	6.0	11476.0	68856.1	0.202E+09
Wearplate	700.0	20.0	22.0	14000.0	308000.0	0.191E+09



Web	25.0	226.0	145.0	5650.0	819249.4	0.243E+08
BasePlate	528.0	42.0	279.0	22176.0	6187102.0	0.441E+09
Totals	...	...	...	53302.0	7383207.5	0.857E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7383208/53302.0$$

$$= 138.517 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 150.0/2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 1178539$$

$$= 305680.3438 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 7.3090 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2550.0 - 2250.0 \cdot \sin(150.0/2) / 1.309$$

$$= 889.696 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 272072800.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 43.9405 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(1178539 + 26849) \cdot 528.0 / (2 \cdot 3980.0 \cdot 173.33)}$$

$$= 37.202 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (2250.0 + 12.0 + 20.0)$$

$$= 4408.485 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 4408.485 / (5 - 1)$$

$$= 1102.121 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1102.1213 \cdot 528.0/2$$

$$= 290960.031 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1205387 / (3980.0 \cdot 528.0)$$

$$= 0.574 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 290960.0 \cdot 0.57$$

$$= 166894.859 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8125.0 + 13776.518$$

$$= 21901.518 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 166894.9/21901.5176$$

$$= 7.621 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.170E+09
Web	551.1	25.0	12.5	13776.5	172206.5	0.588E+08
Totals	...	...	...	21901.5	1695643.9	0.229E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350.0 - 25.0$$

$$= 325.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1695644/21901.518$$

$$= 77.421 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(77.421, 350.0 - 77.421)$$

$$= 272.579 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 0.22875\text{E}+09/21901.518 }$$

$$= 102.199 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2550.0 - \cos(150.0/2)(2250.0 + 12.0 + 20.0) - 42.0$$

$$= 1917.375 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(150.0/2)(2250.0 + 12.0 + 20.0)$$

$$= 4408.485 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3980.0 - 50.8$$

$$= 3929.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd)/2$$

$$= (4408.485 - 3929.2)/2$$

$$= 239.643 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{ Lv^2 + Dd^2 }$$

$$= \sqrt{ 1917.375^2 + 239.643^2 }$$

$$= 1932.293 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 \cdot \pi^2 \cdot 199955/260.0 }$$

$$= 123.211$$



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1932.293/102.199$$

$$= 18.907$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 11658.6 / (2 \cdot 3980.0) \cdot 1102.121 \cdot 1932.29/2$$

$$= 1560199.500 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (18.9072 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.91)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (18.91) / (8 \cdot 123.21) - (18.91^3) / (8 \cdot 123.21^3))$$

$$= 149.1 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 7.62/149.05 + (1560200 \cdot 272.579 / .22875E+09) / 173.33$$

$$= 0.062$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.220E+09
Web	1102.1	25.0	12.5	27553.0	344412.9	0.452E+08
Totals	...	...	...	35678.0	1867850.4	0.265E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1867850/35678.035$$

$$= 52.353 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(52.353, \text{abs}(350.0 - 52.353))$$

$$= 297.647 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2550.0 - \cos(2282.0/2) (42.0 + 0.0 + 0.0) - 0.0$$

$$= 226.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (0.26511E+09 / 35678.035) }$$

$$= 86.202 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 226.0/86.202$$

$$= 2.622$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 11658.557 / (2 \cdot 3980.0)$$

$$= 1.465 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.465 \cdot 1102.121 \cdot 226.0$$

$$= 364960.125 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 364960.156/890700$$



= 0.410 MPa

Compressive Allowable, $KL/r < C_c$ (2.6218 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (2.62)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (2.62) / (8 \cdot 123.21) - (2.62^3) / (8 \cdot 123.21^3))$$

$$= 155.2 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 8.45 / 155.22 + (364960.16 \cdot 297.647 / .26511E+09) / 173.33$$

$$= 0.057$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	40.0 mm
Nominal Bolt Diameter	Bnd	48.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	1155.3340 mm ²
Saddle Load Q0 (Weight)	Q0	1187061.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	4359.1 N
Maximum Transverse Force	Ft	9534.0 N
Maximum Longitudinal Force	Fl	11658.6 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 11658.6 / (1155.334 \cdot 4)$$

$$= 2.5 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9534.0 / (1155.334 \cdot 4)$$

$$= 2.1 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 11658.56 / (68.4 \cdot 4.0)$$

$$= 42.6161 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2550.0 \cdot 9534.02 + 0.0$$

$$= 24321612.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 24321612 / 1187062$$

$$= 20.48 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 42.6161, 0.0]$$



= 42.6161 mm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	2257.00	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	11300.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2240.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	164.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	1130.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2550.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	1188075.75	N
Pressure used in Analysis	P	0.477 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	42.39	234.00
Long. Stress at Bottom of Midspan	47.36	234.00
Long. Stress at Top of Saddles	55.17	234.00
Long. Stress at Bottom of Saddles	28.85	234.00
Tangential Shear in Shell	16.00	138.58
Circ. Stress at Horn of Saddle	91.08	350.99
Circ. Compressive Stress in Shell	3.75	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(19068.0/2 + 0) \cdot 2550.0/3980.0$$

$$= 18325.4 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(11675.81, 0) \cdot 2550.0/6820.0$$

$$= 4365.6 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 1169750 + \max(4366, 18325, 0, 0)$$

$$= 1188075.8 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(4.524 + \pi/4(4.528 \cdot 1.2)^2)$$

$$= 23352.828 \text{ N}$$

**Summary of Loads at the Base of this Saddle:**

Vertical Load (including saddle weight)	1214921.88	N
Transverse Shear Load	9534.02	N
Longitudinal Shear Load	11675.81	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length

= 3980.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0313	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0243
K7p = 0.0240			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -1188076 \cdot 2240.0 \left[1 - \left(1 - \frac{2240.0}{11300.0} + \frac{(2257.0^2 - 1130.0^2)}{(2 \cdot 2240.0 \cdot 11300.0)} \right) / \left(1 + \frac{(4 \cdot 1130.0)}{(3 \cdot 11300.0)} \right) \right]$$

$$= -601761408.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(R_m^2 - h m^2)/(L^2))}{(1 + (4hm^2)/(3L))} - \frac{4a}{L} \right)$$

$$= 1188076 \cdot 11300.0 / 4 \left(\frac{(1 + 2(2257^2 - 1130^2)/(11300^2))}{(1 + (4 \cdot 1130)/(3 \cdot 11300))} - \frac{4 \cdot 2240}{11300} \right)$$

$$= 477416352.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 0.477 \cdot 2257.0 / (2 \cdot 12.0) - 0.47742E+09 / (\pi \cdot 2257.0^2 \cdot 12.0)$$

$$= 42.39 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.477 \cdot 2257.0 / (2 \cdot 12.0) + 0.47742E+09 / (\pi \cdot 2257.0^2 \cdot 12.0)$$

$$= 47.36 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.477 \cdot 2257.0 / (2 \cdot 14.0) - 0.6E+09 / (0.1607 \cdot \pi \cdot 2257.0^2 \cdot 14.0)$$

$$= 55.17 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.477 \cdot 2257.0 / (2 \cdot 14.0) + 0.6E+09 / (0.2792 \cdot \pi \cdot 2257.0^2 \cdot 14.0)$$

$$= 28.85 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm/3))$$

$$= 1188076(11300.0 - 2 \cdot 2240.0) / (11300.0 + (4 \cdot 1130.0/3))$$

$$= 632692.2 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 632692 / (2257.0 \cdot 14.0)$$

$$= 16.00 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2257.0 \cdot 14.0)}$$

$$= 138.651 \text{ mm}$$



Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{ 2257.0 \cdot 14.0 }, 2 \cdot 2240.0)$$

$$= 627.30 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.6733 \cdot 1188076 \cdot 0.1 / (627.303(14.0 + 1.0 \cdot 20.0))$$

$$= -3.75 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -1188076 / (4(14.0 + 1.0 \cdot 20.0)627.303) -$$

$$12 \cdot 0.031 \cdot 1188076 \cdot 2257.0 / (11300.0(14.0 + 1.0 \cdot 20.0)^2)$$

$$= -91.08 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3980.0000	mm
Baseplate Thickness	Bpthk	42.0000	mm
Baseplate Width	Bpwid	528.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	25.0000	mm
Web Thickness	Webtk	25.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	264.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ 2250.0 \cdot 14.0 } + 700.0) 14.0^3) / 12$$

$$= 223377.984 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 223377.984 + 13676.216(134.105 - 7.0)^2$$

$$= 221172912.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	976.9	14.0	7.0	13676.2	95733.5	0.221E+09
Wearplate	700.0	20.0	24.0	14000.0	336000.0	0.170E+09
Web	25.0	224.0	146.0	5600.0	817600.3	0.242E+08
BasePlate	528.0	42.0	279.0	22176.0	6187104.5	0.469E+09
Totals	...	...	...	55452.2	7436438.0	0.884E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7436438 / 55452.215$$

$$= 134.105 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:



$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.259 \cdot 1188076 \\
 &= 308154.0312 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 7.3770 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 2550.0 - 2250.0 \cdot \sin(150.0/2) / 1.309 \\
 &= 889.696 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 274274528.0000 \text{ N-mm}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 41.5758 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt[3]{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\
 &= \sqrt[3]{(3 \cdot (1188076 + 26846) \cdot 528.0 / (2 \cdot 3980.0 \cdot 173.33))} \\
 &= 37.349 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 150.0/2) (2250.0 + 14.0 + 20.0) \\
 &= 4412.349 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 4412.3486 / (5 - 1) \\
 &= 1103.087 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot \text{Bpwid} / 2 \\
 &= 1103.0872 \cdot 528.0 / 2 \\
 &= 291215.000 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 1214922 / (3980.0 \cdot 528.0) \\
 &= 0.578 \text{ N/mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 291215.0 \cdot 0.58 \\
 &= 168362.406 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of the Rib and Web [Ar]:} \\
 &= \text{Rib Area} + \text{Web Area} \\
 &= 8125.0 + 13788.59 \\
 &= 21913.590 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Compressive Stress [Sc]:} \\
 &= P/Ar \\
 &= 168362.4 / 21913.5898 \\
 &= 7.684 \text{ MPa}
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.170E+09
Web	551.5	25.0	12.5	13788.6	172357.4	0.588E+08
Totals	...	...	...	21913.6	1695794.8	0.229E+09



Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 350.0 - 25.0 \\
 &= 325.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 1695795/21913.59 \\
 &= 77.386 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \max(\text{ytot}, \text{Saddle Width} - \text{ytot}) \\
 &= \max(77.386, 350.0 - 77.386) \\
 &= 272.614 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 0.22880\text{E}+09/21913.59 } \\
 &= 102.182 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2550.0 - \cos(150.0/2) (2250.0 + 14.0 + 20.0) - 42.0 \\
 &= 1916.857 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(150.0/2) (2250.0 + 14.0 + 20.0) \\
 &= 4412.349 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 3980.0 - 50.8 \\
 &= 3929.200 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (\text{Lw} - \text{Bd}) / 2 \\
 &= (4412.349 - 3929.2) / 2 \\
 &= 241.574 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{ \text{Lv}^2 + \text{Dd}^2 } \\
 &= \sqrt{ 1916.857^2 + 241.574^2 } \\
 &= 1932.020 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 199955/260.0 } \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 1932.02/102.182 \\
 &= 18.908
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2 \\
 &= 11675.8 / (2 \cdot 3980.0) \cdot 1103.087 \cdot 1932.02/2 \\
 &= 1563656.875 \text{ N-mm}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($18.9076 < 123.2107$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (18.91)^2 / (2 \cdot 123.21^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (18.91) / (8 \cdot 123.21) - (18.91^3) / (8 \cdot 123.21^3)) \\
 &= 149.1 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 7.68/149.05 + (1563657 \cdot 272.614 / .22880\text{E}+09) / 173.33
 \end{aligned}$$



= 0.062

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.220E+09
Web	1103.1	25.0	12.5	27577.2	344714.8	0.452E+08
Totals	...	...	...	35702.2	1868152.1	0.265E+09

Distance to Centroid from Datum [ytot]:

= AY / A

= 1868152/35702.18

= 52.326 mm

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))

= max(52.326, abs(350.0 - 52.326))

= 297.674 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 2550.0 - cos(2284.0/2)(42.0 + 0.0 + 0.0) - 0.0

= 224.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{0.26515\text{E}+09/35702.18}$

= 86.179 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 224.0/86.179

= 2.599

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 11675.811/(2 • 3980.0)

= 1.467 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 1.467 • 1103.087 • 224.0

= 362583.469 N-mm

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 362583.438/890752

= 0.407 MPa

Compressive Allowable, KL/r < Cc (2.5992 < 123.2107) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (2.6)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (2.6) / (8 \cdot 123.21) - (2.6)^3 / (8 \cdot 123.21^3))$

= 155.2 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I) /Sba

= 8.52/155.23 + (362583.44 • 297.674/.26515E+09)/173.33

= 0.057

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-36	
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm



Distance from Bolts to Edge	Edgedis	40.0 mm
Nominal Bolt Diameter	Bnd	48.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	1155.3340 mm ²
Saddle Load Q0 (Weight)	Q0	1196596.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	4365.6 N
Maximum Transverse Force	Ft	9534.0 N
Maximum Longitudinal Force	Fl	11675.8 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 11675.8 / (1155.334 \cdot 4)$$

$$= 2.5 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9534.0 / (1155.334 \cdot 4)$$

$$= 2.1 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 11675.81 / (68.4 \cdot 4.0)$$

$$= 42.6791 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2550.0 \cdot 9534.02 + 0.0$$

$$= 24321612.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 24321612 / 1196596$$

$$= 20.32 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 42.6791, 0.0]$$

$$= 42.6791 \text{ mm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

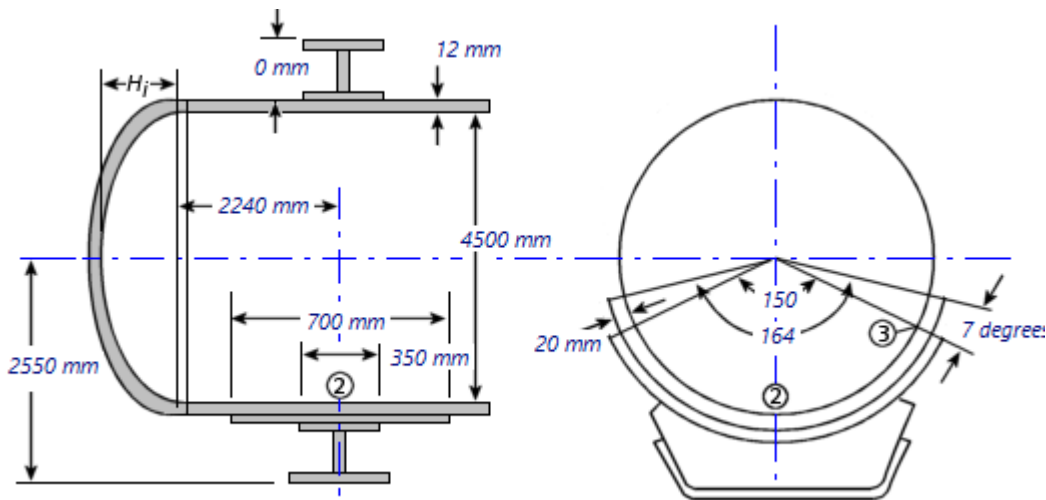
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	2256.80	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	11300.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2240.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	164.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	1130.80	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2550.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	1266969.38	N
Pressure used in Analysis	P	-0.052 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	-8.67	-68.86
Long. Stress at Top of Midspan	8.67	138.00
Long. Stress at Bottom of Midspan	-2.55	-68.86
Long. Stress at Bottom of Midspan	2.55	138.00
Long. Stress at Top of Saddles	18.39	138.00
Long. Stress at Bottom of Saddles	-19.42	-68.86
Long. Stress at Bottom of Saddles	19.42	138.00
Tangential Shear in Shell	22.96	110.40
Circ. Stress at Horn of Saddle	120.61	172.50



Circ. Compressive Stress in Shell	4.76	138.00
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(38136.1/2 + 0) \cdot 2550.0/3980.0$$

$$= 36650.9 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind; Sum of X Forces}) \cdot B / Ls$$

$$= \max(23317.11, 0) \cdot 2550.0/6820.0$$

$$= 8718.3 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq; Sum of X Forces}) \cdot B / Ls$$

$$= \max(145677.48, 0) \cdot 2550.0/6820.0$$

$$= 54468.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(145677/2 + 0) \cdot 2550.0/3980.0$$

$$= 140004.1 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 1126965 + \max(8718, 36651, 54469, 140004)$$

$$= 1266969.4 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(4.524 + \pi/4(4.524 \cdot 1.2)^2)$$

$$= 23318.318 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1293818.12	N
Transverse Shear Load	72838.75	N
Longitudinal Shear Load	145677.48	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3980.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0313	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0243
K7p = 0.0240			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -1266969 \cdot 2240.0 [1 - (1 - 2240.0/11300.0 + (2256.8^2 - 1130.8^2)/(2 \cdot 2240.0 \cdot 11300.0))/(1 + (4 \cdot 1130.8)/(3 \cdot 11300.0))]$$

$$= -642038208.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 1266969 \cdot 11300/4((1 + 2(2257^2 - 1131^2)/(11300^2))/(1 + (4 \cdot 1131)/(3 \cdot 11300)) - 4 \cdot 2240/11300)$$

$$= 508705760.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:



$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= -0.052 \cdot 2256.8 / (2 \cdot 10.4) - 0.50871E+09 / (\pi \cdot 2256.8^2 \cdot 10.4)$$

$$= -8.67 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.052 \cdot 2256.8 / (2 \cdot 10.4) + 0.50871E+09 / (\pi \cdot 2256.8^2 \cdot 10.4)$$

$$= -2.55 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= -0.052 \cdot 2256.8 / (2 \cdot 10.4) - 0.6E+09 / (0.1607 \cdot \pi \cdot 2256.8^2 \cdot 10.4)$$

$$= 18.39 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.052 \cdot 2256.8 / (2 \cdot 10.4) + 0.6E+09 / (0.2792 \cdot \pi \cdot 2256.8^2 \cdot 10.4)$$

$$= -19.42 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 1266969 (11300.0 - 2 \cdot 2240.0) / (11300.0 + (4 \cdot 1130.8 / 3))$$

$$= 674649.6 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 674650 / (2256.8003 \cdot 10.4)$$

$$= 22.96 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2256.8 \cdot 10.4)}$$

$$= 119.497 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{(2256.8 \cdot 10.4)}, 2 \cdot 2240.0)$$

$$= 588.99 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.6733 \cdot 1266969 \cdot 0.1 / (588.994(10.4 + 1.0 \cdot 20.0))$$

$$= -4.76 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma_{7,r*}$]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2)$$

$$= -1266969 / (4(10.4 + 1.0 \cdot 20.0) 588.994) -$$

$$12 \cdot 0.031 \cdot 1266969 \cdot 2256.8 / (11300.0(10.4 + 1.0 \cdot 20.0)^2)$$

$$= -120.61 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3980.0000	mm
Baseplate Thickness	Bpthk	42.0000	mm
Baseplate Width	Bpwid	528.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	25.0000	mm
Web Thickness	Webtk	25.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.6	MPa
Height of Web at Center	Hw,c	264.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(2251.6 \cdot 10.4)} + 700.0)^3) / 12$$

$$= 87994.180 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 87994.18 + 9762.677 (141.469 - 5.2)^2$$

$$= 181373232.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	938.7	10.4	5.2	9762.7	50765.9	0.181E+09
Wearplate	700.0	20.0	20.4	14000.0	285600.0	0.206E+09
Web	25.0	226.0	143.4	5650.0	810209.4	0.241E+08
BasePlate	528.0	42.0	277.4	22176.0	6151620.0	0.413E+09
Totals	...	...	...	51588.7	7298195.5	0.824E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7298196 / 51588.668$$

$$= 141.469 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 1266969$$

$$= 328616.8438 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 7.8574 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.5600 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2550.0 - 2251.6 \cdot \sin(150.0/2) / 1.309$$

$$= 888.515 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 292099360.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 50.1254 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.7334 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(1266969 + 26849) \cdot 528.0) / (2 \cdot 3980.0 \cdot 161.733)}$$

$$= 39.900 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (2250.0 + 12.0 + 20.0)$$

$$= 4408.485 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 4408.4854 / (5 - 1)$$

$$= 1102.121 \text{ mm}$$



Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 1102.1213 \cdot 528.0 / 2$$

$$= 290960.031 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 1293818 / (3980.0 \cdot 528.0)$$

$$= 0.616 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 290960.0 \cdot 0.62$$

$$= 179138.766 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 8125.0 + 13776.518$$

$$= 21901.518 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 179138.8 / 21901.5176$$

$$= 8.180 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.170E+09
Web	551.1	25.0	12.5	13776.5	172206.5	0.588E+08
Totals	...	...	...	21901.5	1695643.9	0.229E+09

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 350.0 - 25.0$$

$$= 325.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1695644 / 21901.518$$

$$= 77.421 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, Saddle \text{ Width} - ytot)$$

$$= \max(77.421, 350.0 - 77.421)$$

$$= 272.579 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (0.22875E+09 / 21901.518) }$$

$$= 102.199 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk$$

$$= 2550.0 - \cos(150.0 / 2)(2250.0 + 12.0 + 20.0) - 42.0$$

$$= 1917.375 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(150.0 / 2)(2250.0 + 12.0 + 20.0)$$

$$= 4408.485 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 3980.0 - 50.8$$



$$= 3929.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (4408.485 - 3929.2) / 2$$

$$= 239.643 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1917.375^2 + 239.643^2}$$

$$= 1932.293 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 242.6)}$$

$$= 127.552$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1932.293 / 102.199$$

$$= 18.907$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 145677.5 / (2 \cdot 3980.0) \cdot 1102.121 \cdot 1932.29 / 2$$

$$= 19495206.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($18.9072 < 127.5515$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.91)^2 / (2 \cdot 127.55^2)) 243 /$$

$$(5/3 + 3 \cdot (18.91) / (8 \cdot 127.55) - (18.91^3) / (8 \cdot 127.55^3))$$

$$= 139.3 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.18 / 139.35 + (19495206 \cdot 272.579 / .22875E+09) / 161.73$$

$$= 0.202$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.220E+09
Web	1102.1	25.0	12.5	27553.0	344412.9	0.452E+08
Totals	...	...	...	35678.0	1867850.4	0.265E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1867850 / 35678.035$$

$$= 52.353 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(52.353, \text{abs}(350.0 - 52.353))$$

$$= 297.647 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2550.0 - \cos(2282.0/2) (42.0 + 0.0 + 0.0) - 0.0$$

$$= 226.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{0.26511E+09 / 35678.035}$$

$$= 86.202 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



$$= 1 \cdot 226.0/86.202$$

$$= 2.622$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 145677.484 / (2 \cdot 3980.0)$$

$$= 18.301 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 18.301 \cdot 1102.121 \cdot 226.0$$

$$= 4560297.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4560297/890700$$

$$= 5.118 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($2.6218 < 127.5515$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (2.62)^2 / (2 \cdot 127.55^2)) 243 / (5/3 + 3 \cdot (2.62) / (8 \cdot 127.55) - (2.62^3) / (8 \cdot 127.55^3))$$

$$= 144.9 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 9.07/144.86 + (4560297 \cdot 297.647 / 1.26511E+09) / 161.73$$

$$= 0.094$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	2257.80	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	11300.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2240.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	164.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	1130.80	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2550.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		1276506.62	N
Pressure used in Analysis	P	-0.052	MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top	of Midspan	-8.69	-68.86
Long. Stress at Top	of Midspan	8.69	138.00
Long. Stress at Bottom	of Midspan	-2.54	-68.86
Long. Stress at Bottom	of Midspan	2.54	138.00
Long. Stress at Top	of Saddles	15.55	138.00



Long. Stress at Bottom of Saddles	-16.37	-82.06
Long. Stress at Bottom of Saddles	16.37	138.00
Tangential Shear in Shell	19.40	110.40
Circ. Stress at Horn of Saddle	107.39	172.50
Circ. Compressive Stress in Shell	4.34	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(38136.1/2 + 0) \cdot 2550.0/3980.0$
 $= 36650.9 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(23351.62, 0) \cdot 2550.0/6820.0$
 $= 8731.2 \text{ N}$

Saddle Reaction Force due to Earthquake FI [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(145677.48, 0) \cdot 2550.0/6820.0$
 $= 54468.9 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(145677/2 + 0) \cdot 2550.0/3980.0$
 $= 140004.1 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 1136503 + \max(8731, 36651, 54469, 140004)$
 $= 1276506.6 \text{ N}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08(4.524 + \pi/4(4.528 \cdot 1.2)^2)$
 $= 23352.828 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1303352.62	N
Transverse Shear Load	72838.75	N
Longitudinal Shear Load	145677.48	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 3980.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0313	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0243
K7p = 0.0240			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -1276507 \cdot 2240.0 [1 - (1 - 2240.0/11300.0 + (2257.8^2 - 1130.8^2)/(2 \cdot 2240.0 \cdot 11300.0))/(1 + (4 \cdot 1130.8)/(3 \cdot 11300.0))]$
 $= -646646144.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$



$$= 1276507 \cdot 11300 / 4 \left((1 + 2(2258^2 - 1131^2) / (11300^2)) / (1 + (4 \cdot 1131) / (3 \cdot 11300)) - 4 \cdot 2240 / 11300 \right)$$

$$= 512760256.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.052 \cdot 2257.8 / (2 \cdot 10.4) - 0.51276 \text{E}+09 / (\pi \cdot 2257.8^2 \cdot 10.4)$$

$$= -8.69 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.052 \cdot 2257.8 / (2 \cdot 10.4) + 0.51276 \text{E}+09 / (\pi \cdot 2257.8^2 \cdot 10.4)$$

$$= -2.54 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.052 \cdot 2257.8 / (2 \cdot 12.4) - 0.6 \text{E}+09 / (0.1607 \cdot \pi \cdot 2257.8^2 \cdot 12.4)$$

$$= 15.55 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.052 \cdot 2257.8 / (2 \cdot 12.4) + 0.6 \text{E}+09 / (0.2792 \cdot \pi \cdot 2257.8^2 \cdot 12.4)$$

$$= -16.37 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 1276507(11300.0 - 2 \cdot 2240.0) / (11300.0 + (4 \cdot 1130.8 / 3))$$

$$= 679728.1 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 679728 / (2257.8 \cdot 12.4)$$

$$= 19.40 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{2257.8 \cdot 12.4}$$

$$= 130.511 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{2257.8 \cdot 12.4}, 2 \cdot 2240.0)$$

$$= 611.02 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.6733 \cdot 1276507 \cdot 0.1 / (611.023(12.4 + 1.0 \cdot 20.0))$$

$$= -4.34 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma_{7,r*}$]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -1276507 / (4(12.4 + 1.0 \cdot 20.0)611.023) -$$

$$12 \cdot 0.031 \cdot 1276507 \cdot 2257.8 / (11300.0(12.4 + 1.0 \cdot 20.0)^2)$$

$$= -107.39 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3980.0000	mm
Baseplate Thickness	Bpthk	42.0000	mm
Baseplate Width	Bpwid	528.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	25.0000	mm
Web Thickness	Webtk	25.0000	mm
Web Location	Webloc	Side	



Saddle Design Temperature 21.1 °C
Saddle Yield Stress Sy 242.6 MPa
Height of Web at Center Hw,c 264.0 mm
Friction Coefficient mu 0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(2251.6 \cdot 12.4)} + 700.0)^3) / 12$
 $= 152635.297 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 152635.297 + 11912.234 (136.859 - 6.2)^2$
 $= 203515232.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	960.7	12.4	6.2	11912.2	73855.9	0.204E+09
Wearplate	700.0	20.0	22.4	14000.0	313600.0	0.184E+09
Web	25.0	224.0	144.4	5600.0	808638.9	0.237E+08
BasePlate	528.0	42.0	277.4	22176.0	6151618.5	0.441E+09
Totals	...	...	...	53688.2	7347713.0	0.852E+09

Distance to Centroid [C1]:
 $= AY / A$
 $= 7347713 / 53688.227$
 $= 136.859 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 150.0 / 2$
 $= 105.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$
 $= 0.2594$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.259 \cdot 1276507$
 $= 331090.5312 \text{ N}$

Tension Stress, St = (Fh/As) = 7.9261 MPa
Allowed Stress, Sa = 0.6 * Yield Str = 145.5600 MPa

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 2550.0 - 2251.6 \cdot \sin(150.0/2) / 1.309$
 $= 888.515 \text{ mm}$

Bending Moment, M = Fh * d = 294298144.0000 N-mm

Bending Stress, Sb = (M * C1 / I) = 47.2364 MPa
Allowed Stress, Sa = 2/3 * Yield Str = 161.7334 MPa

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(1276507 + 26846)528.0) / (2 \cdot 3980.0 \cdot 161.733)}$
 $= 40.047 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 150.0/2) (2250.0 + 14.0 + 20.0)$



$$= 4412.349 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 4412.3486 / (5 - 1)$$

$$= 1103.087 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1103.0872 \cdot 528.0 / 2$$

$$= 291215.000 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1303353 / (3980.0 \cdot 528.0)$$

$$= 0.620 \text{ N/mm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 291215.0 \cdot 0.62$$

$$= 180617.031 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8125.0 + 13788.59$$

$$= 21913.590 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 180617.0 / 21913.5898$$

$$= 8.243 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.170E+09
Web	551.5	25.0	12.5	13788.6	172357.4	0.588E+08
Totals	...	...	...	21913.6	1695794.8	0.229E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350.0 - 25.0$$

$$= 325.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 1695795 / 21913.59$$

$$= 77.386 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(77.386, 350.0 - 77.386)$$

$$= 272.614 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(0.22880E+09 / 21913.59)}$$

$$= 102.182 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2550.0 - \cos(150.0/2) (2250.0 + 14.0 + 20.0) - 42.0$$

$$= 1916.857 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$



$$= 2 \cdot \sin(150.0/2) (2250.0 + 14.0 + 20.0)$$

$$= 4412.349 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3980.0 - 50.8$$

$$= 3929.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d)/2$$

$$= (4412.349 - 3929.2)/2$$

$$= 241.574 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(1916.857^2 + 241.574^2)}$$

$$= 1932.020 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/242.6)}$$

$$= 127.552$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1932.02/102.182$$

$$= 18.908$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 145677.5 / (2 \cdot 3980.0) \cdot 1103.087 \cdot 1932.02/2$$

$$= 19509530.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($18.9076 < 127.5515$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.91)^2 / (2 \cdot 127.55^2)) 243 /$$

$$(5/3 + 3 \cdot (18.91) / (8 \cdot 127.55) - (18.91^3) / (8 \cdot 127.55^3))$$

$$= 139.3 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.24/139.35 + (19509530 \cdot 272.614 / .22880E+09) / 161.73$$

$$= 0.203$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	25.0	325.0	187.5	8125.0	1523437.5	0.220E+09
Web	1103.1	25.0	12.5	27577.2	344714.8	0.452E+08
Totals	...	...	...	35702.2	1868152.1	0.265E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1868152/35702.18$$

$$= 52.326 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(52.326, \text{abs}(350.0 - 52.326))$$

$$= 297.674 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2550.0 - \cos(2284.0/2) (42.0 + 0.0 + 0.0) - 0.0$$

$$= 224.000 \text{ mm}$$

Radius of Gyration [r]:



$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(0.26515\text{E}+09 / 35702.18)}$$

$$= 86.179 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 224.0 / 86.179$$

$$= 2.599$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 145677.484 / (2 \cdot 3980.0)$$

$$= 18.301 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 18.301 \cdot 1103.087 \cdot 224.0$$

$$= 4523903.500 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4523904 / 890752$$

$$= 5.077 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (2.5992 < 127.5515) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (2.6)^2 / (2 \cdot 127.55^2)) 243 /$$

$$(5/3 + 3 \cdot (2.6) / (8 \cdot 127.55) - (2.6^3) / (8 \cdot 127.55^3))$$

$$= 144.9 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 9.14 / 144.87 + (4523904 \cdot 297.674 / .26515\text{E}+09) / 161.73$$

$$= 0.094$$

*Review notes about nozzle loadings and supports in the Warnings report.***PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.**



Input, Nozzle Desc: M1-24"

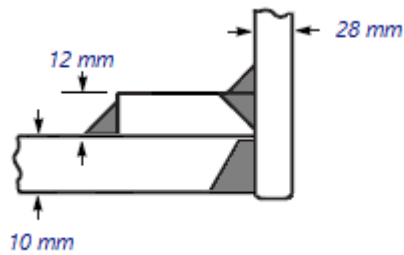
From: 10

Pressure for Reinforcement Calculations	P	0.4160	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	4500.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm
Head Internal Corrosion Allowance	c	1.6000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	909.6000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		150.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M1-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.102 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.42 \cdot 4503.2 \cdot 0.8994 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.42)$
 $= 6.1068 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.42 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.42)$
 $= 0.9178 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0288 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1113.6000 mm
Parallel to Vessel Wall, opening length	d	556.8000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		21.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

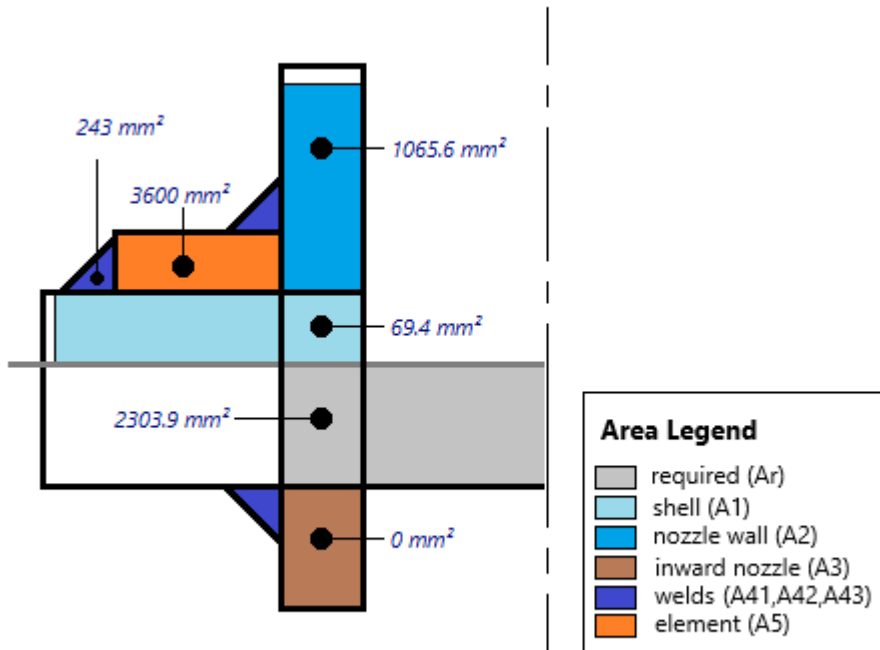


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3400.289	2303.877	NA
Area in Shell	A1	1276.831	69.364	NA
Area in Nozzle Wall	A2	1070.254	1065.592	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	243.000	243.000	NA
Area in Element	A5	3600.000	3600.000	NA
TOTAL AREA AVAILABLE	Atot	6190.084	4977.956	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 686.7601 12.0000 mm
Based on given Pad Diameter: 909.6000 3.0864 mm
Based on Shell or Nozzle Thickness: 702.1921 10.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(556.8 \cdot 8.2754 \cdot 1 + 2 \cdot 26.4 \cdot 8.2754 \cdot 1(1 - 1.0))$
 $= 2303.877 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 556.8(1.0 \cdot 8.4 - 1.0 \cdot 8.275) - 2 \cdot 26.4$
 $(1.0 \cdot 8.4 - 1.0 \cdot 8.2754) \cdot (1 - 1.0)$
 $= 69.364 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 21.0) \cdot (26.4 - 1.03) \cdot 1.0$$

$$= 1065.592 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot Fr_3 + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= (99.0) \cdot 1.0 + (0.0) \cdot 1.0 + 12.0^2 \cdot 1.0$$

$$= 243.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (909.6 - 609.6) \cdot 12.0 \cdot 1.0$$

$$= 3600.000 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:Wall Thickness for Internal/External pressures $t_a = 2.6288 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 2.6288 \text{ mm}$$

Available Nozzle Neck Thickness = 28.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)**

Govrn. thk, $t_g = 28.0$, $t_r = 0.918$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 22 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 12.0$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.727$, Temp. Reduction = 15 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 12.0$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.727$, Temp. Reduction = 15 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:D, Curve:B)

Govrn. thk, $t_g = 10.0$, $t_r = 6.107$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.727$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:D, Curve:B)

Govrn. thk, $t_g = 10.0$, $t_r = 6.107$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.727$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Gov. MDMT of the Nozzle : -44 °C
 Gov. MDMT of the Reinforcement Pad : -44 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.42/1.96 = 0.212

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 10.4000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	5.2000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (2303.8774 - 69.3643 + 2 \cdot 26.4 \cdot 1.0 \cdot (1.0 \cdot 8.4 - 8.2754)) \cdot 138)$
 $= 309238.69 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1065.5916 + 3600.0 + 243.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 677315.81 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1065.5916 + 0.0 + 100.0 + (443.52)) \cdot 138$
 $= 222034.52 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1065.5916 + 0.0 + 243.0 + 3600.0 + (443.52)) \cdot 138$
 $= 738515.25 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 647445. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 909.6 \cdot 12.0 \cdot 0.49 \cdot 138$
 $= 1159263. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 291.6) \cdot (28.0 - 1.6) \cdot 0.7 \cdot 138$
 $= 2336046. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 12.0 \cdot 0.74 \cdot 138$
 $= 1173329. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (10.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 821331. \text{ N}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1159263 + 2336047) = 3495310 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (647445 + 1173329 + 821331 + 0) = 2642105 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (1159263 + 821331 + 0) = 1980594 \text{ N}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 3495309 \text{ N} , \text{ must exceed } W = 309238 \text{ N} \quad \text{or } W1 = 677315 \text{ N}$$

$$\text{Path 2-2} = 2642105 \text{ N} , \text{ must exceed } W = 309238 \text{ N} \quad \text{or } W2 = 222034 \text{ N}$$

$$\text{Path 3-3} = 1980593 \text{ N} , \text{ must exceed } W = 309238 \text{ N} \quad \text{or } W3 = 738515 \text{ N}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 11.4633 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 271.4634 mm

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 * t_{nom} / R_f * (1 - R_f / R_o)) \quad 4.814 \%$$

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Input, Nozzle Desc: K2B-2"

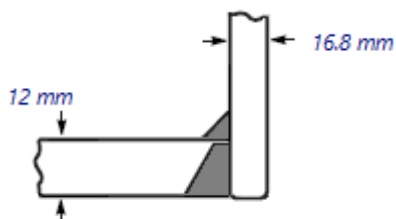
From: 20

Pressure for Reinforcement Calculations	P	0.4378	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	1775.0001	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		650.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2B-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.44 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.44)$
= 7.1575 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{0.44}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0858 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2975 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	108.0000	mm
Parallel to Vessel Wall, opening length	d	54.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

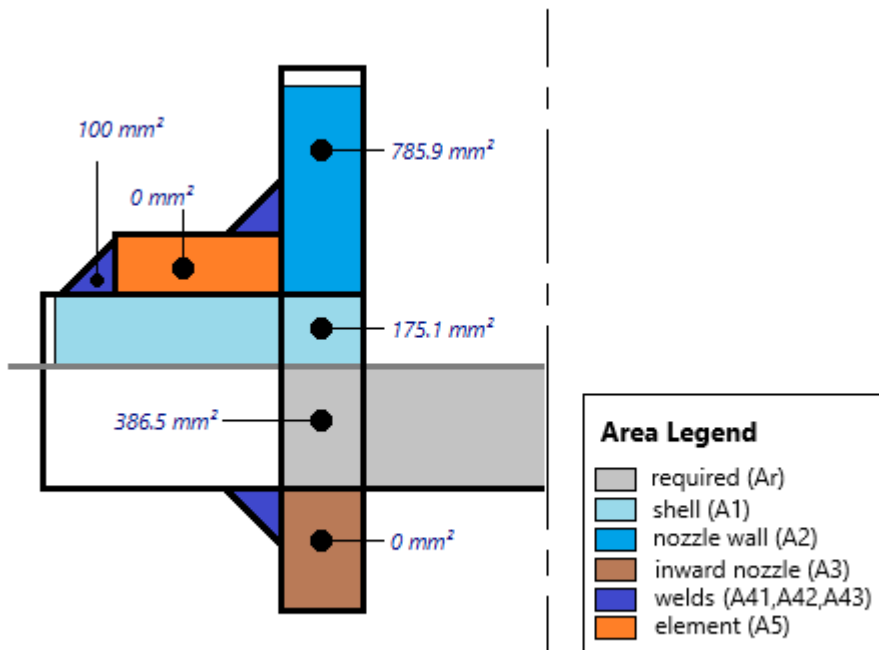


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	386.503	202.427	NA
Area in Shell	A1	175.097	156.745	NA
Area in Nozzle Wall	A2	785.938	774.931	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1061.035	1031.676	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (54.0 \cdot 7.1575 \cdot 1.0 + 2 \cdot 15.2 \cdot 7.1575 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 386.503 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 54.0(1.0 \cdot 10.4 - 1.0 \cdot 7.157) - 2 \cdot 15.2 \\
 &\quad (1.0 \cdot 10.4 - 1.0 \cdot 7.1575) \cdot (1 - 1.0) \\
 &= 175.097 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 26.0)(15.2 - 0.09)1.0 \\
 &= 785.938 \text{ mm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0
 \end{aligned}$$



= 100.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 1.8975$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.7575$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.7575$ mm
 Wall Thickness, shell/head, external pressure $trb_2 = 2.4440$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000$ mm
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$
 = $\min[6.4, \max(8.7575, 4.1)]$
 = 6.4000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$
 = $\max(1.8975, 6.4)$
 = 6.4000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 12.0$, $t_r = 7.157$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -20 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $0.44/5.10 = 0.086$

Weld Size Calculations, Description: K2B-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 10.4000$ mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= $\max(0, (A-A1+2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 = $\max(0, (386.5033 - 175.0967 + 2 \cdot 15.2 \cdot 1.0 \cdot (1.0 \cdot 10.4 - 7.1575)) \cdot 138)$
 = 42772.76 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= $(A2+A5+A4-(W_i \cdot Can/.707)^2 \cdot fr_2) \cdot Sv$
 = $(785.9383 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 = 122246.88 N

Weld Load [W2]:

= $(A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$



$$= (785.9383 + 0.0 + 100.0 + (316.16)) \cdot 138$$

$$= 165872.47 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t_{fr1})) \cdot S$$

$$= (785.9383 + 0.0 + 100.0 + 0.0 + (316.16)) \cdot 138$$

$$= 165872.47 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 89638. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 159591. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 140789. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (89638 + 159591) = 249230 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (89638 + 0 + 140789 + 0) = 230427 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (89638 + 140789 + 0) = 230427 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 249229 N	, must exceed W = 42772 N	or W1 = 122246 N
Path 2-2 = 230427 N	, must exceed W = 42772 N	or W2 = 165872 N
Path 3-3 = 230427 N	, must exceed W = 42772 N	or W3 = 165872 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.3957 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.7708 mm

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Input, Nozzle Desc: K2A-2"

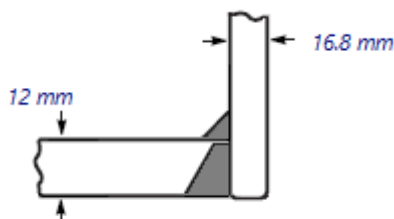
From: 20

Pressure for Reinforcement Calculations	P	0.3991	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	1775.0001	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		550.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2A-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.4 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
= 6.5224 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{0.4}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0782 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4258 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	108.0000	mm
Parallel to Vessel Wall, opening length	d	54.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

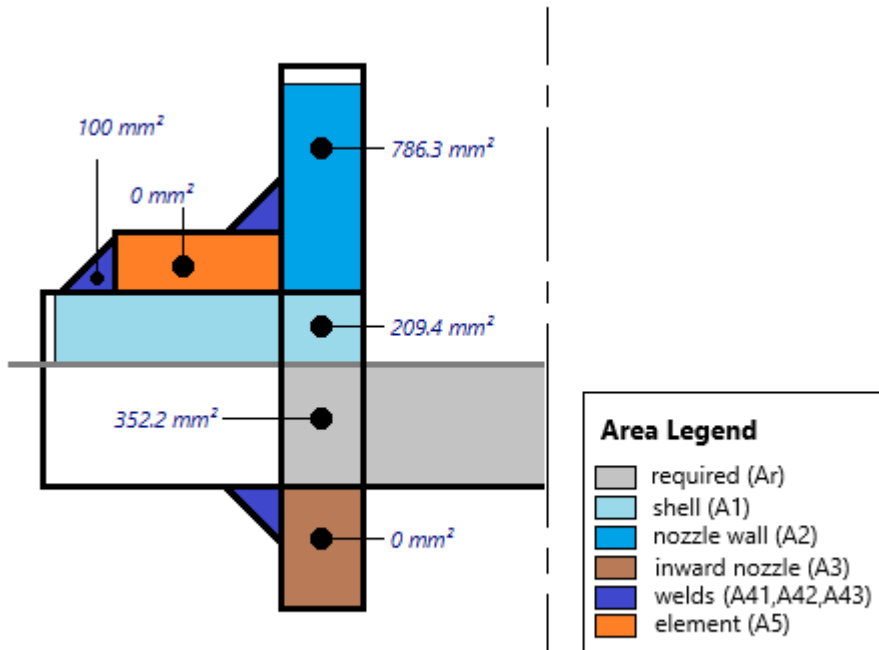


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	352.212	202.427	NA
Area in Shell	A1	209.388	156.745	NA
Area in Nozzle Wall	A2	786.334	768.260	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1095.722	1025.006	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (54.0 \cdot 6.5224 \cdot 1.0 + 2 \cdot 15.2 \cdot 6.5224 \cdot 1.0 \cdot (1 - 1.0))$
 $= 352.212 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 54.0(1.0 \cdot 10.4 - 1.0 \cdot 6.522) - 2 \cdot 15.2$
 $(1.0 \cdot 10.4 - 1.0 \cdot 6.5224) \cdot (1 - 1.0)$
 $= 209.388 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 26.0)(15.2 - 0.08)1.0$
 $= 786.334 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$



= 100.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0258$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.1224$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.1224$ mm
 Wall Thickness, shell/head, external pressure $trb_2 = 2.4440$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000$ mm
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$
 = $\min[6.4, \max(8.1224, 4.1)]$
 = 6.4000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$
 = $\max(2.0258, 6.4)$
 = 6.4000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 12.0$, $t_r = 6.522$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.627$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $0.40/5.10 = 0.078$

Weld Size Calculations, Description: K2A-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 10.4000$ mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= $\max(0, (A-A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 = $\max(0, (352.2119 - 209.3881 + 2 \cdot 15.2 \cdot 1.0 \cdot (1.0 \cdot 10.4 - 6.5224)) \cdot 138)$
 = 35973.10 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= $(A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$
 = $(786.3339 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 = 122301.48 N

Weld Load [W2]:

= $(A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$



$$= (786.3339 + 0.0 + 100.0 + (316.16)) \cdot 138$$
$$= 165927.06 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$
$$= (786.3339 + 0.0 + 100.0 + 0.0 + (316.16)) \cdot 138$$
$$= 165927.06 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$
$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$
$$= 89638. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$
$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$
$$= 159591. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$
$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$
$$= 140789. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (89638 + 159591) = 249230 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (89638 + 0 + 140789 + 0) = 230427 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (89638 + 140789 + 0) = 230427 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 249229 N	, must exceed W = 35973 N	or W1 = 122301 N
Path 2-2 = 230427 N	, must exceed W = 35973 N	or W2 = 165927 N
Path 3-3 = 230427 N	, must exceed W = 35973 N	or W3 = 165927 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.3957 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 490.0438 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 54.00 mm

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Input, Nozzle Desc: N3-4"

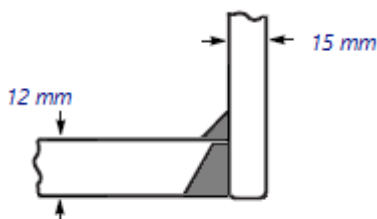
From: 20

Pressure for Reinforcement Calculations	P	0.3991	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	1775.0001	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1050.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	414.1480	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N3-4"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.500 in.
 Actual Thickness Used in Calculation 0.590 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.4 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
 $= 6.5224 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.4 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.4)$
 $= 0.1930 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4751 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	175.0814	mm
Parallel to Vessel Wall, opening length	d	87.5407	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000	mm

Weld Strength Reduction Factor [fr1]:

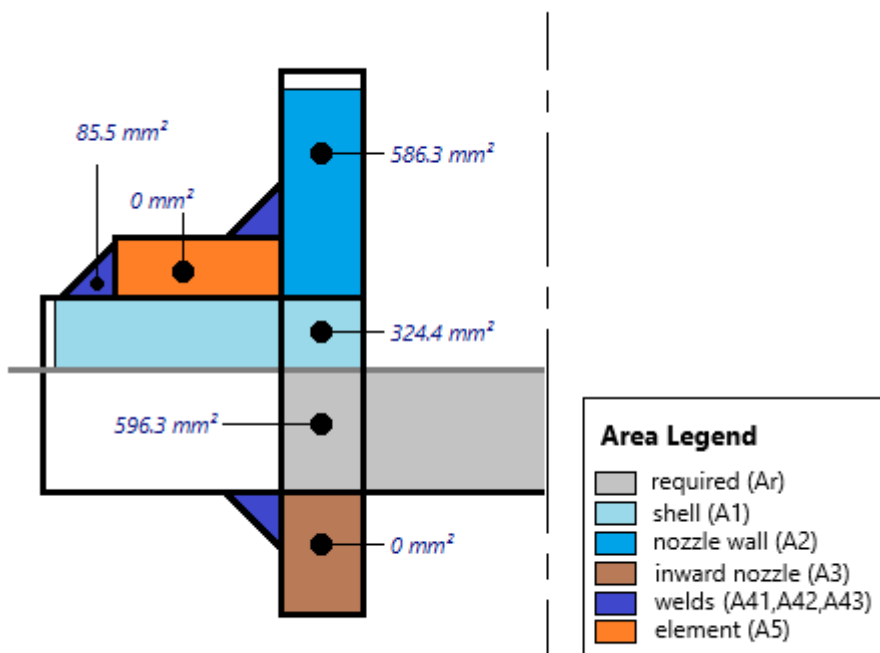
$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

**Figure UG 37.1 Nozzle Areas**

Results of Nozzle Reinforcement Area Calculations: (mm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	596.271	342.696	NA
Area in Shell	A1	324.408	242.848	NA
Area in Nozzle Wall	A2	586.338	573.795	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.509	85.509	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	996.255	902.151	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (87.5407 \cdot 6.5224 \cdot 1.0 + 2 \cdot 13.3797 \cdot 6.5224 \cdot 1.0 \cdot (1 - 0.86))$
 $= 596.271 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 87.541(1.0 \cdot 10.4 - 1.0 \cdot 6.522) - 2 \cdot 13.38$
 $(1.0 \cdot 10.4 - 1.0 \cdot 6.522) \cdot (1 - 0.855)$
 $= 324.408 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 26.0)(13.38 - 0.19)0.8551$
 $= 586.338 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 85.509 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.0751 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.1000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 8.1224 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 8.1224 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.4440 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.1000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.858, \max(8.1224, 4.1)]$
 $= 6.8578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.0751, 6.8578)$
 $= 6.8578 \text{ mm}$

Available Nozzle Neck Thickness = 14.9797 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 22.9,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 272.1 MPa	Passed
Occasional	: 0.6,	Allowable	: 156.9 MPa	Passed
Shear	: 16.3,	Allowable	: 82.6 MPa	Passed



The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 14.98$, $t_r = 0.193$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.014$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 6.522$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.627$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.40/5.10 = 0.078$

Weld Size Calculations, Description: N3-4"

Intermediate Calc. for nozzle/shell Welds T_{min} 10.4000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (596.2715 - 324.408 + 2 \cdot 13.3797 \cdot 0.8551 \cdot (1.0 \cdot 10.4 - 6.5224)) \cdot 138)$
 $= 49756.02 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (586.3383 + 0.0 + 85.5088 - 0.0 \cdot 0.86) \cdot 138$
 $= 92705.33 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (586.3383 + 0.0 + 85.5088 + (237.9682)) \cdot 138$
 $= 125541.55 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (586.3383 + 0.0 + 85.5088 + 0.0 + (237.9682)) \cdot 138$
 $= 125541.55 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\
 &= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118 \\
 &= 103802. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.1416 \cdot 50.4602) \cdot (14.9797 - 1.6) \cdot 0.7 \cdot 118 \\
 &= 175181. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 114.3 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138 \\
 &= 190663. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (103802 + 175181) = 278983 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (103802 + 0 + 190663 + 0) = 294465 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (103802 + 190663 + 0) = 294465 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 278983 N	, must exceed W = 49756 N	or W1 = 92705 N
Path 2-2 = 294464 N	, must exceed W = 49756 N	or W2 = 125541 N
Path 3-3 = 294464 N	, must exceed W = 49756 N	or W3 = 125541 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.7259 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.8740 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 87.54 mm

Input Echo, WRC297 Item 1, Description: N3-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4500.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	14.9797	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	



Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	0.40	MPa
Radial Load	P	7500.00	N
Circumferential Shear	Vc	5000.00	N
Longitudinal Shear	Vl	5000.00	N
Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.630) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2256.8(12.0 - 1.6))}$$

$$= 619.865 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4513.601 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	10.400 mm
Nozzle Thickness used	(t) =	13.380 mm
T / t	=	0.777
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.528$$

Nr/P	=	0.156
Mr/P	=	0.292
M0/P	=	0.095
N0/P	=	0.170
MrD/Mc	=	0.300
NrDL/Mc	=	0.115
M0d/Mc	=	0.109
N0DL/Mc	=	-0.067
MrD/Ml	=	0.261
NrDL/Ml	=	0.115
M0D/Ml	=	0.094
N0DL/Ml	=	0.019

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-39	39	-39	39
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-2	-2	2	2
Outplane Bending (Ml)	-63	63	63	-63
Normal Pressure Stress	86	86	86	86



Outplane Stress Summary -31 176 101 52

Vessel Stresses (MPa):

LONGITUDINAL PLANE
(Stresses parallel to
longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	-10	-10	-10	-10
Inplane Bending (P)	-121	121	-121	121
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-13	-13	13	13
Inplane Bending (ML)	-177	177	177	-177
Inplane Pressure Stress	43	43	43	43

Inplane Stress Summary -279 318 100 -10

Vessel Stresses (MPa):

LONGITUDINAL PLANE
(Shear stress normal to
longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	9	9	9	9

Shear Stress Summary 12 12 6 6

Vessel Stresses (MPa):

LONGITUDINAL PLANE
(Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 280 319 107 64

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses Normal to
circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-39	39	-39	39
Outplane Membrane (Mc)	7	7	-7	-7
Outplane Bending (Mc)	-74	74	74	-74
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43

Outplane Stress Summary -74 152 58 -11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses parallel to
circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	-10	-10	-10	-10
Inplane Bending (P)	-121	121	-121	121
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-203	203	203	-203
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	86	86	86	86

Inplane Stress Summary -263 388 170 6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Shear stress normal to
circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	9	9	9	9

Shear Stress Summary	6	6	12	12
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	263	388	171	29
------------------------	-----	-----	-----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1

Hoop Stress Summary	-12	-12	-8	-7
---------------------	-----	-----	----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-53	53	-53	53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-83	83	83	-83
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-152	121	42	-16
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	153	134	52	18
------------------------	-----	-----	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	7	7	-7	-7
Hoop Bending (Mc)	0	0	0	0



Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1

Hoop Stress Summary	-2	-2	-18	-17
---------------------	----	----	-----	-----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-53	53	-53	53
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-99	99	99	-99
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-168	137	59	-32
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary	5	5	9	9
----------------------	---	---	---	---

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	169	140	79	36
------------------------	-----	-----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	86	86	86	86	86	86	86	86
Circ. Pl (SUS)	-13	-13	-9	-9	-23	-23	3	3
Circ. Q (SUS)	-102	102	24	-24	-324	324	82	-82
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-23	-23	3	3	-4	-4	-18	-18
Long. Q (SUS)	-298	298	56	-56	-113	113	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	86.0	86.0	86.0	86.0	86.0	86.0	86.0	86.0
Pm+Pl (SUS)	73.1	73.1	77.1	77.1	63.2	63.2	89.1	89.1
Pm+Pl+Q (Total)	278.5	318.8	108.5	64.5	261.3	387.2	172.1	27.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	86.00	138.00	Passed



Pm+Pl (SUS)	89.06	207.00	Passed
Pm+Pl+Q (TOTAL)	387.21	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	1	1	1	1	1	1	1
Circ. Pl (SUS)	-13	-13	-9	-9	-4	-4	-18	-18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-15	-15	13	13	-15	-15	13	13
Long. Q (SUS)	-136	136	30	-30	-152	152	46	-46
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm+Pl (SUS)	16.0	16.0	21.4	21.4	15.3	15.3	30.3	30.3
Pm+Pl+Q (Total)	151.6	134.2	52.0	19.2	167.2	140.4	78.1	37.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1.00	118.00	Passed
Pm+Pl (SUS)	30.27	177.00	Passed
Pm+Pl+Q (TOTAL)	167.15	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N4-4"

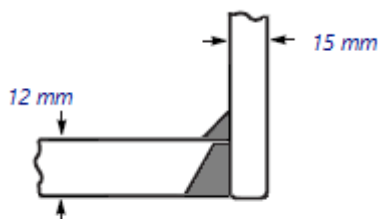
From: 20

Pressure for Reinforcement Calculations	P	0.3991	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	2799.9995	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2550.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	414.1480	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N4-4"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.500 in.
Actual Thickness Used in Calculation 0.590 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.4 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
= 6.5224 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.4 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.4)$
= 0.1930 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4751 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	175.0814	mm
Parallel to Vessel Wall, opening length	d	87.5407	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000	mm

Weld Strength Reduction Factor [fr1]:

= min(1, S_n / S_v)
= min(1, 118.0/138.0)
= 0.855

Weld Strength Reduction Factor [fr2]:

= min(1, S_n / S_v)
= min(1, 118.0/138.0)
= 0.855

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(0.855, 1.0)
= 0.855

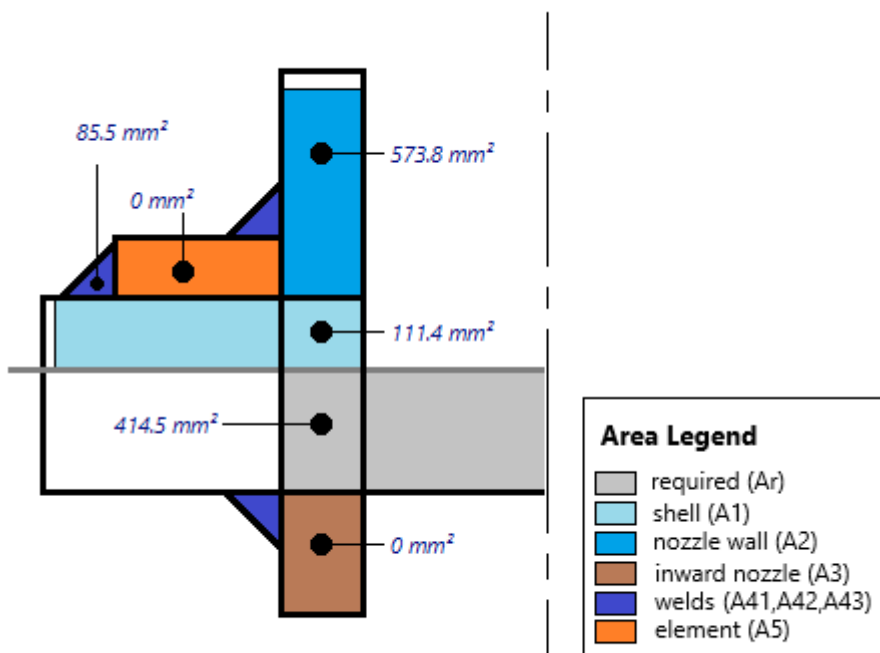


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	596.271	414.530	NA
Area in Shell	A1	324.408	111.368	NA
Area in Nozzle Wall	A2	586.338	573.795	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.509	85.509	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	996.255	770.672	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(87.5407 \cdot 9.0688 \cdot 1 + 2 \cdot 13.3797 \cdot 9.0688 \cdot 1(1-0.86))$
 $= 414.530 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 87.541(1.0 \cdot 10.4 - 1.0 \cdot 9.069) - 2 \cdot 13.38$
 $(1.0 \cdot 10.4 - 1.0 \cdot 9.0688) \cdot (1 - 0.855)$
 $= 111.368 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 26.0)(13.38 - 0.48) 0.8551$
 $= 573.795 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 85.509 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.0751 \text{ mm}$
Wall Thickness per UG16(b), $tr16b = 4.1000 \text{ mm}$
Wall Thickness, shell/head, internal pressure $trb1 = 8.1224 \text{ mm}$
Wall Thickness $tb1 = \max(trb1, tr16b) = 8.1224 \text{ mm}$
Wall Thickness, shell/head, external pressure $trb2 = 2.4440 \text{ mm}$
Wall Thickness $tb2 = \max(trb2, tr16b) = 4.1000 \text{ mm}$
Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.858, \max(8.1224, 4.1)]$
 $= 6.8578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.0751, 6.8578)$
 $= 6.8578 \text{ mm}$

Available Nozzle Neck Thickness = 14.9797 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	22.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	272.1 MPa	Passed
Occasional	:	0.6,	Allowable	:	156.9 MPa	Passed
Shear	:	16.3,	Allowable	:	82.6 MPa	Passed



The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 14.98, tr = 0.193, c = 1.6 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.014$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 6.522, c = 1.6 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.627$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.40/5.10 = 0.078$

Weld Size Calculations, Description: N4-4"

Intermediate Calc. for nozzle/shell Welds Tmin 10.4000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (414.5298 - 111.3683 + 2 \cdot 13.3797 \cdot 0.8551 \cdot (1.0 \cdot 10.4 - 9.0688)) \cdot 138)$
 $= 46034.88 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (573.7946 + 0.0 + 85.5088 - 0.0 \cdot 0.86) \cdot 138$
 $= 90974.48 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (573.7946 + 0.0 + 85.5088 + (237.9682)) \cdot 138$
 $= 123810.70 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (573.7946 + 0.0 + 85.5088 + 0.0 + (237.9682)) \cdot 138$
 $= 123810.70 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 103802. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 50.4602) \cdot (14.9797 - 1.6) \cdot 0.7 \cdot 118$$

$$= 175181. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 190663. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (103802 + 175181) = 278983 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (103802 + 0 + 190663 + 0) = 294465 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (103802 + 190663 + 0) = 294465 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 278983 N	, must exceed W = 46034 N	or W1 = 90974 N
Path 2-2 = 294464 N	, must exceed W = 46034 N	or W2 = 123810 N
Path 3-3 = 294464 N	, must exceed W = 46034 N	or W3 = 123810 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.7259 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.8740 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 87.54 mm

Input Echo, WRC297 Item 1, Description: N4-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4500.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	14.9797	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	



Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	0.40	MPa
Radial Load	P	7500.00	N
Circumferential Shear	Vc	5000.00	N
Longitudinal Shear	Vl	5000.00	N
Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.630) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2256.8(12.0 - 1.6))}$$

$$= 619.865 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4513.601 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	10.400 mm
Nozzle Thickness used	(t) =	13.380 mm
T / t	=	0.777
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.528$$

Nr/P	=	0.156
Mr/P	=	0.292
M0/P	=	0.095
N0/P	=	0.170
MrD/Mc	=	0.300
NrDL/Mc	=	0.115
M0d/Mc	=	0.109
N0DL/Mc	=	-0.067
MrD/Ml	=	0.261
NrDL/Ml	=	0.115
M0D/Ml	=	0.094
N0DL/Ml	=	0.019

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-39	39	-39	39
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-63	63	63	-63
Normal Pressure Stress	86	86	86	86



Outplane Stress Summary -31 176 101 52

Vessel Stresses (MPa):

LONGITUDINAL PLANE Au Al Bu Bl
(Stresses parallel to Top Top Bottom Bottom
longitudinal plane) Outside Inside Outside Inside

Inplane Membrane (P)	-10	-10	-10	-10
Inplane Bending (P)	-121	121	-121	121
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-13	-13	13	13
Inplane Bending (ML)	-177	177	177	-177
Inplane Pressure Stress	43	43	43	43

Inplane Stress Summary -279 318 100 -10

Vessel Stresses (MPa):

LONGITUDINAL PLANE Au Al Bu Bl
(Shear stress normal to Top Top Bottom Bottom
longitudinal plane) Outside Inside Outside Inside

Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	9	9	9	9

Shear Stress Summary 12 12 6 6

Vessel Stresses (MPa):

LONGITUDINAL PLANE Au Al Bu Bl
(Stress Intensities Top Top Bottom Bottom
Outside Inside Outside Inside

Two * Max Shear Stress 280 319 107 64

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
(Stresses Normal to Left Left Right Right
circumferential plane) Outside Inside Outside Inside

Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-39	39	-39	39
Outplane Membrane (Mc)	7	7	-7	-7
Outplane Bending (Mc)	-74	74	74	-74
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43

Outplane Stress Summary -74 152 58 -11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
(Stresses parallel to Left Left Right Right
circumferential plane) Outside Inside Outside Inside

Inplane Membrane (P)	-10	-10	-10	-10
Inplane Bending (P)	-121	121	-121	121
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-203	203	203	-203
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	86	86	86	86

Inplane Stress Summary -263 388 170 6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
(Shear stress normal to Left Left Right Right
circumferential plane) Outside Inside Outside Inside



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	9	9	9	9

Shear Stress Summary	6	6	12	12
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	263	388	171	29
------------------------	-----	-----	-----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1

Hoop Stress Summary	-12	-12	-8	-7
---------------------	-----	-----	----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-53	53	-53	53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-83	83	83	-83
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-152	121	42	-16
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	153	134	52	18
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	7	7	-7	-7
Hoop Bending (Mc)	0	0	0	0



Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1

Hoop Stress Summary	-2	-2	-18	-17
---------------------	----	----	-----	-----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-53	53	-53	53
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-99	99	99	-99
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-168	137	59	-32
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary	5	5	9	9
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	169	140	79	36
------------------------	-----	-----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	86	86	86	86	86	86	86	86
Circ. Pl (SUS)	-13	-13	-9	-9	-23	-23	3	3
Circ. Q (SUS)	-102	102	24	-24	-324	324	82	-82
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-23	-23	3	3	-4	-4	-18	-18
Long. Q (SUS)	-298	298	56	-56	-113	113	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	86.0	86.0	86.0	86.0	86.0	86.0	86.0	86.0
Pm+Pl (SUS)	73.1	73.1	77.1	77.1	63.2	63.2	89.1	89.1
Pm+Pl+Q (Total)	278.5	318.8	108.5	64.5	261.3	387.2	172.1	27.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	86.00	138.00	Passed



Pm+Pl (SUS)	89.06	207.00	Passed
Pm+Pl+Q (TOTAL)	387.21	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	1	1	1	1	1	1	1
Circ. Pl (SUS)	-13	-13	-9	-9	-4	-4	-18	-18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-15	-15	13	13	-15	-15	13	13
Long. Q (SUS)	-136	136	30	-30	-152	152	46	-46
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm+Pl (SUS)	16.0	16.0	21.4	21.4	15.3	15.3	30.3	30.3
Pm+Pl+Q (Total)	151.6	134.2	52.0	19.2	167.2	140.4	78.1	37.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1.00	118.00	Passed
Pm+Pl (SUS)	30.27	177.00	Passed
Pm+Pl+Q (TOTAL)	167.15	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K3-2"

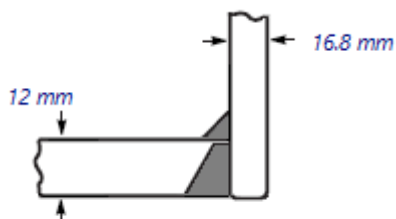
From: 30

Pressure for Reinforcement Calculations	P	0.3991	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	2799.9995	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3450.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.4 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
= 6.5224 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{0.4}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0782 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4258 mm

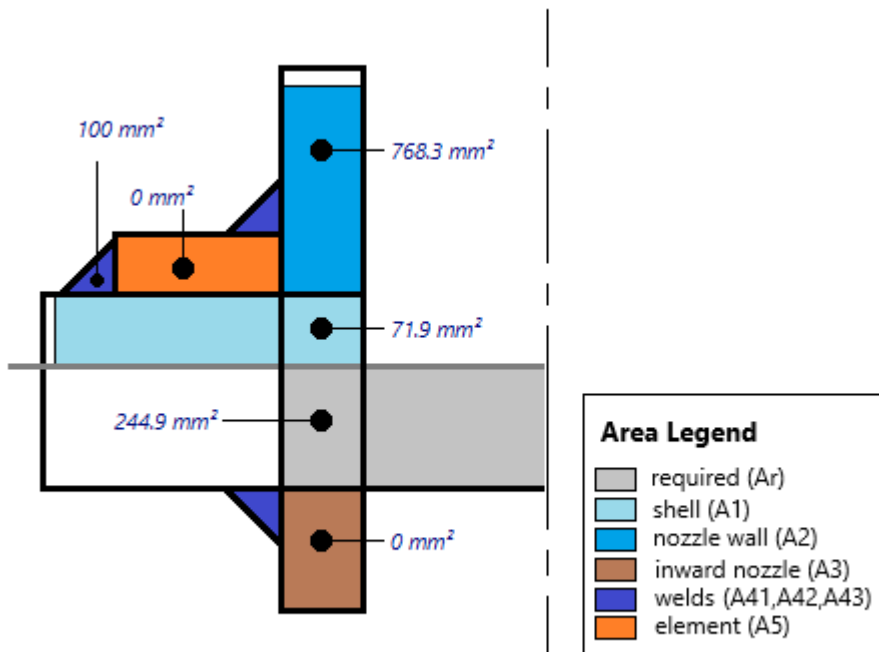
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	108.0000	mm
Parallel to Vessel Wall, opening length	d	54.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

**Figure UG 37.1 Nozzle Areas****Results of Nozzle Reinforcement Area Calculations: (mm²)**

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	352.212	244.859	NA
Area in Shell	A1	209.388	71.882	NA
Area in Nozzle Wall	A2	786.334	768.260	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1095.722	940.143	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(54.0 \cdot 9.0688 \cdot 1 + 2 \cdot 15.2 \cdot 9.0688 \cdot 1(1 - 1.0))$$

$$= 244.859 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 54.0(1.0 \cdot 10.4 - 1.0 \cdot 9.069) - 2 \cdot 15.2$$

$$(1.0 \cdot 10.4 - 1.0 \cdot 9.0688) \cdot (1 - 1.0)$$

$$= 71.882 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 26.0)(15.2 - 0.43)1.0$$

$$= 768.260 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



= 100.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0258$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.1224$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.1224$ mm
 Wall Thickness, shell/head, external pressure $trb_2 = 2.4440$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000$ mm
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$
 = $\min[6.4, \max(8.1224, 4.1)]$
 = 6.4000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$
 = $\max(2.0258, 6.4)$
 = 6.4000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 12.0$, $t_r = 6.522$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.627$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $0.40/5.10 = 0.078$

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 10.4000$ mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= $\max(0, (A-A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 = $\max(0, (244.8588 - 71.8823 + 2 \cdot 15.2 \cdot 1.0 \cdot (1.0 \cdot 10.4 - 9.0688)) \cdot 138)$
 = 29452.18 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= $(A_2 + A_5 + A_4 - (W_i - Can / 7.07)^2 \cdot fr_2) \cdot S_v$
 = $(768.2603 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 = 119807.55 N

Weld Load [W2]:

= $(A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$



$$= (768.2603 + 0.0 + 100.0 + (316.16)) \cdot 138$$

$$= 163433.12 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t_{fr1})) \cdot S$$

$$= (768.2603 + 0.0 + 100.0 + 0.0 + (316.16)) \cdot 138$$

$$= 163433.12 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 89638. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 159591. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 140789. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (89638 + 159591) = 249230 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (89638 + 0 + 140789 + 0) = 230427 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (89638 + 140789 + 0) = 230427 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 249229 N	, must exceed W = 29452 N	or W1 = 119807 N
Path 2-2 = 230427 N	, must exceed W = 29452 N	or W2 = 163433 N
Path 3-3 = 230427 N	, must exceed W = 29452 N	or W3 = 163433 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.3957 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 490.0438 mm

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Input, Nozzle Desc: K4-2"

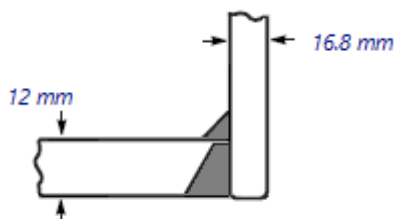
From: 30

Pressure for Reinforcement Calculations	P	0.3991	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	2799.9995	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4050.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K4-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.4 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
= 6.5224 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{0.4}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0782 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4258 mm

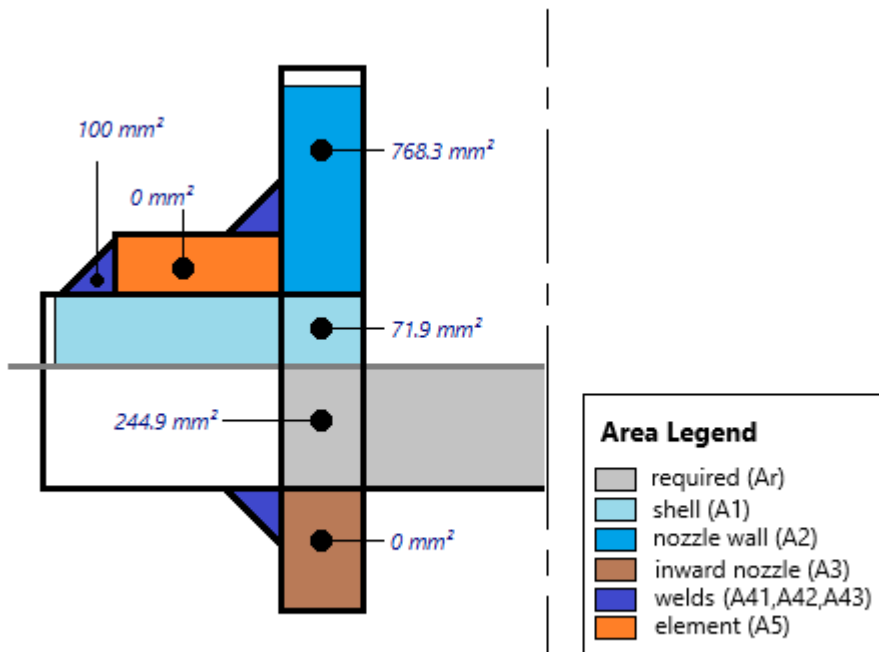
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	108.0000	mm
Parallel to Vessel Wall, opening length	d	54.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

**Figure UG 37.1 Nozzle Areas****Results of Nozzle Reinforcement Area Calculations: (mm²)**

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	352.212	244.859	NA
Area in Shell	A1	209.388	71.882	NA
Area in Nozzle Wall	A2	786.334	768.260	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1095.722	940.143	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(54.0 \cdot 9.0688 \cdot 1 + 2 \cdot 15.2 \cdot 9.0688 \cdot 1(1 - 1.0))$$

$$= 244.859 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 54.0(1.0 \cdot 10.4 - 1.0 \cdot 9.069) - 2 \cdot 15.2$$

$$(1.0 \cdot 10.4 - 1.0 \cdot 9.0688) \cdot (1 - 1.0)$$

$$= 71.882 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 26.0)(15.2 - 0.43)1.0$$

$$= 768.260 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



= 100.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0258$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.1224$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.1224$ mm
 Wall Thickness, shell/head, external pressure $trb_2 = 2.4440$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000$ mm
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$
 = $\min[6.4, \max(8.1224, 4.1)]$
 = 6.4000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$
 = $\max(2.0258, 6.4)$
 = 6.4000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 12.0$, $t_r = 6.522$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.627$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $0.40/5.10 = 0.078$

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 10.4000$ mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= $\max(0, (A-A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 = $\max(0, (244.8588 - 71.8823 + 2 \cdot 15.2 \cdot 1.0 \cdot (1.0 \cdot 10.4 - 9.0688)) \cdot 138)$
 = 29452.18 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= $(A_2 + A_5 + A_4 - (W_i - Can / 7.07)^2 \cdot fr_2) \cdot S_v$
 = $(768.2603 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 = 119807.55 N

Weld Load [W2]:

= $(A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$



$$= (768.2603 + 0.0 + 100.0 + (316.16)) \cdot 138$$

$$= 163433.12 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t_{fr1})) \cdot S$$

$$= (768.2603 + 0.0 + 100.0 + 0.0 + (316.16)) \cdot 138$$

$$= 163433.12 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 89638. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 159591. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 140789. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (89638 + 159591) = 249230 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (89638 + 0 + 140789 + 0) = 230427 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (89638 + 140789 + 0) = 230427 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 249229 N	, must exceed W = 29452 N	or W1 = 119807 N
Path 2-2 = 230427 N	, must exceed W = 29452 N	or W2 = 163433 N
Path 3-3 = 230427 N	, must exceed W = 29452 N	or W3 = 163433 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.3957 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 490.0438 mm

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Input, Nozzle Desc: N1-24

From: 40

Pressure for Reinforcement Calculations	P	0.4378	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	2849.9998	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	1000.0000	mm
Distance from Bottom/Left Tangent		7550.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

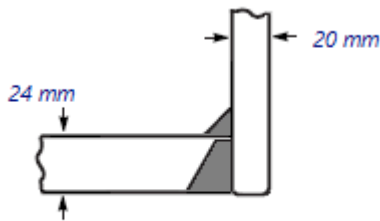
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	750.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1-24**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.44 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.44)$
 $= 7.1575 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.44 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.44)$
 $= 0.9658 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6110 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1000.0000 mm
Parallel to Vessel Wall, opening length	d	500.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

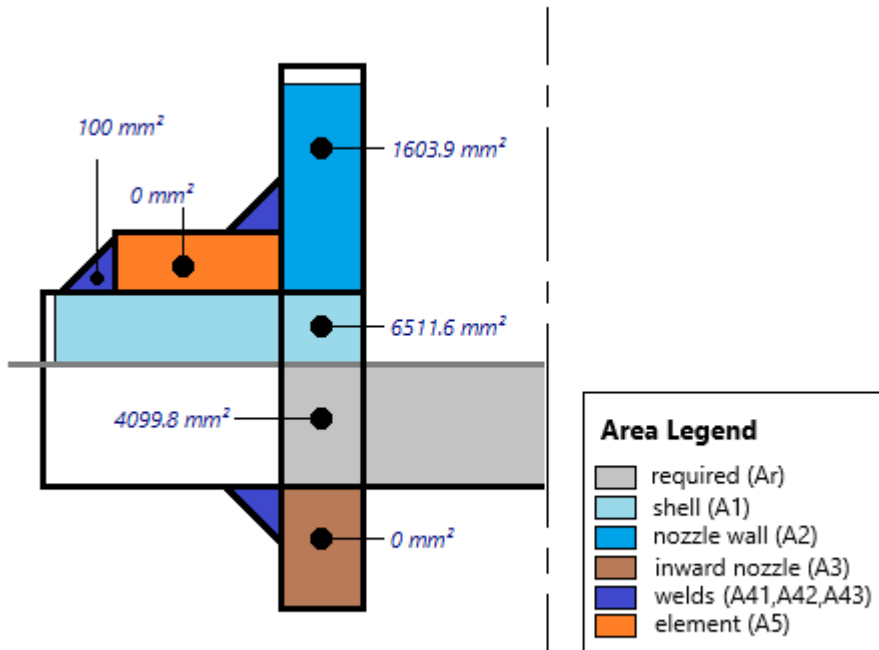


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	4099.797	2614.523	NA
Area in Shell	A1	6511.609	5669.405	NA
Area in Nozzle Wall	A2	1603.944	1544.586	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	8215.554	7313.992	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (572.8 \cdot 7.1575 \cdot 1.0 + 2 \cdot 18.4 \cdot 7.1575 \cdot 1.0 \cdot (1 - 1.0))$
 $= 4099.797 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 427.2(1.0 \cdot 22.4 - 1.0 \cdot 7.157) - 2 \cdot 18.4$
 $(1.0 \cdot 22.4 - 1.0 \cdot 7.1575) \cdot (1 - 1.0)$
 $= 6511.609 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 46.0)(18.4 - 0.97)1.0$
 $= 1603.944 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$



= 100.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.2110$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.7575$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.7575$ mm
 Wall Thickness, shell/head, external pressure $trb_2 = 2.4440$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000$ mm
 Wall Thickness per table UG-45 $tb_3 = 9.9312$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$
 = $\min[9.931, \max(8.7575, 4.1)]$
 = 8.7575 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$
 = $\max(3.211, 8.7575)$
 = 8.7575 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	16.5,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	328.5 MPa	Passed
Occasional	:	3.3,	Allowable	:	183.5 MPa	Passed
Shear	:	10.2,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 20.0$, $tr = 0.966$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:D, Curve:B)

Govrn. thk, $t_g = 20.0$, $tr = 0.966$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.44/1.96 = 0.223$

Weld Size Calculations, Description: N1-24

Intermediate Calc. for nozzle/shell Welds	T_{min}	18.4000 mm
---	-----------	------------

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (4099.7974 - 6511.6089 + 2 \cdot 18.4 \cdot 1.0 \cdot (1.0 \cdot 22.4 - 7.1575)) \cdot 138)$$

$$= \max(0, -255395.98) \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (1603.944 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 235120.03 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (1603.944 + 0.0 + 100.0 + (824.3199)) \cdot 138$$

$$= 348864.47 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (1603.944 + 0.0 + 100.0 + 0.0 + (824.3199)) \cdot 138$$

$$= 348864.47 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 647434. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 295.6) \cdot (20.0 - 1.6) \cdot 0.7 \cdot 138$$

$$= 1650488. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{g_{nvi}} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 1173329. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (647434 + 1650488) = 2297921 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (647434 + 0 + 1173329 + 0) = 1820763 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (647434 + 1173329 + 0) = 1820763 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2297921 N	, must exceed W = 0 N	or W1 = 235120 N
Path 2-2 = 1820762 N	, must exceed W = 0 N	or W2 = 348864 N
Path 3-3 = 1820762 N	, must exceed W = 0 N	or W3 = 348864 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 20.7408 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 794.7408 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.392 %

Input Echo, WRC297 Item 1, Description: N1-24 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.6000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	4500.000	mm	
Shell Thickness	24.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	1.6000	mm	
Nozzle Diameter	609.600	mm	
Nozzle Thickness	20.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-516 70		
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	0.44	MPa
Radial Load	P	-44999.97	N
Circumferential Shear	Vc	29988.50	N
Longitudinal Shear	Vl	29988.50	N
Circumferential Moment	Mc	0.504E+08	N-mm
Longitudinal Moment	Ml	0.504E+08	N-mm
Torsional Moment	Mt	0.720E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{mean}(t - ca))}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{(2262.8(24.0 - 1.6))}$$

$$= 1352.552 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4525.601 mm
Nozzle Outside Diameter	(d) =	609.600 mm
Vessel Thickness used	(T) =	22.400 mm
Nozzle Thickness used	(t) =	18.400 mm
T / t	=	1.217
d / t	=	33.130

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.915$$



Nr/P = 0.057
 Mr/P = 0.124
 M0/P = 0.046
 N0/P = 0.130
 MrD/Mc = 0.242
 NrDL/Mc = 0.087
 M0d/Mc = 0.100
 N0DL/Mc = 0.135
 MrD/Ml = 0.124
 NrDL/Ml = 0.056
 M0D/Ml = 0.046
 N0DL/Ml = 0.209

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	24	-24	24	-24
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-34	-34	34	34
Outplane Bending (ML)	-45	45	45	-45
Normal Pressure Stress	43	44	43	44
Outplane Stress Summary	0	42	159	20

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	66	-66	66	-66
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-122	122	122	-122
Inplane Pressure Stress	21	21	21	21
Inplane Stress Summary	-37	72	225	-152

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	4	4

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	40	74	225	173

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	24	-24	24	-24



Outplane Membrane (Mc)	-22	-22	22	22
Outplane Bending (Mc)	-98	98	98	-98
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21

Outplane Stress Summary	-62	84	178	-67
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	66	-66	66	-66
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-238	238	238	-238
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	43	44	43	44

Inplane Stress Summary	-137	206	368	-242
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	137	206	369	242
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-34	-34	34	34
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8

Hoop Stress Summary	-14	-14	54	54
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	76	-76	76	-76
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-139	139	139	-139
Axial Pressure Stress	3	3	3	3



Axial Stress Summary -68 58 231 -200

Nozzle Stresses (MPa):

LONGITUDINAL PLANE Au Al Bu Bl
(Shear stress) Outside Inside Outside Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary 8 8 4 4

Nozzle Stresses (MPa):

LONGITUDINAL PLANE Au Al Bu Bl
(Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 69 74 232 255

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
(Stresses in the Left Left Right Right
hoop direction) Outside Inside Outside Inside

Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-22	-22	22	22
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8

Hoop Stress Summary -2 -2 41 42

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
(Stresses Normal to Left Left Right Right
pipe cross-section) Outside Inside Outside Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	76	-76	76	-76
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-290	290	290	-290
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary -218 208 382 -351

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
(Shear stress) Outside Inside Outside Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary 4 4 8 8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
(Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 218 210 382 393

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of

Stress Values at



Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	43	44	43	44	43	44	43	44
Circ. Pl (SUS)	-23	-23	45	45	-9	-9	19	19
Circ. Q (SUS)	-21	21	69	-69	-172	172	304	-304
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-4	-4	14	14	-11	-11	33	33
Long. Q (SUS)	-56	56	188	-188	-74	74	122	-122
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	43.0	44.0	43.0	44.0	43.0	44.0	43.0	44.0
Pm+Pl (SUS)	20.3	21.2	88.0	89.0	34.0	35.0	62.1	63.1
Pm+Pl+Q (Total)	39.9	74.1	223.2	173.2	138.2	207.1	366.2	241.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	44.00	138.00	Passed
Pm+Pl (SUS)	89.02	207.00	Passed
Pm+Pl+Q (TOTAL)	366.19	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	8	7	8	7	8	7	8
Circ. Pl (SUS)	-23	-23	45	45	-11	-11	33	33
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-63	63	215	-215	-214	214	366	-366
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	7.0	8.0	7.0	8.0	7.0	8.0	7.0	8.0
Pm+Pl (SUS)	16.1	15.1	52.0	53.0	6.4	6.3	40.0	41.0
Pm+Pl+Q (Total)	69.9	73.3	229.1	254.2	220.1	211.2	380.1	393.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	53.03	207.00	Passed
Pm+Pl+Q (TOTAL)	393.25	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

CCW Expansion Vessel (V8801)

PV Elite 28 Licensee: SL User

FileName : CCW Expansion Vessel (V8801)-7%

Nozzle Calcs.: N1-24



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Nozl: 36 3:01pm Sep 16,2026



HEXAGON

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**Input, Nozzle Desc: N2-8"****Parent: N1-24**

Pressure for Reinforcement Calculations	P	0.4378	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	569.60	mm
Design Length of Section	L	750.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm

User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

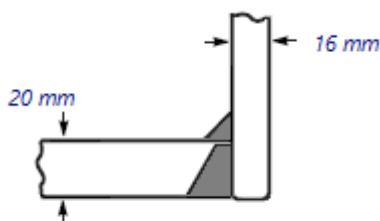
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N2-8"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 8.625 in.
Actual Thickness Used in Calculation 0.629 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.44 \cdot 286.4 / (138 \cdot 1.0 - 0.6 \cdot 0.44)$
= 0.9104 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.44 \cdot 109.5375 / (118 \cdot 1.0 + 0.4 \cdot 0.44)$
= 0.4058 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5132 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	380.6309	mm
Parallel to Vessel Wall, opening length	d	190.3154	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	35.9494	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855

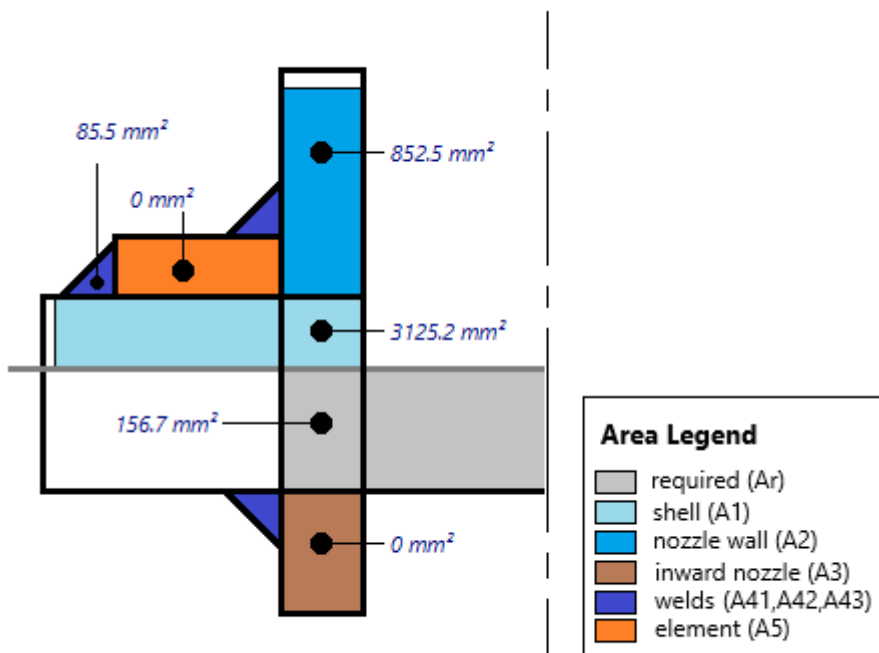


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
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Area Required	Ar	177.058	156.658	NA
Area in Shell	A1	3255.643	3125.225	NA
Area in Nozzle Wall	A2	859.100	852.502	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.507	85.507	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	4200.250	4063.234	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1)) \text{ per UG-37(d)}$$

$$= 0.5(190.3154 \cdot 1.611 \cdot 1 + 2 \cdot 14.3798 \cdot 1.611 \cdot 1(1-0.86))$$

$$= 156.658 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 190.315(1.0 \cdot 18.4 - 1.0 \cdot 1.611) - 2 \cdot 14.38$$

$$(1.0 \cdot 18.4 - 1.0 \cdot 1.611) \cdot (1 - 0.855)$$

$$= 3125.225 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn) fr2$$

$$= (2 \cdot 35.95)(14.38 - 0.51) 0.8551$$

$$= 852.502 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 85.507 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.1132 mm
Wall Thickness per UG16(b),	tr16b = 4.1000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 2.5104 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 4.1000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.1000 mm
Wall Thickness per table UG-45	tb3 = 8.7600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.76, \max(4.1, 4.1)]$$

$$= 4.1000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.1132, 4.1)$$

$$= 4.1000 \text{ mm}$$

Available Nozzle Neck Thickness = 15.9798 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17.5,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 277.5 MPa	Passed
Occasional	: 1.3,	Allowable	: 156.9 MPa	Passed
Shear	: 13.1,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, tg = 15.98, tr = 0.406, c = 1.6 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.028$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 15.98, tr = 0.406, c = 1.6 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.028$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.44/1.96 = 0.223$

Weld Size Calculations, Description: N2-8"

Intermediate Calc. for nozzle/shell Welds Tmin 14.3798 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (156.6581 - 3125.2249 + 2 \cdot 14.3798 \cdot 0.8551 \cdot (1.0 \cdot 18.4 - 1.611)) \cdot 138)$
 $= \max(0, -352656.72) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (852.5022 + 0.0 + 85.5072 - 0.0 \cdot 0.86) \cdot 138$
 $= 129434.30 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (852.5022 + 0.0 + 85.5072 + (452.4836)) \cdot 138$
 $= 191871.72 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (852.5022 + 0.0 + 85.5072 + 0.0 + (452.4836)) \cdot 138$
 $= 191871.72 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 198955. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 102.3476) \cdot (15.9798 - 1.6) \cdot 0.7 \cdot 118$$

$$= 381876. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 421665. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (198955 + 381876) = 580831 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198955 + 0 + 421665 + 0) = 620620 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (198955 + 421665 + 0) = 620620 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 580830 N , must exceed W = 0 N or W1 = 129434 N

Path 2-2 = 620619 N , must exceed W = 0 N or W2 = 191871 N

Path 3-3 = 620619 N , must exceed W = 0 N or W3 = 191871 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 21.9073 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.9073 mm

Input Echo, WRC297 Item 1, Description: N2-8" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	569.600	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm
Nozzle Diameter	219.075	mm
Nozzle Thickness	15.9798	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	0.44	MPa
Radial Load	P	-14999.99	N
Circumferential Shear	Vc	9999.95	N
Longitudinal Shear	Vl	9999.95	N
Circumferential Moment	Mc	0.560E+07	N-mm
Longitudinal Moment	Ml	0.560E+07	N-mm
Torsional Moment	Mt	0.800E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{295.6(20.0 - 1.6)}$$

$$= 462.449 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	591.200 mm
Nozzle Outside Diameter	(d) =	219.075 mm
Vessel Thickness used	(T) =	18.400 mm
Nozzle Thickness used	(t) =	14.380 mm
T / t	=	1.280
d / t	=	15.235

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.100$$

Nr/P	=	0.079
Mr/P	=	0.112
M0/P	=	0.044
N0/P	=	0.121
MrD/Mc	=	0.235
NrDL/Mc	=	0.131
M0d/Mc	=	0.104
N0DL/Mc	=	0.086
MrD/Ml	=	0.123
NrDL/Ml	=	0.077
M0D/Ml	=	0.057
N0DL/Ml	=	0.165

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-12	-12	12	12
Outplane Bending (Ml)	-25	25	25	-25
Normal Pressure Stress	6	7	6	7
Outplane Stress Summary	-14	14	61	-12

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	3	3	3	3

Inplane Stress Summary	-24	26	98	-72
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	7	7	4	4
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	28	30	98	73
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-47	47	47	-47
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	3	3	3	3

Outplane Stress Summary	-33	37	73	-43
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-106	106	106	-106
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	6	7	6	7

Inplane Stress Summary	-76	77	156	-115
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	7	7
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 76 77 157 116

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3
Hoop Stress Summary	-3	-3	21	21

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-12	-12	12	12
Axial Bending (ML)	-62	62	62	-62
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-40	21	110	-78

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7
Shear Stress Summary	9	9	5	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	42	31	110	100

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3
Hoop Stress Summary	2	2	15	15

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-125	125	125	-125
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-103	84	173	-141

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7
Shear Stress Summary	5	5	9	9

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	106	85	173	158

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	-7	-7	17	17	-6	-6	12	12
Circ. Q (SUS)	-14	14	36	-36	-77	77	135	-135
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-2	-2	8	8	-1	-1	11	11
Long. Q (SUS)	-26	26	84	-84	-36	36	58	-58
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	2.8	2.2	23.1	24.1	2.8	2.6	18.2	19.2
Pm+Pl+Q (Total)	27.8	29.3	95.4	73.3	77.4	78.4	153.4	116.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+Pl (SUS)	24.08	207.00	Passed
Pm+Pl+Q (TOTAL)	153.44	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	3	3	3	3	3	3	3
Circ. Pl (SUS)	-7	-7	17	17	-1	-1	11	11
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-11	-11	13	13	-11	-11	13	13
Long. Q (SUS)	-31	31	93	-93	-94	94	156	-156
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm+Pl (SUS)	10.6	10.6	20.6	20.6	12.6	12.6	16.0	16.0
Pm+Pl+Q (Total)	43.1	30.8	107.3	99.5	106.5	84.3	170.5	157.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	118.00	Passed
Pm+Pl (SUS)	20.61	177.00	Passed
Pm+Pl+Q (TOTAL)	170.52	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N5-4"

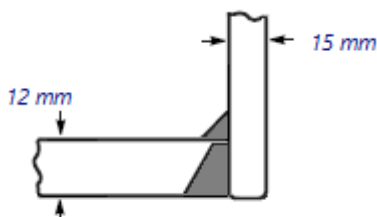
From: 40

Pressure for Reinforcement Calculations	P	0.3991	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	2849.9998	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7250.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	414.1480	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N5-4"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.500 in.
 Actual Thickness Used in Calculation 0.590 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.4 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
 $= 6.5224 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.4 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.4)$
 $= 0.1930 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4751 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	175.0814	mm
Parallel to Vessel Wall, opening length	d	87.5407	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000	mm

Weld Strength Reduction Factor [fr1]:

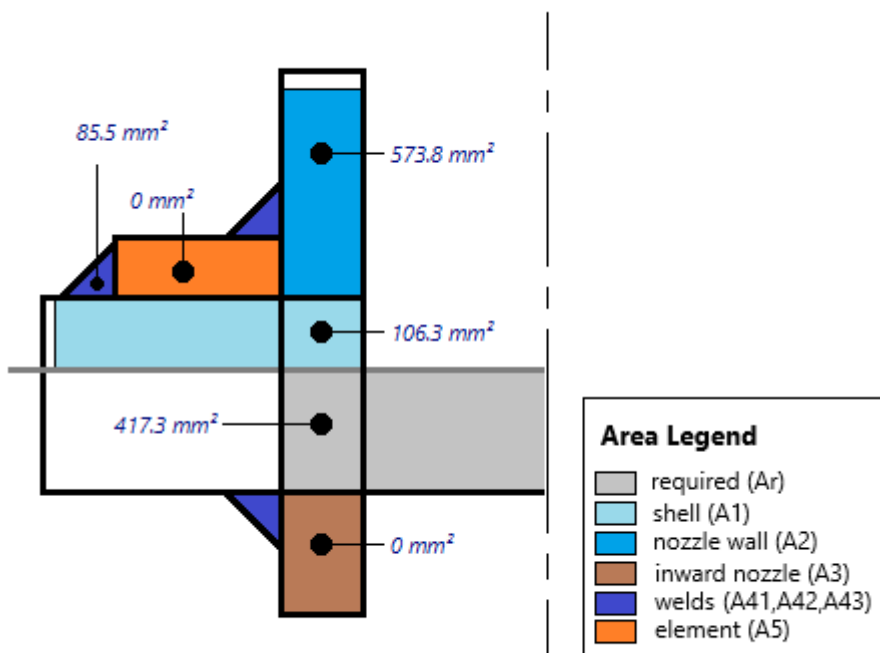
$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

**Figure UG 37.1 Nozzle Areas**

Results of Nozzle Reinforcement Area Calculations: (mm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	596.271	417.276	NA
Area in Shell	A1	324.408	106.342	NA
Area in Nozzle Wall	A2	586.338	573.795	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.509	85.509	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	996.255	765.646	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(87.5407 \cdot 9.1289 \cdot 1 + 2 \cdot 13.3797 \cdot 9.1289 \cdot 1(1-0.86))$
 $= 417.276 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 87.541(1.0 \cdot 10.4 - 1.0 \cdot 9.129) - 2 \cdot 13.38$
 $(1.0 \cdot 10.4 - 1.0 \cdot 9.1289) \cdot (1 - 0.855)$
 $= 106.342 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 26.0)(13.38 - 0.48) 0.8551$
 $= 573.795 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 85.509 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.0751 \text{ mm}$
Wall Thickness per UG16(b), $tr16b = 4.1000 \text{ mm}$
Wall Thickness, shell/head, internal pressure $trb1 = 8.1224 \text{ mm}$
Wall Thickness $tb1 = \max(trb1, tr16b) = 8.1224 \text{ mm}$
Wall Thickness, shell/head, external pressure $trb2 = 2.4440 \text{ mm}$
Wall Thickness $tb2 = \max(trb2, tr16b) = 4.1000 \text{ mm}$
Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.858, \max(8.1224, 4.1)]$
 $= 6.8578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.0751, 6.8578)$
 $= 6.8578 \text{ mm}$

Available Nozzle Neck Thickness = 14.9797 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	22.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	272.1 MPa	Passed
Occasional	:	0.6,	Allowable	:	156.9 MPa	Passed
Shear	:	16.3,	Allowable	:	82.6 MPa	Passed



The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 14.98, tr = 0.193, c = 1.6 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.014$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 6.522, c = 1.6 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.627$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.40/5.10 = 0.078$

Weld Size Calculations, Description: N5-4"

Intermediate Calc. for nozzle/shell Welds Tmin 10.4000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (417.2758 - 106.3423 + 2 \cdot 13.3797 \cdot 0.8551 \cdot (1.0 \cdot 10.4 - 9.1289)) \cdot 138)$$

$$= 46917.62 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (573.7946 + 0.0 + 85.5088 - 0.0 \cdot 0.86) \cdot 138$$

$$= 90974.48 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (573.7946 + 0.0 + 85.5088 + (237.9682)) \cdot 138$$

$$= 123810.70 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (573.7946 + 0.0 + 85.5088 + 0.0 + (237.9682)) \cdot 138$$

$$= 123810.70 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 103802. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 50.4602) \cdot (14.9797 - 1.6) \cdot 0.7 \cdot 118$$

$$= 175181. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 190663. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (103802 + 175181) = 278983 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (103802 + 0 + 190663 + 0) = 294465 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (103802 + 190663 + 0) = 294465 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 278983 N	, must exceed W = 46917 N	or W1 = 90974 N
Path 2-2 = 294464 N	, must exceed W = 46917 N	or W2 = 123810 N
Path 3-3 = 294464 N	, must exceed W = 46917 N	or W3 = 123810 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.7259 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.8740 mm

Input Echo, WRC297 Item 1, Description: N5-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4500.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	14.9797	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.40	MPa
Radial Load	P	7500.00	N
Circumferential Shear	Vc	5000.00	N
Longitudinal Shear	Vl	5000.00	N
Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.630) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2256.8(12.0 - 1.6))}$$

$$= 619.865 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4513.601 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	10.400 mm
Nozzle Thickness used	(t) =	13.380 mm
T / t	=	0.777
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.528$$

Nr/P	=	0.156
Mr/P	=	0.292
M0/P	=	0.095
N0/P	=	0.170
MrD/Mc	=	0.300
NrDL/Mc	=	0.115
M0d/Mc	=	0.109
N0DL/Mc	=	-0.067
MrD/Ml	=	0.261
NrDL/Ml	=	0.115
M0D/Ml	=	0.094
N0DL/Ml	=	0.019

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-39	39	-39	39
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-2	-2	2	2
Outplane Bending (Ml)	-63	63	63	-63
Normal Pressure Stress	86	86	86	86
Outplane Stress Summary	-31	176	101	52

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-10	-10	-10	-10
Inplane Bending (P)	-121	121	-121	121
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-13	-13	13	13
Inplane Bending (ML)	-177	177	177	-177
Inplane Pressure Stress	43	43	43	43
Inplane Stress Summary	-279	318	100	-10

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
(Shear stress normal to longitudinal plane)				
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	9	9	9	9
Shear Stress Summary	12	12	6	6

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
(Stress Intensities)				
Two * Max Shear Stress	280	319	107	64

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
(Stresses Normal to circumferential plane)				
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-39	39	-39	39
Outplane Membrane (Mc)	7	7	-7	-7
Outplane Bending (Mc)	-74	74	74	-74
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43
Outplane Stress Summary	-74	152	58	-11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
(Stresses parallel to circumferential plane)				
Inplane Membrane (P)	-10	-10	-10	-10
Inplane Bending (P)	-121	121	-121	121
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-203	203	203	-203
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	86	86	86	86
Inplane Stress Summary	-263	388	170	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
(Shear stress normal to circumferential plane)				
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	9	9	9	9



Shear Stress Summary	6	6	12	12
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	263	388	171	29
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1

Hoop Stress Summary	-12	-12	-8	-7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-53	53	-53	53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-83	83	83	-83
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-152	121	42	-16
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	153	134	52	18
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	7	7	-7	-7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1



Hoop Stress Summary -2 -2 -18 -17

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses Normal to
pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-53	53	-53	53
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-99	99	99	-99
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary -168 137 59 -32

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary 5 5 9 9

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress 169 140 79 36

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	86	86	86	86	86	86	86	86
Circ. Pl (SUS)	-13	-13	-9	-9	-23	-23	3	3
Circ. Q (SUS)	-102	102	24	-24	-324	324	82	-82
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-23	-23	3	3	-4	-4	-18	-18
Long. Q (SUS)	-298	298	56	-56	-113	113	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	86.0	86.0	86.0	86.0	86.0	86.0	86.0	86.0
Pm+Pl (SUS)	73.1	73.1	77.1	77.1	63.2	63.2	89.1	89.1
Pm+Pl+Q (Total)	278.5	318.8	108.5	64.5	261.3	387.2	172.1	27.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	86.00	138.00	Passed
Pm+Pl (SUS)	89.06	207.00	Passed
Pm+Pl+Q (TOTAL)	387.21	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:****Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	1	1	1	1	1	1	1
Circ. Pl (SUS)	-13	-13	-9	-9	-4	-4	-18	-18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-15	-15	13	13	-15	-15	13	13
Long. Q (SUS)	-136	136	30	-30	-152	152	46	-46
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm+Pl (SUS)	16.0	16.0	21.4	21.4	15.3	15.3	30.3	30.3
Pm+Pl+Q (Total)	151.6	134.2	52.0	19.2	167.2	140.4	78.1	37.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1.00	118.00	Passed
Pm+Pl (SUS)	30.27	177.00	Passed
Pm+Pl+Q (TOTAL)	167.15	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N6-4"

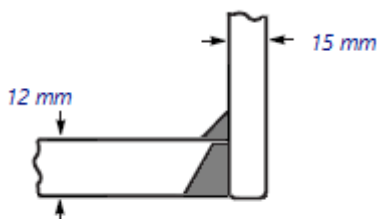
From: 40

Pressure for Reinforcement Calculations	P	0.3991	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	2849.9998	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7650.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	414.1480	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N6-4"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.500 in.
 Actual Thickness Used in Calculation 0.590 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.4 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
 $= 6.5224 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.4 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.4)$
 $= 0.1930 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4751 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	175.0814	mm
Parallel to Vessel Wall, opening length	d	87.5407	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000	mm

Weld Strength Reduction Factor [fr1]:

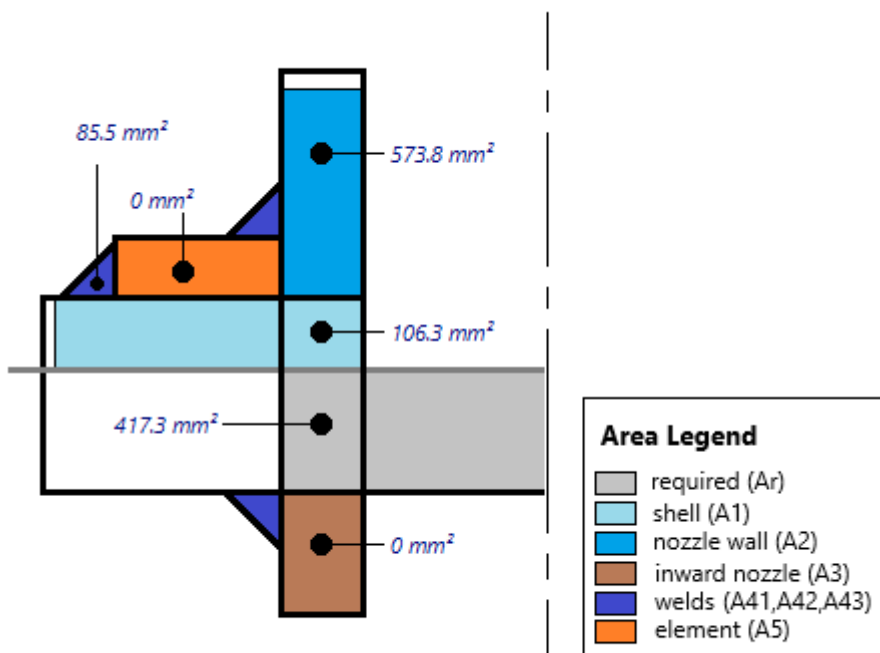
$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

**Figure UG 37.1 Nozzle Areas**

Results of Nozzle Reinforcement Area Calculations: (mm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	596.271	417.276	NA
Area in Shell	A1	324.408	106.342	NA
Area in Nozzle Wall	A2	586.338	573.795	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.509	85.509	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	996.255	765.646	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(87.5407 \cdot 9.1289 \cdot 1 + 2 \cdot 13.3797 \cdot 9.1289 \cdot 1(1-0.86))$
 $= 417.276 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 87.541(1.0 \cdot 10.4 - 1.0 \cdot 9.129) - 2 \cdot 13.38$
 $(1.0 \cdot 10.4 - 1.0 \cdot 9.1289) \cdot (1 - 0.855)$
 $= 106.342 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 26.0)(13.38 - 0.48) 0.8551$
 $= 573.795 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 85.509 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.0751 \text{ mm}$
Wall Thickness per UG16(b), $tr16b = 4.1000 \text{ mm}$
Wall Thickness, shell/head, internal pressure $trb1 = 8.1224 \text{ mm}$
Wall Thickness $tb1 = \max(trb1, tr16b) = 8.1224 \text{ mm}$
Wall Thickness, shell/head, external pressure $trb2 = 2.4440 \text{ mm}$
Wall Thickness $tb2 = \max(trb2, tr16b) = 4.1000 \text{ mm}$
Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.858, \max(8.1224, 4.1)]$
 $= 6.8578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.0751, 6.8578)$
 $= 6.8578 \text{ mm}$

Available Nozzle Neck Thickness = 14.9797 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	22.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	272.1 MPa	Passed
Occasional	:	0.6,	Allowable	:	156.9 MPa	Passed
Shear	:	16.3,	Allowable	:	82.6 MPa	Passed



The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 14.98$, $t_r = 0.193$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.014$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 6.522$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.627$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.40/5.10 = 0.078$

Weld Size Calculations, Description: N6-4"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 10.4000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (417.2758 - 106.3423 + 2 \cdot 13.3797 \cdot 0.8551 \cdot (1.0 \cdot 10.4 - 9.1289)) \cdot 138)$
 $= 46917.62 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (573.7946 + 0.0 + 85.5088 - 0.0 \cdot 0.86) \cdot 138$
 $= 90974.48 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (573.7946 + 0.0 + 85.5088 + (237.9682)) \cdot 138$
 $= 123810.70 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (573.7946 + 0.0 + 85.5088 + 0.0 + (237.9682)) \cdot 138$
 $= 123810.70 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 103802. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 50.4602) \cdot (14.9797 - 1.6) \cdot 0.7 \cdot 118$$

$$= 175181. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 190663. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (103802 + 175181) = 278983 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (103802 + 0 + 190663 + 0) = 294465 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (103802 + 190663 + 0) = 294465 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 278983 N	, must exceed W = 46917 N	or W1 = 90974 N
Path 2-2 = 294464 N	, must exceed W = 46917 N	or W2 = 123810 N
Path 3-3 = 294464 N	, must exceed W = 46917 N	or W3 = 123810 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.7259 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.8740 mm

Input Echo, WRC297 Item 1, Description: N6-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4500.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	14.9797	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.40	MPa
Radial Load	P	7497.10	N
Circumferential Shear	Vc	4998.08	N
Longitudinal Shear	Vl	4998.08	N
Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.630) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2256.8(12.0 - 1.6))}$$

$$= 619.865 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4513.601 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	10.400 mm
Nozzle Thickness used	(t) =	13.380 mm
T / t	=	0.777
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.528$$

Nr/P	=	0.156
Mr/P	=	0.292
M0/P	=	0.095
N0/P	=	0.170
MrD/Mc	=	0.300
NrDL/Mc	=	0.115
M0d/Mc	=	0.109
N0DL/Mc	=	-0.067
MrD/Ml	=	0.261
NrDL/Ml	=	0.115
M0D/Ml	=	0.094
N0DL/Ml	=	0.019

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-39	39	-39	39
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-2	-2	2	2
Outplane Bending (Ml)	-63	63	63	-63
Normal Pressure Stress	86	86	86	86
Outplane Stress Summary	-31	176	101	52

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-10	-10	-10	-10
Inplane Bending (P)	-121	121	-121	121
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-13	-13	13	13
Inplane Bending (ML)	-177	177	177	-177
Inplane Pressure Stress	43	43	43	43
Inplane Stress Summary	-279	317	100	-10

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
(Shear stress normal to longitudinal plane)				
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	9	9	9	9
Shear Stress Summary	12	12	6	6

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
(Stress Intensities)				
Two * Max Shear Stress	280	318	107	64

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
(Stresses Normal to circumferential plane)				
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-39	39	-39	39
Outplane Membrane (Mc)	7	7	-7	-7
Outplane Bending (Mc)	-74	74	74	-74
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43
Outplane Stress Summary	-74	152	58	-11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
(Stresses parallel to circumferential plane)				
Inplane Membrane (P)	-10	-10	-10	-10
Inplane Bending (P)	-121	121	-121	121
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-203	203	203	-203
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	86	86	86	86
Inplane Stress Summary	-263	388	170	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
(Shear stress normal to circumferential plane)				
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	9	9	9	9



Shear Stress Summary	6	6	12	12
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	263	388	172	29
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1

Hoop Stress Summary	-12	-12	-8	-7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-53	53	-53	53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-83	83	83	-83
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-152	121	43	-16
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	153	134	52	18
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	7	7	-7	-7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	1	1	1



Hoop Stress Summary -2 -2 -18 -17

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses Normal to
pipe cross-section)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-53	53	-53	53
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-99	99	99	-99
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary -168 137 59 -32

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Shear stress)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary 5 5 9 9

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Two * Max Shear Stress 169 140 79 36

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	86	86	86	86	86	86	86	86
Circ. Pl (SUS)	-13	-13	-9	-9	-23	-23	3	3
Circ. Q (SUS)	-102	102	24	-24	-324	324	82	-82
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-23	-23	3	3	-4	-4	-18	-18
Long. Q (SUS)	-298	298	56	-56	-113	113	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	86.0	86.0	86.0	86.0	86.0	86.0	86.0	86.0
Pm+Pl (SUS)	73.1	73.1	77.1	77.1	63.2	63.2	89.1	89.1
Pm+Pl+Q (Total)	278.5	318.8	108.5	64.5	261.3	387.2	172.1	27.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	86.00	138.00	Passed
Pm+Pl (SUS)	89.06	207.00	Passed
Pm+Pl+Q (TOTAL)	387.21	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:****Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	1	1	1	1	1	1	1
Circ. Pl (SUS)	-13	-13	-9	-9	-4	-4	-18	-18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-15	-15	13	13	-15	-15	13	13
Long. Q (SUS)	-136	136	30	-30	-152	152	46	-46
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm+Pl (SUS)	16.0	16.0	21.4	21.4	15.3	15.3	30.3	30.3
Pm+Pl+Q (Total)	151.6	134.2	52.0	19.2	167.2	140.4	78.1	37.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1.00	118.00	Passed
Pm+Pl (SUS)	30.27	177.00	Passed
Pm+Pl+Q (TOTAL)	167.15	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K1A-2"

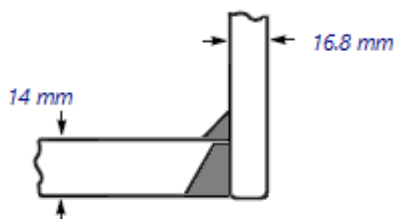
From: 50

Pressure for Reinforcement Calculations	P	0.3991	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	1825.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		10150.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1A-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.4 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
= 6.5224 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{0.4}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0782 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4258 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	109.2000 mm
Parallel to Vessel Wall	Rn+tn+t	54.6000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.0000 mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

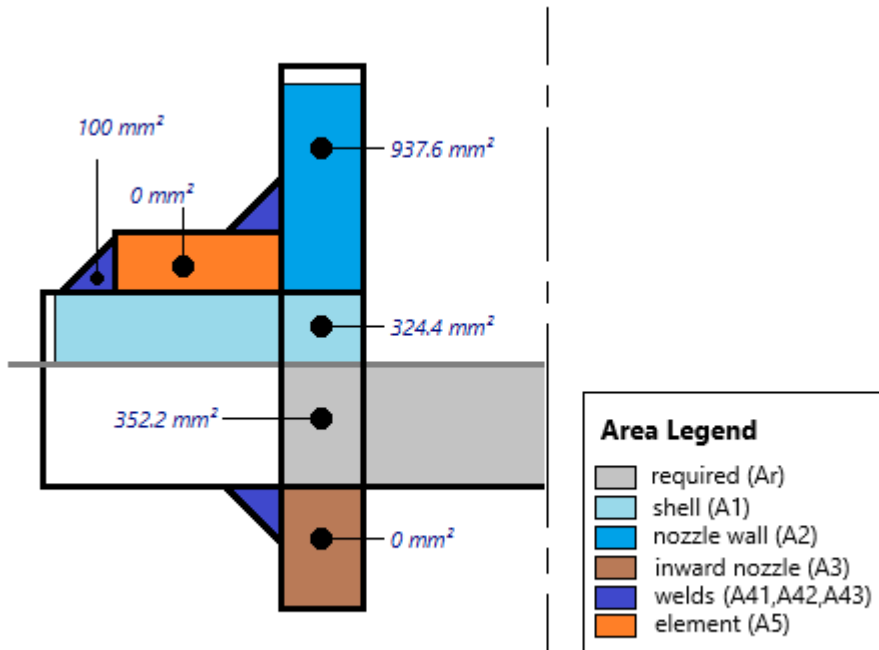


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	352.212	204.936	NA
Area in Shell	A1	324.441	265.500	NA
Area in Nozzle Wall	A2	937.552	916.003	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1361.993	1281.503	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (54.0 \cdot 6.5224 \cdot 1.0 + 2 \cdot 15.2 \cdot 6.5224 \cdot 1.0 \cdot (1 - 1.0))$
 $= 352.212 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 55.2(1.0 \cdot 12.4 - 1.0 \cdot 6.522) - 2 \cdot 15.2$
 $(1.0 \cdot 12.4 - 1.0 \cdot 6.5224) \cdot (1 - 1.0)$
 $= 324.441 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 31.0)(15.2 - 0.08)1.0$
 $= 937.552 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$



= 100.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0258$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.1224$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.1224$ mm
 Wall Thickness, shell/head, external pressure $trb_2 = 2.4440$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000$ mm
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$
 = $\min[6.4, \max(8.1224, 4.1)]$
 = 6.4000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$
 = $\max(2.0258, 6.4)$
 = 6.4000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 14.0$, $t_r = 6.522$, $c = 1.6$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.526$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -27 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.40/5.10 = 0.078$

Weld Size Calculations, Description: K1A-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 12.4000$ mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= $\max(0, (A-A_1+2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 = $\max(0, (352.2119 - 324.4412 + 2 \cdot 15.2 \cdot 1.0 \cdot (1.0 \cdot 12.4 - 6.5224))) \cdot 138$
 = 28486.94 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= $(A_2+A_5+A_4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 = $(937.5521 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 = 143167.44 N

Weld Load [W2]:

= $(A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 = $(937.5521 + 0.0 + 100.0 + (376.96)) \cdot 138$



$$= 195182.55 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (937.5521 + 0.0 + 100.0 + 0.0 + (376.96)) \cdot 138$$

$$= 195182.55 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 89638. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 159591. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 162449. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (89638 + 159591) = 249230 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (89638 + 0 + 162449 + 0) = 252087 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (89638 + 162449 + 0) = 252087 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 249229 N	, must exceed W = 28486 N	or W1 = 143167 N
Path 2-2 = 252087 N	, must exceed W = 28486 N	or W2 = 195182 N
Path 3-3 = 252087 N	, must exceed W = 28486 N	or W3 = 195182 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.3957 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 492.0436 mm

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Input, Nozzle Desc: K1B-2"

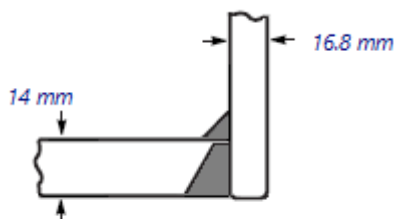
From: 50

Pressure for Reinforcement Calculations	P	0.4378	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4500.00	mm
Design Length of Section	L	1825.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	1.6000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		10150.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1B-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.44 \cdot 2251.6001 / (138 \cdot 1.0 - 0.6 \cdot 0.44)$
= 7.1575 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{0.44}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0858 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2975 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	109.2000	mm
Parallel to Vessel Wall	Rn+tn+t	54.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

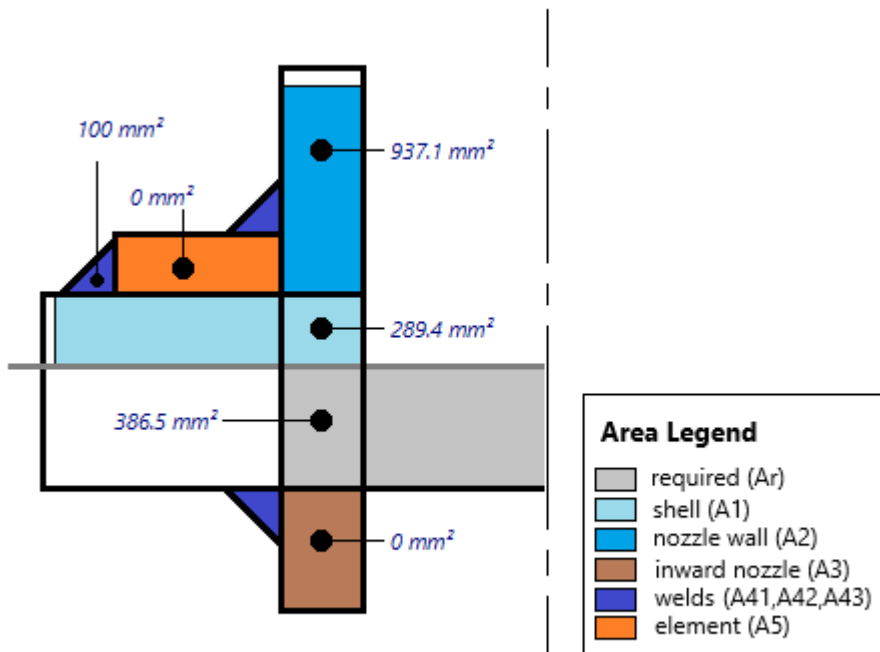


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	386.503	204.936	NA
Area in Shell	A1	289.388	265.500	NA
Area in Nozzle Wall	A2	937.080	923.956	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1326.468	1289.456	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (54.0 \cdot 7.1575 \cdot 1.0 + 2 \cdot 15.2 \cdot 7.1575 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 386.503 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 55.2(1.0 \cdot 12.4 - 1.0 \cdot 7.157) - 2 \cdot 15.2$$

$$(1.0 \cdot 12.4 - 1.0 \cdot 7.1575) \cdot (1 - 1.0)$$

$$= 289.388 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 31.0)(15.2 - 0.09)1.0$$

$$= 937.080 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



= 100.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.8975 mm
Wall Thickness per UG16(b),	tr16b = 4.1000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.7575 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.7575 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.4440 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.1000 mm
Wall Thickness per table UG-45	tb3 = 6.4000 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
 = min[6.4, max(8.7575, 4.1)]
 = 6.4000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(1.8975, 6.4)
 = 6.4000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 14.0, tr = 7.157, c = 1.6 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.577, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-22 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -22 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.44/5.10 = 0.086

Weld Size Calculations, Description: K1B-2"

Intermediate Calc. for nozzle/shell Welds Tmin 12.4000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
 = max(0, (386.5033 - 289.3878 + 2 * 15.2 * 1.0 *
 (1.0 * 12.4 - 7.1575))138)
 = 35391.76 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= (A2+A5+A4-(Wi-Can/.707)^2*fr2)*Sv
 = (937.0804 + 0.0 + 100.0 - 0.0 * 1.0) * 138
 = 143102.34 N

Weld Load [W2]:

= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
 = (937.0804 + 0.0 + 100.0 + (376.96)) * 138



$$= 195117.45 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (937.0804 + 0.0 + 100.0 + 0.0 + (376.96)) \cdot 138$$

$$= 195117.45 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 89638. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 159591. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 162449. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (89638 + 159591) = 249230 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (89638 + 0 + 162449 + 0) = 252087 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (89638 + 162449 + 0) = 252087 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 249229 N	, must exceed W = 35391 N	or W1 = 143102 N
Path 2-2 = 252087 N	, must exceed W = 35391 N	or W2 = 195117 N
Path 3-3 = 252087 N	, must exceed W = 35391 N	or W3 = 195117 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.3957 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.7706 mm

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Input, Nozzle Desc: M2-24"

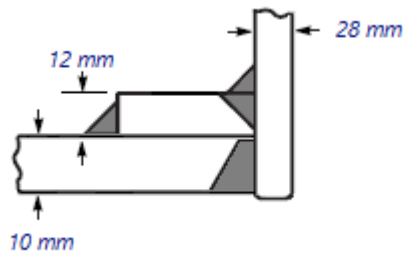
From: 70

Pressure for Reinforcement Calculations	P	0.4160	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	4500.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm
Head Internal Corrosion Allowance	c	1.6000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	909.6000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		150.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M2-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.102 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.42 \cdot 4503.2 \cdot 0.8994 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.42)$
 $= 6.1068 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.42 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.42)$
 $= 0.9178 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0288 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1113.6000 mm
Parallel to Vessel Wall, opening length	d	556.8000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		21.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

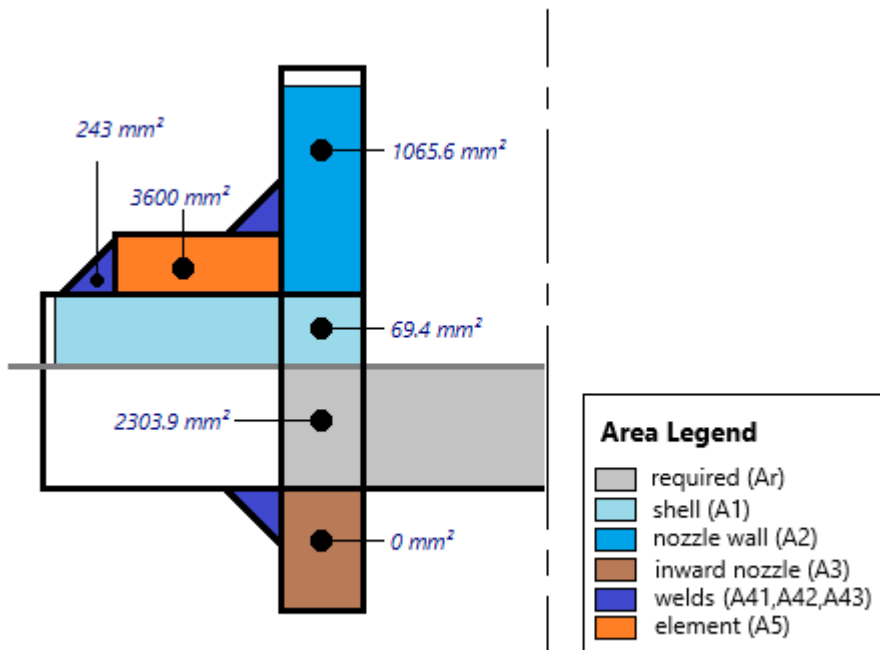


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3400.289	2303.877	NA
Area in Shell	A1	1276.831	69.364	NA
Area in Nozzle Wall	A2	1070.254	1065.592	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	243.000	243.000	NA
Area in Element	A5	3600.000	3600.000	NA
TOTAL AREA AVAILABLE	Atot	6190.084	4977.956	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	686.7601	12.0000 mm
Based on given Pad Diameter:	909.6000	3.0864 mm
Based on Shell or Nozzle Thickness:	702.1921	10.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(556.8 \cdot 8.2754 \cdot 1 + 2 \cdot 26.4 \cdot 8.2754 \cdot 1(1 - 1.0))$
 $= 2303.877 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 556.8(1.0 \cdot 8.4 - 1.0 \cdot 8.275) - 2 \cdot 26.4$
 $(1.0 \cdot 8.4 - 1.0 \cdot 8.2754) \cdot (1 - 1.0)$
 $= 69.364 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 21.0) \cdot (26.4 - 1.03) \cdot 1.0$$

$$= 1065.592 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot Fr3 + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot fr2 + Wp^2 \cdot fr4$$

$$= (99.0) \cdot 1.0 + (0.0) \cdot 1.0 + 12.0^2 \cdot 1.0$$

$$= 243.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr4$$

$$= (909.6 - 609.6) \cdot 12.0 \cdot 1.0$$

$$= 3600.000 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 2.6288 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 2.6288 \text{ mm}$$

Available Nozzle Neck Thickness = 28.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 28.0$, $t_r = 0.918$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A $22 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 12.0$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.727$, Temp. Reduction = 15 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D $-48 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 12.0$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.727$, Temp. Reduction = 15 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D $-48 \text{ }^\circ\text{C}$

Shell to Pad Weld Junction at Pad OD, min(Curve:D, Curve:B)

Govrn. thk, $t_g = 10.0$, $t_r = 6.107$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.727$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B $-29 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-44 \text{ }^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:D, Curve:B)

Govrn. thk, $t_g = 10.0$, $t_r = 6.107$, $c = 1.6 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.727$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B $-29 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-44 \text{ }^\circ\text{C}$

Gov. MDMT of the Nozzle : $-44 \text{ }^\circ\text{C}$
 Gov. MDMT of the Reinforcement Pad : $-44 \text{ }^\circ\text{C}$
 Gov. MDMT of the nozzle to shell joint welded assembly : $-44 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.42/1.96 = 0.212

Weld Size Calculations, Description: M2-24"

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 10.4000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	5.2000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (2303.8774 - 69.3643 + 2 \cdot 26.4 \cdot 1.0 \cdot (1.0 \cdot 8.4 - 8.2754)) \cdot 138)$
 $= 309238.69 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1065.5916 + 3600.0 + 243.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 677315.81 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1065.5916 + 0.0 + 100.0 + (443.52)) \cdot 138$
 $= 222034.52 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1065.5916 + 0.0 + 243.0 + 3600.0 + (443.52)) \cdot 138$
 $= 738515.25 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 647445. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 909.6 \cdot 12.0 \cdot 0.49 \cdot 138$
 $= 1159263. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 291.6) \cdot (28.0 - 1.6) \cdot 0.7 \cdot 138$
 $= 2336046. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 12.0 \cdot 0.74 \cdot 138$
 $= 1173329. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (10.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 821331. \text{ N}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1159263 + 2336047) = 3495310 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (647445 + 1173329 + 821331 + 0) = 2642105 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (1159263 + 821331 + 0) = 1980594 \text{ N}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 3495309 \text{ N} , \text{ must exceed } W = 309238 \text{ N} \quad \text{or } W1 = 677315 \text{ N}$$

$$\text{Path 2-2} = 2642105 \text{ N} , \text{ must exceed } W = 309238 \text{ N} \quad \text{or } W2 = 222034 \text{ N}$$

$$\text{Path 3-3} = 1980593 \text{ N} , \text{ must exceed } W = 309238 \text{ N} \quad \text{or } W3 = 738515 \text{ N}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 11.4633 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 271.4634 mm

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 * t_{nom} / R_f * (1 - R_f / R_o)) \quad 4.814 \%$$

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: